

Low Power Voltage Reference Circuits for Internet of Things Applications

by

U Chi Wa

Doctor of Philosophy in Electrical and Computer Engineering

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Abstract

The bandgap voltage reference (BGR) is an important building block in voltage regulators and data converters. It should provide an accurate and stable voltage reference despite process, voltage, and temperature (PVT) variations. For Internet-of-Things (IoT) applications, due to power budget limitations, the development trend of BGR design is to reduce current consumption while maintaining high accuracy [i.e., temperature coefficient (TC) and σ/μ].

The BGR is a well-known structure that is widely used to provide voltage reference in integrated circuits (ICs). However, the supply voltage of the BGR is limited by the base-emitter voltage (V_{EB}) of the BJT, which must be greater than 0.75V. This limitation is undesirable for IoT applications, as harvesting sources that can extend system runtime typically provide only low voltage supplies. Furthermore, the current gain of PNP bipolar junction transistors (BJTs), which is very sensitive to process drift, becomes the dominant error source of BGR design in modern complementary metal-oxide-semiconductor (CMOS) processes. Current gain compensation should be applied to eliminate this error, but it requires a lot of power. An alternative solution is two-point trimming after IC fabrication, but this requires a lot of time and labor costs.

To achieve a BGR design with low power consumption and high accuracy, this thesis focuses on the design of the BGR structure, and the characteristics of the device used. Among the existing literatures, the following problems and issues still need to

be further explored and studied:

1. The large area overhead problem of the BGR design for the low power application.
2. The suppression of nonlinearity and parasitic effects in switched-capacitor BGRs.
3. The effect of the process-related errors in PNP BJT and its compensation solutions.
4. Auto-calibration control techniques for reducing trimming time and labor cost.

In the thesis, we first present a switched-capacitor network (SCN) based BGR design that can achieve low power consumption without using a large amount of resistance or chip area. A comprehensive study of the second-order effect of the SCN is presented and eliminated during the design process. A 2x charge pump (CP), where the ratio of the flying capacitor and the loading capacitor is 5:3 to minimize the CP output ripple, is employed to supply the core in order to relax V_{DD} from 0.9V to 0.5V. A proportional-to-absolute-temperature (PTAT) current source is proposed to bias the BJT in order to reduce its nonlinearity. A voltage divider SCN with low leakage consideration is designed to form the complementary-to-absolute-temperature (CTAT) voltage and reduce the nonlinearity of its coefficient, thereby improving the TC and extend the temperature range.

To further improve the first SCN based BGR design, in our second design, we employ a dual PTAT clock topology to achieve both high precision and low power

consumption over a wide temperature range, while relaxing the capacitor size requirements. Specifically, the fast PTAT clock assists in reducing the output voltage ripple of the dual-branch interleaved $2\times\text{CP}$. Meanwhile, the slow PTAT clock controls the voltage divider SCN to simultaneously relax the settling error at low temperature and the leakage-induced error at high temperature, resulting in lower power consumption and a smaller TC. Additionally, a replica generation branch in the series-parallel SCN is proposed to improve the BGR output TC due to the finite settling time during switching.

In our third design, we present an auto-calibration technique for the current-based BGR based on a digital assisted auto-calibration loop for calibration cost reduction. We first present the theoretical study on the process variation induced V_{EB} and ΔV_{EB} variations in the BJT, which contribute to the residual errors in the reference voltage (V_{REF}) and its TC after applying the conventional one-point trimming. Based on the study, we further propose an automatic one-point trimming methodology using a current digital-to-analog converter (IDAC), which can simultaneously relax the V_{EB} and ΔV_{EB} variations, resulting in a small drift in both V_{REF} and its TC after auto-calibration.

Declaration

I declare that the thesis here submitted is original except for the source materials explicitly acknowledged and that this thesis as a whole, or any part of this thesis has not been previously submitted for the same degree or a different degree.

I also acknowledge that I have read and understood the Rules on Handling Student Academic Dishonesty and the Regulations of the Student Discipline of the University of Macau.

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List of Abbreviation

BGR	Bandgap voltage reference
CMP	Chopped comparator
CP	Charge pump
CTAT	Complementary to absolute temperature
EN	Enable signal
IC	Integrated circuit
IDAC	Current digital-to-analog converter
IoT	Internet-of-Things
LNR	Line regulation
LSB	Least significant bit
NROC	Normalized rate of change
PSRR	Power supply ripple rejection
PTAT	Proportional to absolute temperature
PVT	Process-voltage-temperature
SCN	Switched-capacitor network
SoC	System on a chip
TC	Temperature coefficient
ULP	Ultra-low-power

Chapter 1 INTRODUCTION

1.1 BACKGROUND OF VOLTAGE REFERENCE

The Internet-of-Things (IoT) is a communication pattern that has emerged in recent years with the aim of improving the utilization of public resources, providing higher quality services to residents, and reducing service costs. This pattern involves the integration of microcontrollers and transceivers for digital communication into everyday devices, along with appropriate protocol stacks that enable communication with other devices and users. As a result, these devices become an integral part of the internet [1.1]. In 2020, there was over 50 billion devices, including smartphones and PCs, connected to the internet, equating to an average of more than six internet-connected devices per person [1.2]. Figure 1-1 illustrates the forecasted market size of IoT in North America. The global IoT market is projected to grow from \$478.36 billion in 2022 to \$2,465.26 billion by 2029 [1.3].

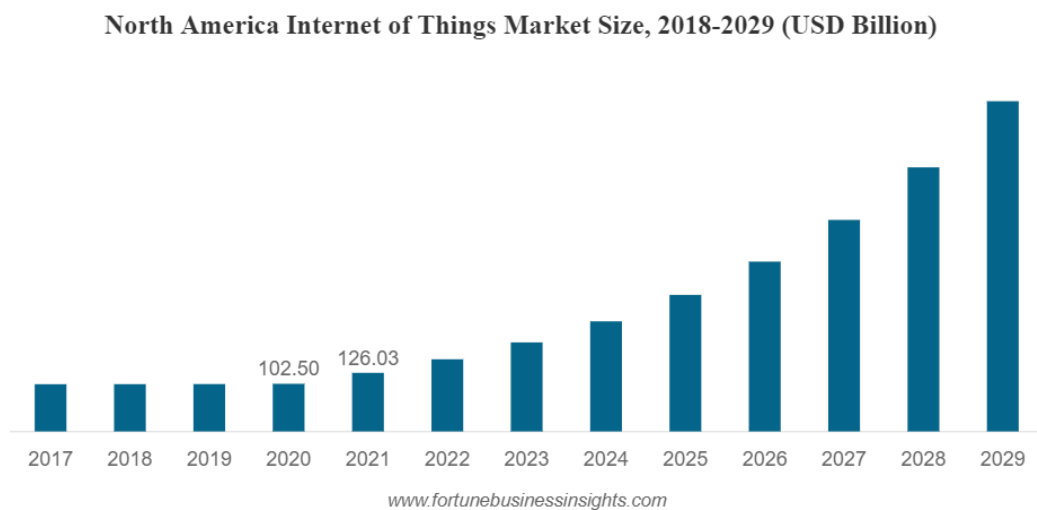


Figure 1-1 The market size forecast of internet of things in North America.

A bandgap voltage reference (BGR) is a crucial component in many analog blocks. As illustrated in Figure 1.2, voltage references are utilized in various fields such as test and measurement, medical, automotive, smart grid and power automation, and industrial applications [1.4]. Given that the accuracy of the system is constrained by the BGR, it is imperative for the BGR to deliver a stable reference voltage that remains constant in the face of variations in process, voltage, and temperature (PVT). Due to the variability of our environment, the placement of IoT devices is often constrained by the location of the measurement signal. This can make it difficult for devices in remote locations to access power. As a result, low power consumption is a critical requirement for IoT applications, as frequent battery replacement would significantly increase maintenance costs [1.6]. Therefore, designing an accurate BGR that operates within limited power constraints is a significant challenge for the BGR design in IoT applications.



Figure 1-2 The voltage reference application [1.4].

1.1.1 SPECIFIC BACKGROUND

The BGR voltage takes a vital role in the performance of the analog and mixed-signal systems, which design specification relies on several factors [1.5].

Temperature coefficient (TC) is the standard metric used for quantifying the performance of BGR voltage against temperature variation, which should be typically expressed in parts-per million per degree Celsius (ppm/°C),

$$TC = \frac{1}{V_{Nominal}} \frac{\Delta V_{REF}}{\Delta T} \times 1,000,000 \quad (1-1)$$

where ΔT is the difference between the lower and upper limit of the temperature range, ΔV_{REF} is the difference of the maximum and minimum value of the reference voltage within the temperature range, and the $V_{Nominal}$ is nominal value of the reference voltage. Also, the temperature range is an essential parameter since the performance of the BGR would become worse in extreme high and low temperature, hence the temperature range should be mentioned together with the TC.

Line regulation (LNR) is a parameter defined as the dc change of output voltage (i.e., reference voltage) due to the change of the input voltage (i.e., supply voltage) and is expressed in percent per volt in general,

$$LNR = \frac{\Delta V_{OUT}}{\Delta V_{IN}} \times 100\% \quad (1-2)$$

Power supply ripple rejection (PSRR) is another parameter that describes the variation of the output voltage because of the change of the input voltage, but the difference is that the PSRR is defined as the variation of the ac ripple in the output

voltage (δV_{OUT}) due to the ac ripple in input voltage (δV_{IN}).

$$PSRR = 20\log\left(\frac{\delta V_{OUT}}{\delta V_{IN}}\right) \quad (1-3)$$

Another critical parameter describing the performance of BGR is the supply voltage range since the supply voltage of a battery would drop as the battery life loss.

The 3σ inaccuracy of a reference is the deviation of the actual output voltage from the desired specification and quantifies the effect of random process variations, mismatch, and package stresses on the dc accuracy of the reference voltage. Since this is a random effect, 3σ inaccuracy can only be specified with statistical analysis of enough sample size [1.7]. The 3σ inaccuracy is defined as,

$$3\sigma \text{ inaccuracy} = \frac{3\sigma_{V_REF}}{\mu_{V_REF}} \quad (1-4)$$

where σ_{V_REF} is standard deviation and μ_{V_REF} is mean value of the voltage reference. One common method for achieving a desired level of accuracy is by trim bits. Trimming is typically performed at room temperature (27°C) for convenience, so the initial accuracy is usually specified at this temperature. For the designs that require high accuracy, a 3σ inaccuracy is specified across the entire temperature range.

In addition, due to the long lifespan required for the IoT device applications, power consumption is a primary concern for the BGRs. This necessitates the use of circuits that operate with low current consumption and low supply voltage. Low current consumption is constrained by the performance of BGR components such as the gain and bandwidth of operational amplifiers as well as the linearity of BJTs. Low power supply voltage is limited by the minimum supply voltage required for

components such as BJTs, which must operate at 700mV at low temperatures.

1.2 LITERATURE REVIEW OF BANDGAP VOLTAGE

REFERENCE DESIGN

Bandgap references can be classified as either voltage-based or current-based. In a current-based BGR, two or more currents are summed together. In contrast, a voltage-based BGR sums two or more voltages together. Generally, current-based BGR are more accurate, while voltage-based BGR consume less power. Both types of bandgap references are widely used in different application scenarios within IoT applications. In the following section, we will review both current-based and voltage-based bandgap references.

1.2.1 CONVENTIONAL BANDGAP VOLTAGE REFERENCE

From the conventional current-based BGR as shown in Figure 1-3, here, we first assume that there is no offset in the amplifier A_1 and no mismatch in the current mirror $M_{P1} - M_{P3}$ for simplicity. With $V_X = V_Y$ and considering the reverse Early Effect [1.8], the current flow through resistor R_1 (i.e., the PTAT current I_{PTAT}) can be expressed as,

$$I_{PTAT} = \frac{\Delta V_{EB}}{R_1} = \frac{V_{EB1} - V_{EB2}}{R_1} = \frac{\eta_F V_T}{R_1} \ln(N) \quad (1-5)$$

where $V_T = kT/q$ is the thermal voltage, η_F is the BJT forward emission coefficient, k is the Boltzmann constant, T is the absolute temperature in Kelvin, q is the electron charge, and N is the area ratio of BJTs Q_1 and Q_2 . I_{PTAT} is copied by the current mirror M_{P3} to bias R_2 and Q_3 to produce the reference voltage V_{REF} , which can be elaborated

as,

$$V_{REF} = V_{EB3} + \gamma \Delta V_{EB} = V_{EB3} + \frac{R_2}{R_1} \eta_F V_T \ln(N) \quad (1-6)$$

There are several limitations to the conventional BGR design. Firstly, the supply voltage must be greater than bandgap voltage of the silicon, resulting in a minimum V_{DD} of $\sim 1.25V$. Due to its high supply voltage requirement, the conventional BGRs are not suitable for IoT applications, which often rely on low voltage power sources such as energy harvesting or solar power. Secondly, the conventional BGR is also known as a first-order compensated reference voltage. In an ideal case, its best TC is greater than $15\text{-ppm}/^\circ\text{C}$, as higher-order components of the CTAT voltage are not compensated. This makes it unsuitable for high-accuracy applications such as sensors. Finally, since ΔV_{EB} is typically considered as a process-insensitive parameter, it is possible to perform PTAT trimming by adjusting γ in equation (1.6) to eliminate voltage drift in V_{EB3} caused by mismatch and process variation. However, this approach only controls the PTAT voltage, which leads to residual error in V_{REF} even after applying the PTAT trimming. Due to the small beta in advanced process, this problem becomes worse.

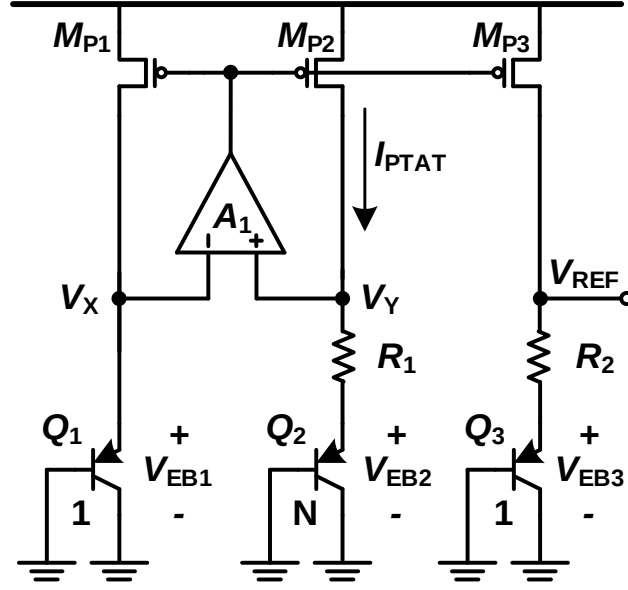


Figure 1-3 Circuit configuration of a conventional current-based BGR [1.9].

1.2.2 SUB-1V BANDGAP VOLTAGE REFERENCES

To overcome the limitations of silicon's bandgap voltage, current-based BGRs are commonly used. Figure 1-4 shows the circuit design for a sub-1V BGR with one amplifier [1.10]. Here, we first assume that there is no offset in amplifier A_1 and no mismatch in the current mirror $M_{P1} - M_{P3}$. Amplifier A_1 ensure that the voltage at its input node is equal to V_{EB1} . As a result, the current flowing through M_{P1} and M_{P2} (i.e., I_{PTAT} and I_{CTAT}) can be expressed as:

$$I_{PTAT} = \frac{\Delta V_{EB}}{R_2} = \frac{V_{EB1} - V_{EB2}}{R_2} = \frac{\eta_F V_T}{R_2} \ln(N) \quad (1-7)$$

$$I_{CTAT} = \frac{V_{EB1}}{R_1} \quad (1-8)$$

where $V_T = kT/q$ is the thermal voltage, η_F is the BJT forward emission coefficient, k is the Boltzmann constant, T is the absolute temperature in Kelvin, q is the electron

charge, N is the area ratio of BJTs Q_1 and Q_2 , and $R_1 = R_{1a} = R_{1b}$.

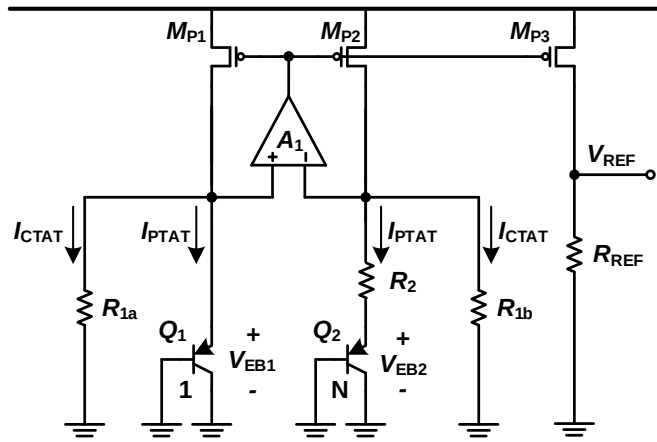


Figure 1-4 Circuit configuration of the sub-1V BGR with one amplifier [1.10].

These two currents are copied by M_{P3} to bias R_{REF} to generate the reference voltage

V_{REF} , which can be evaluated as,

$$V_{REF} = (I_{PTAT} + I_{CTAT})R_{REF} = \alpha V_{EB1} + \gamma \Delta V_{EB} = \left(\frac{\eta_F V_T \ln(N)}{R_2} + \frac{V_{EB1}}{R_1} \right) R_{REF} \quad (1-9)$$

Since there is a coefficient α that reduce the voltage level of V_{EB1} , this structure can overcome the limitation of silicon's bandgap voltage (i.e., $\alpha < 0$). Alternatively, the sub-1V BGR with two amplifiers, as shown in Figure 1-5, is also another popular design typology. Here, we first assume that there is no offset in amplifier A_1 and A_2 and no mismatch in the current mirror $M_{P1} - M_{P5}$. Amplifier A_1 and A_2 ensure that the voltage at their input nodes is equal to V_{EB1} and produces I_{PTAT} and I_{CTAT} . These currents can be expressed as,

$$I_{PTAT} = \frac{\Delta V_{EB}}{R_2} = \frac{V_{EB1} - V_{EB2}}{R_2} = \frac{\eta_F V_T}{R_2} \ln(N) \quad (1-10)$$

(1-11)

which can be evaluated as,

(1-12)

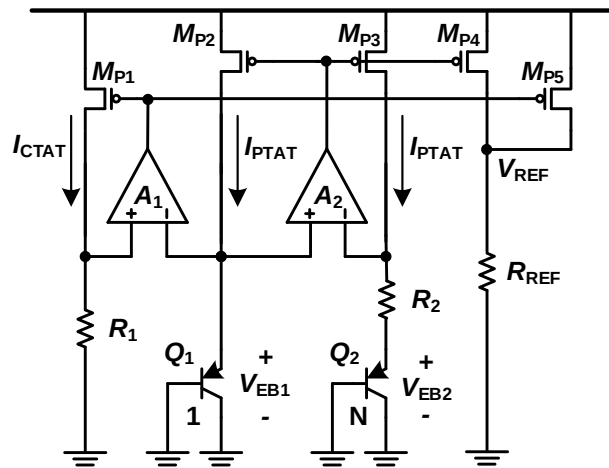


Figure 1-5 Circuit configuration of the sub-1V BGR with two amplifiers [1.11].

From observation, the expression for V_{REF} in the BGR structures shown in Figure 1-4 and Figure 1-5 is the same [i.e., (1-9) and

(1-12)]. However, they are subject to different limitations. For the structure shown in Figure 1-4, which uses only one amplifier and fewer current mirrors, the error caused by mismatch is small compared to the structure shown in Figure 1-5. However, the structure shown in Figure 1-4 uses an extra resistor R_1 , which is much larger than R_2 . This is not favored by sub- μW BGRs because they require a large resistance to reduce current consumption. However, to reduce power consumption to several tens of nW, these resistor-based BGRs require a resistance of around several hundred M Ω ,

which can be difficult to achieve.

1.2.3 SWITCHED-CAPACITOR BANDGAP VOLTAGE REFERENCES

To solve the problem of large resistors in nW resistor-based BGR designs, using a switched-capacitor network (SCN) is a promising solution. For instance, an SCN-based BGR requires only a few pF of capacitance to achieve nW operation, while the resistor-based BGR designs require resistor of around several hundred M Ω . This can save a lot of chip area. Figure 1-6 shows a schematic of the 0.5V SCN-based BGR using 2x charge pump (CP). The 2x CP boosts the input voltage to drive the BJTs Q_1 and Q_2 and provide them with charge to generate V_{EB} . There is an N-fold difference in the area of BJTs Q_1 and Q_2 to generate a PTAT voltage ΔV_{EB} that depends only on physical constants. The SCN scales V_{EB} and ΔV_{EB} by the ratios α and γ to generate V_{REF} .

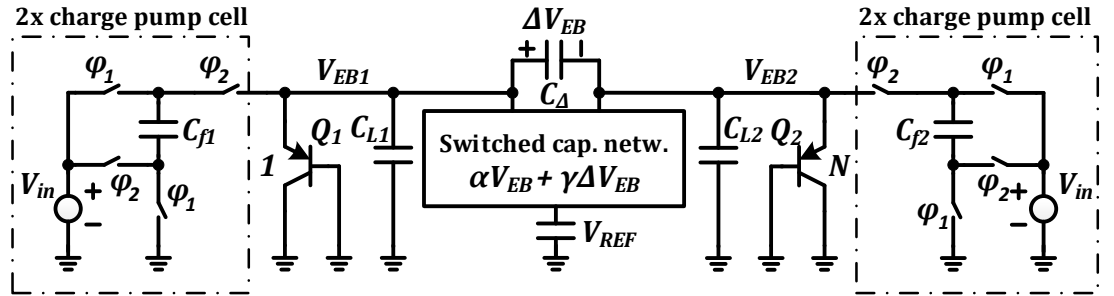


Figure 1-6 Bandgap reference generation from 0.5V V_{in} using 2x CP cell [1.12]

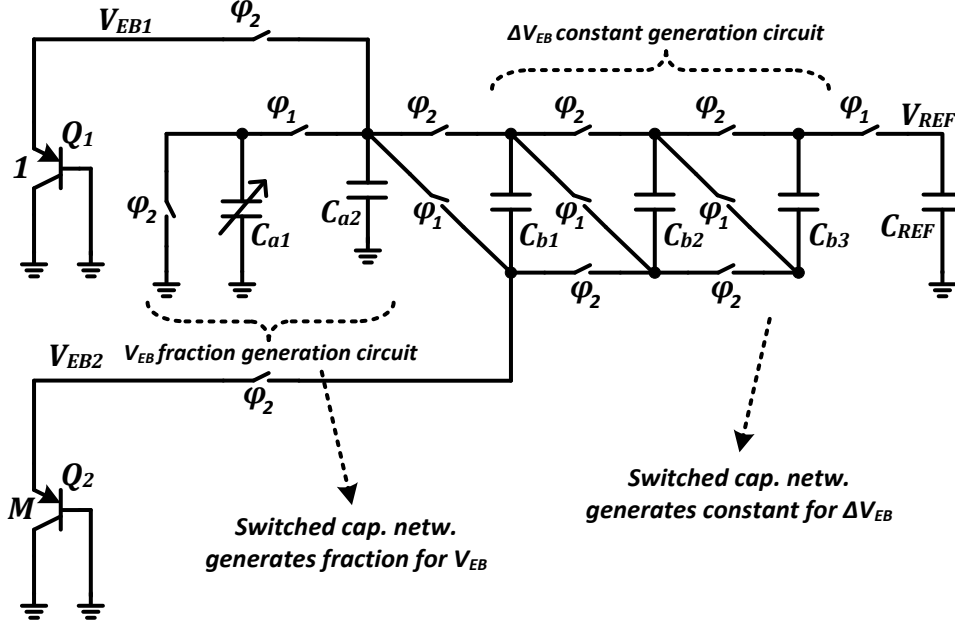


Figure 1-7 Bandgap reference generation from 0.5V V_{in} using 2x CP cell [1.12].

The SCN is shown in Figure 1-7 and uses non-overlapping phases ϕ_1 and ϕ_2 to generate the reference voltage. During phase ϕ_2 , capacitor C_{a1} discharges to ground, V_{EB1} charges capacitor C_{a2} , and the voltage difference ΔV_{EB} charges capacitors C_{b1} , C_{b2} , and C_{b3} . During phase ϕ_1 , capacitors C_{b1} , C_{b2} , and C_{b3} are connected in series to generate $3\Delta V_{EB}$. Capacitors C_{a1} and C_{a2} are also connected in parallel to share the charge and provide the ratio α (i.e., $C_{a2}/(C_{a1}+C_{a2})$). By connecting capacitors C_{a1} and C_{a2} in parallel and capacitors C_{b1} , C_{b2} , and C_{b3} in series, the reference voltage V_{REF} can be generated, which can be expressed as,

$$V_{REF} = \alpha V_{EB1} + \gamma \Delta V_{EB} = \frac{C_{a2}}{C_{a1} + C_{a2}} V_{EB1} + 3\Delta V_{EB} \quad (1-13)$$

where $\gamma=3$ is the number of the serial-parallel capacitor.

There are some limitations to this circuit design. Firstly, because the source of current for the BJT is a discharging capacitor, it results in a non-linear current. This

causes non-linearity in the BJT, resulting in a large TC in its V_{REF} . Secondly, due to the large variation in current leakage from MOSFETs at extremely high and low temperatures, the temperature range of the circuit is narrow. Lastly, several second-order effects, such as bottom-plate parasitic and charge injection in the SCN, may degrade the performance of V_{REF} (such as TC and σ inaccuracy).

1.2.4 HIGH ACCURACY BANDGAP VOLTAGE REFERENCE

As discussed in Section 1.2.1, and 1.2.2, the conventional BGR only cancels the first-order term of the V_{EB} . As a result, its achievable TC is limited to $>15\text{ppm}/^\circ\text{C}$ due to the residual curvature of the V_{EB} [1.13]. Typically, compensation techniques such as curvature compensation (as shown in Figure 1-8) or piecewise curvature correction (as shown in Figure 1-9) are employed to reduce the TC of V_{REF} to $<15\text{ppm}/^\circ\text{C}$.

By using the compensation techniques, a V_{REF} with small TC can be produced. These types of BGR are known as high-order compensated BGRs. However, as shown in Figure 1-10, due to process corner variations, the first-order compensated V_{REF} can drift, resulting in a TC of several ten $\text{ppm}/^\circ\text{C}$. This limits the achievable TC of the high-order compensated BGRs.

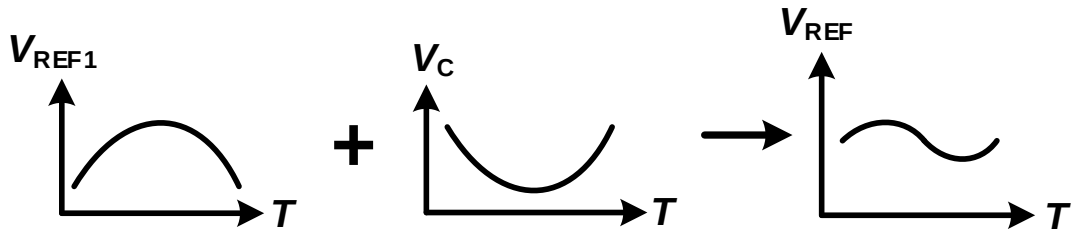


Figure 1-8 Operating principle of the curvature compensation [1.14].

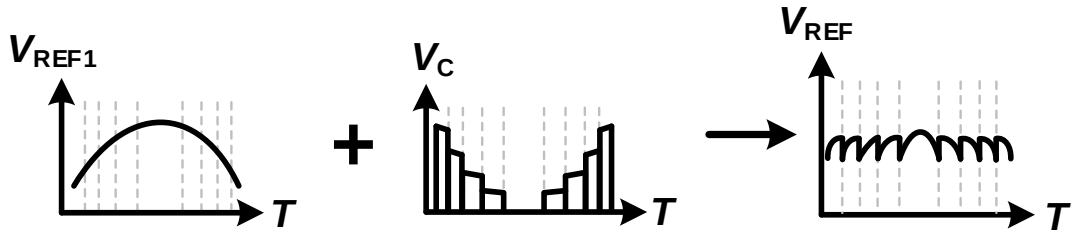


Figure 1-9 Operating principle of the piecewise curvature correction [1.15].

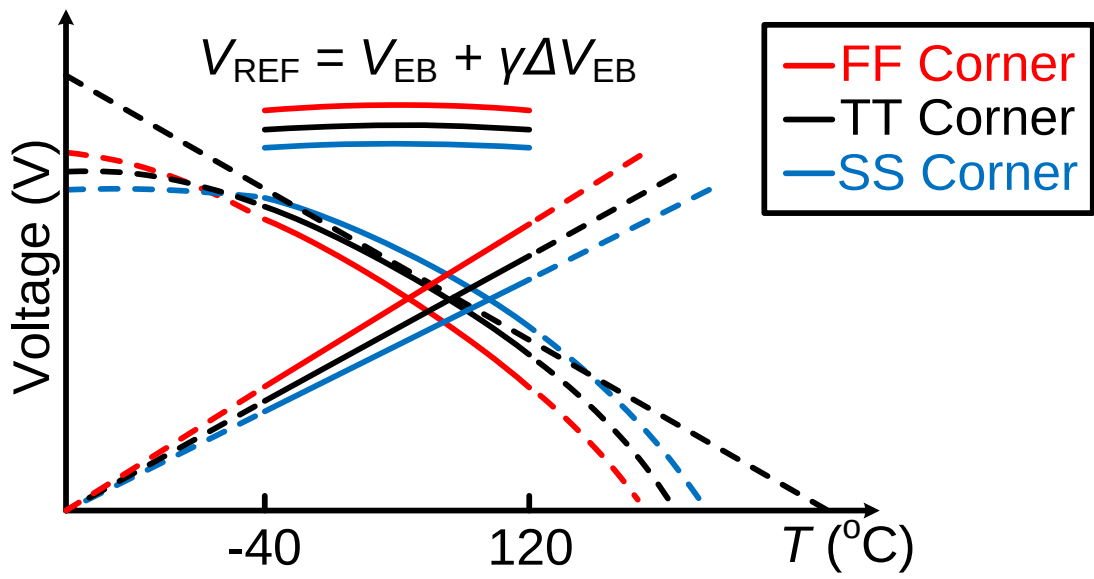


Figure 1-10 Conceptual waveform of first-order compensated V_{REF} across process corner.

To address this issue, two-point trimming [1.15] is usually applied to the high-order compensated BGRs to improve their accuracy. However, this inevitably requires a time-consuming calibration process after wafer-level fabrication, thus significantly increase the calibration labor costs.

1.2.5 SUMMARY

The structures presented in Sections 1.2.1 and 1.2.3 are voltage-based BGR, while

those presented in Sections 1.2.2 and 1.2.4 are current-based BGR. The general characteristics of both current-based and voltage-based BGR are summarized in Table 1-1.

Table 1-1 Comparison of the Current-Based and Voltage-Based BGR		
	Current-based BGR	Voltage-based BGR
Power Consumption	Higher	Lower
TC	Smaller	Larger
3σ inaccuracy	Smaller	Larger
Area	Larger	Smaller

1.3 RESEARCH OBJECTIVES AND CHALLENGES

Existing BGR designs are unable to simultaneously achieve small area consumption, low power consumption, and high accuracy. Additionally, achieving high accuracy or precision using one-point trimming (i.e., adjusting both V_{REF} and its TC) remains a challenging problem in the literature.

To develop a BGR design with low-power consumption and high accuracy properties, the thesis focuses on the design of the BGR structure and the characteristics of BJT. Among the existing literature, the following problems and issues still need to be further exploration and study:

1. The large area consumption overhead of the BGR designs for the low-power applications.
2. The suppression of nonlinearity and parasitic effects in switched-capacitor BGR.

3. A systematic analysis of the effects of process variation-induced voltage and TC drifts in V_{REF} , and its compensation solutions.
4. An automatic calibration control technique to reduce the time-consuming trimming process.

1.4 STRUCTURE OF THE THESIS

This thesis is organized into five chapters. The first chapter provides an overview of the thesis background and reviews the literature on voltage reference structures. It also outlines the motivation, research objectives, and design challenges. Additionally, this chapter includes the Statement of Originality and a list of Respective Publications. Chapters 2 through 4 discuss low power and high accuracy BGR designs. The final chapter presents the conclusions and discusses the possible future work.

Chapter 2 presents a switched-capacitor BGR with nonlinearity suppression, which achieves low power consumption, small area, and small TC. This chapter explains the principle of nonlinearity suppression and presents the circuit architecture and the design consideration. Representative simulation and measurement results are also provided.

Chapter 3 presents a switched-capacitor BGRs with a dual-frequency clock for further reducing TC, power, and area compared with the first design in Chapter 2. This chapter also introduces a replication structure for generating V_{EB} to suppress the settling error and improve TC. The circuit architecture and the design consideration are presented, along with representative simulation and measurement results.

Chapter 4 presents the current-based BGR with auto-calibration. A digital-assisted loop is used to control the current digital-to-analog converter for trimming. This chapter explains the principle of the auto-calibration technique and presents the circuit architecture and design considerations. Simulation and measurement results are also provided.

Finally, **Chapter 5** presents the conclusions of the thesis and discusses the possible future work.

1.5 STATEMENT OF ORIGINALITY AND RESPECTIVE PUBLICATIONS

The research works conducted in this thesis led to the following original contributions:

Patent:

- [1] **Chi-Wa U**, Chon-Fai Lee, Man-Kay Law, Chi-Seng Lam, Rui P. Martins, “Calibration Circuit of Current Mode Bandgap Voltage Reference” ***Chinese Patent for Invention, Granted***, No. 202111304187.0, 31 Jan. 2023.

International Journal Papers:

- [2] **Chi-Wa U**, Man-Kay Law, Rui P. Martins, Chi-Seng Lam, “Sub- μ W Auto-Calibration Bandgap Voltage Reference with 1σ inaccuracy of $\pm 0.12\%$ within -40 to 120°C ”, ***IEEE Journal of Solid-State Circuits***, under major revision.

[SCI]

- [3] **Chi-Wa U**, Man-Kay Law, Chi-Seng Lam, Rui P. Martins, “Switched-Capacitor Bandgap Voltage Reference for IoT Applications”, *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 69, no. 1, pp. 16-29, Jan. 2022. [SCI]
- [4] **Chi-Wa U**, Wen-Liang Zeng, Man-Kay Law, Chi-Seng Lam, Rui P. Martins, “A 0.5-V Supply, 36nW Bandgap Reference with 42ppm/°C Average Temperature Coefficient Within -40°C to 120°C”, *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 67, no. 11, pp. 3656-3669, Nov. 2020. [SCI]
- [5] **Chi-Wa U**, Wen-Liang Zeng, Rui P. Martins, Chi-Seng Lam, “A 2V to 12V Supply Bandgap Reference with 44.9ppm/°C Average Temperature Coefficient within -40°C to 150°C”, *IEEE Transactions on Circuits and Systems II: Express Briefs*, under review. [SCI]
- [6] Cong Liu, **Chi-Wa U**, Rui P. Martins, Chi-Seng Lam, “An 1V Supply, 740nW, 8.7ppm/°C Bandgap Voltage Reference Employing Segmented Curvature Compensation”, *IEEE Transactions on Circuits and Systems I: Regular Papers*, under review. [SCI]
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- [9] Tianzheng Yan, **Chi-Wa U**, Man-Kay Law, Chi-Seng Lam, “A -40°C to 125°C, 1.08 ppm/°C, 918 nW bandgap voltage reference with segmented curvature compensation”, *Microelectronics Journal*, vol. 105, pp.104897, Nov. 2020. [SCI]

International Conference Papers:

- [10] **Chi-Wa U**, Man-Kay Law, Chi-Seng Lam, and Rui P. Martins, “Auto-Calibration Technique for Current-Based Bandgap Voltage Reference”, *2021 IEEE Asian Solid-State Circuits Conference (A-SSCC)*, Busan, Korea, Nov. 2021, pp. 1-3. [Highlighted Paper]
- [11] **Chi-Wa U**, Wen-Liang Zeng, Chi-Seng Lam, “Review and Comparison of Integrated Inductive-Based Hybrid Step-Down DC-DC Converter Under CCM Operation”, *2019 IEEE Asia Pacific Conference on Circuits and Systems (APCCAS)*, Bangkok, Thailand, Nov. 2019, pp. 49-52.
- [12] Wen-Liang Zeng, **Chi-Wa U**, Chi-Seng Lam, “Review and Comparison of Integrated Inductive-Based Hybrid Step-Up DC-DC Converter under CCM”,

2019 IEEE 62nd International Midwest Symposium on Circuits and Systems (MWSCAS), Dallas, USA, Oct. 2019, pp. 730-733.

Awards:

- [13] Wen-Liang Zeng, **Chi-Wa U**, “Control and Design of Integrated KY DC-DC Converter”, **The Second Prize of South China District**, The China Graduate Electronic Design Competition, China, 08/2020.
- [14] **Chi-Wa U**, “Auto-Calibration Technique for Current-Based Bandgap Voltage Reference”, **Best Poster Award of 2nd AMSV Industrial Group Forum**, Hengqin, China, 02/12/2022.

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Chapter 2 A 0.5-V SUPPLY, 36nW BANDGAP REFERENCE WITH 42ppm/°C AVERAGE TEMPERATURE COEFFICIENT WITHIN -40°C to 120°C

2.1 INTRODUCTION

Due to the inconsistency of our environment, the location of the IoT device is limited by the position of the measurement signal. In this case, the device is difficult to access power at its remote location. Consequently, low energy consumption is the most general constraint in the IoT application as frequent battery replacement will significantly increase the maintenance cost [2.1]. As a result, the self-powered device has attracted more interest over the last decade, in which energy harvesting techniques are considered as a promising and achievable technology. In addition, the start-up voltage of an ULP system is always limited by the minimum voltage at which the BGR starts operation [2.2]. Hence, the runtime of the ULP system can be extended by lowering the supply voltage of BGR, such that the harvested voltage as low as 0.5V [2.3] can be used to supply the BGR.

The SCN based BGR [2.2], [2.4] – [2.6] is suitable for the IoT application due to its low power consumption characteristics. The work in [2.4] employed the reverse bandgap concept to reduce the supply voltage down to 0.75V; however, due to the emitter-base voltage V_{EB} of the PNP BJT and the drain-source saturation voltage V_{SAT} of the MOS transistor ($V_{EB} + V_{SAT}$) constrain, its temperature range is narrow. The studies in [2.2], [2.5] – [2.6] employed a 2x CP SCN to reduce the supply voltage by

half. However, due to the nonlinearity of the BJT biasing [2.2] and the leakage current in the SCN [2.2], [2.5], the operating temperature range is still narrow. A large capacitor is used to reduce the leakage effect, thus causing a settling problem especially at low temperature. In [2.6], it employed a buffer connection to form the PTAT voltage but suffers from a large TC due to the extra non-linearity in V_{TH} . Consequently, second-order curvature compensation becomes necessary, resulting in an additional power consumption. Also, although the SCN-based BGR can achieve low power consumption, the SCN suffers from additional nonlinearity, such as significant current leakage especially at high temperature, charge injection, clock feedthrough, and capacitor parasitic. Therefore, limiting its accuracy over a wide temperature range.

This chapter presents an SCN-based BGR circuit designed and implemented in a 65nm standard CMOS process with a wide temperature range and high precision while keeping low supply voltage as well as low power consumption. The proposed BGR sums up a CTAT voltage αV_{EB} together with a PTAT voltage $\gamma \Delta V_{EB}$ to generate a 495mV reference voltage by employing two SCNs, thus reducing the power consumption. A 2x CP, where the ratio of the flying capacitor and the loading capacitor is 5:3 in order to minimize the CP output ripple, is employed to supply the V_{EB} generator in order to relax V_{DD} from 0.9V to 0.5V. A PTAT current source is proposed to bias the PNP BJT in order to reduce the nonlinearity of the V_{EB} . A voltage divider SCN with low leakage consideration is designed to form the CTAT voltage αV_{EB} and

reduce the nonlinearity of its coefficient α , so as to improve the TC and extend the temperature range. A series-parallel SCN with adjusted clock swing is designed to form the PTAT voltage γV_{EB} , in order to improve the LNR of the BGR. This works achieves a 1.45x improvement in temperature range when compared with [2.5], while consuming about 54% less power when compared with [2.6].

This chapter is organized as follows. Section 2.2 presents the principle of the nonlinearity reduction of the proposed BGR. Section 2.3 presents the proposed BGR architecture and its analysis. Section 2.4 presents the simulation and measurement results. The summary is provided in Section 2.5.

2.2 PRINCIPLE OF NONLINEARITY REDUCTION

Figure 2-1 shows an SCN-based BGR proposed in [2.2], which mainly suffers from two problems: 1) the biasing current of two PNP BJT Q_3 and Q_4 is not well defined, thus increasing the nonlinearity of V_{EB} and significantly degrade the TC and limit the temperature range, 2) long holding time is employed in order to reduce the power consumption, but this causes significant current leakage especially at high temperatures.

In the following Sections 2.2.1 and 2.2.2, the biasing current and the long holding time problems and their corresponding solutions will be discussed in detail.

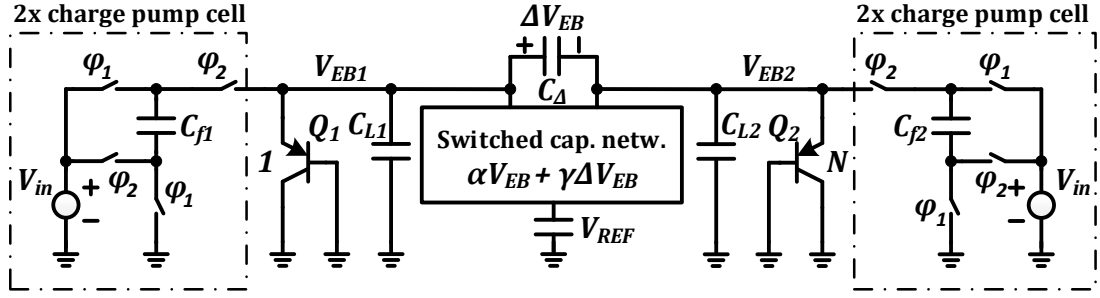


Figure 2-1 A SCN-based BGR [2.2]

2.2.1 BJT'S NONLINEAR TERM $c(T)$ REDUCTION

The temperature characteristic of the V_{EB} of the BJT is well defined in [2.7], which can be expressed as the summation of a constant term, a linear term, and a nonlinear term:

$$V_{EB}(T) = [V_{G0} + (\eta - \zeta) \frac{kT_R}{q}] - \lambda T + c(T) \quad (2-1)$$

$$c(T) = (\eta - \zeta) \frac{k}{q} [T - T_R - T \ln \left(\frac{T}{T_R} \right)] \quad (2-2)$$

where V_{G0} is the extrapolated bandgap voltage at 0K (temperature in Kelvin), η is a process dependent parameter, ζ is the temperature exponent of the BJT collector current [i.e., $I_C \propto (T/T_R)^\zeta$], T is the absolute temperature in Kelvin, T_R is a reference temperature, k is the Boltzmann constant, q is the electron charge, λ is the extrapolated $V_{EB}(T)$ slope at T_R , and $c(T)$ is the nonlinear term.

In 65nm standard CMOS process, it only offers low-gain ($\beta \approx 1$) PNP BJT. Thus, the collector current I_C cannot be approximated by the emitter current I_E directly, and hence the collector current should be expressed as:

$$I_C = \frac{\beta_F(T)}{\beta_F(T) + 1} I_E \quad (2-3)$$

where $\beta_F(T)$ is the forward beta. Since $\beta_F(T)$ depends exponentially on the process

parameter X_{TB} , $\beta_F/(\beta_F+1)$ can be approximated by the power function as:

$$\frac{\beta_F(T)}{\beta_F(T)+1} \approx \frac{\beta_{f0}}{\beta_{f0}+1} \left(\frac{T}{T_R} \right)^{X_{TB}} \quad (2-4)$$

where β_{f0} is $\beta_F(T)$'s nominal value at T_R and X_{TB} is its temperature exponent.

Figure 2-2 shows the simulated $\beta_F/(\beta_F+1)$ of PNP BJT from emitter area A_E equals to $2 \times 2 \mu\text{m}^2$, $5 \times 5 \mu\text{m}^2$, and $10 \times 10 \mu\text{m}^2$. By setting the T_R equals 300K, we can estimate the temperature exponent from Figure 2-2, where exponent X_{TB} in (2-4) increases with A_E .

By substitute (2-4) into (2-3), the collector current can be expressed as:

$$I_C \approx \frac{\beta_{f0}}{\beta_{f0}+1} \left(\frac{T}{T_R} \right)^{X_{TB}} I_E \quad (2-5)$$

Assumed the I_E is provided by an ideal current source, the temperature exponent ζ of I_C can be approximated by X_{TB} . Therefore, the ζ in (2-2) is approximately equal to X_{TB} , which shows that the nonlinear term $c(T)$ effect on the V_{EB} in (2-1) decreases as the A_E increases.

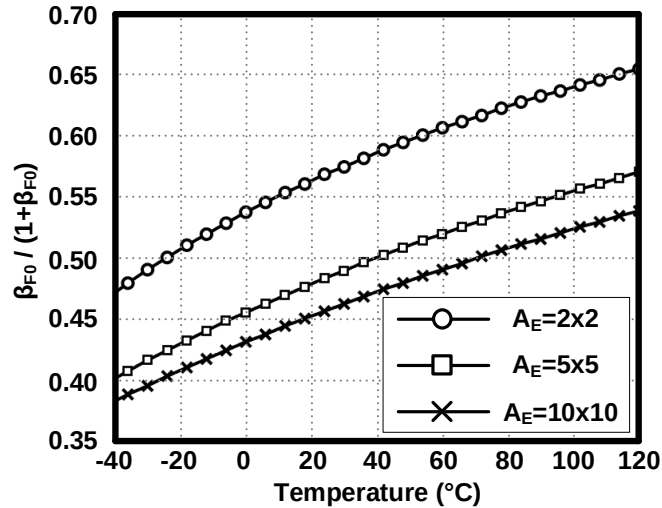


Figure 2-2 Simulated $\beta_F/(\beta_F+1)$ of PNP BJT from emitter area A_E equals to

$2 \times 2 \mu\text{m}^2$, $5 \times 5 \mu\text{m}^2$, and $10 \times 10 \mu\text{m}^2$.

Figure 2-3 shows the simulated residual curvature (nonlinear term) $c(T)$ of V_{EB} , which shows that the residual curvature can be reduced by 24% if the A_E is increased from $2 \times 2 \mu\text{m}^2$ to $10 \times 10 \mu\text{m}^2$. Besides, similar to the study in [2.8], if the PNP BJT is biased by a PTAT current instead of an ideal current source, the ζ in (2-2) becomes $I + X_{TB}$ and (2-2) can be revised as:

$$c'(T) = (\eta - 1 - X_{TB}) \frac{k}{q} [T - T_R - T \ln \left(\frac{T}{T_R} \right)] \quad (2-6)$$

Figure 2-3 shows the simulated residual curvature $c(T)$ and $c'(T)$ of V_{EB} for PNP BJT under different A_E , which shows that the residual curvature can be reduced by 24%, 30% and 30% for $A_E = 2 \times 2 \mu\text{m}^2$, $A_E = 5 \times 5 \mu\text{m}^2$, and $A_E = 10 \times 10 \mu\text{m}^2$, respectively if the PNP BJT is biased by a PTAT current instead of an ideal current source.

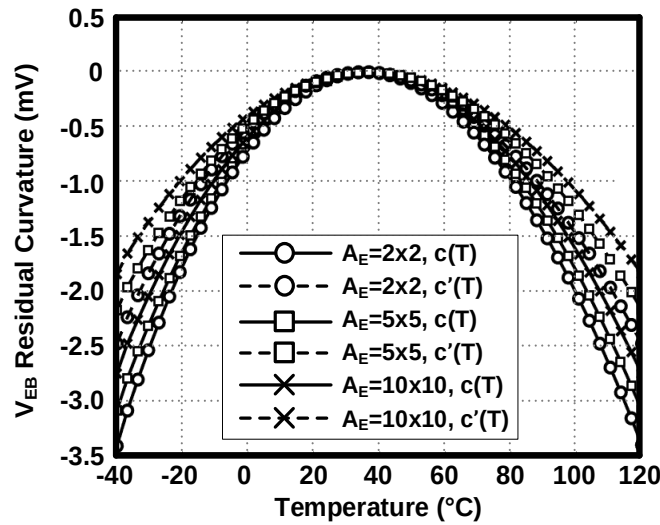


Figure 2-3 Simulated $\beta_F/(\beta_F + 1)$ of PNP BJT from emitter area A_E equals to

$2 \times 2 \mu\text{m}^2$, $5 \times 5 \mu\text{m}^2$, and $10 \times 10 \mu\text{m}^2$.

After considering both the accuracy and the area, the PNP BJT with $A_E = 5 \times 5 \mu\text{m}^2$ is employed in this work. As shown in Figure 2-3, the nonlinearity of a PNP BJT with

an emitter area of $5 \times 5 \mu\text{m}^2$ under PTAT current biasing from -40 to 120°C is about 2.5mV .

2.2.2 MOS TRANSISTOR'S LEAKAGE DEPENDENT NONLINEARITY

Similar to [2.2], this work requires storing portions of V_{BE} and ΔV_{BE} onto capacitors for generating V_{REF} , and the associated switches can inevitably cause charge leakage due to the MOS transistor leakage current. The current of the MOS transistor operating in subthreshold region is well defined in [2.9], which can be expressed as:

$$I_{\text{SUB}} = \mu C_{\text{ox}} \frac{W}{L} (m - 1) V_T^2 \exp\left(\frac{V_{\text{GS}} - V_{\text{TH}}}{m V_T}\right) [1 - \exp\left(\frac{-V_{\text{DS}}}{V_T}\right)] \quad (2-7)$$

where μ is the mobility, C_{ox} is oxide capacitance, W is transistor width, L is transistor length, m is subthreshold slope factor ($m = 1 + C_{\text{d}} / C_{\text{ox}}$, where C_{d} is depletion capacitance), V_T is thermal voltage kT/q , V_{GS} is gate-to-source voltage, V_{TH} is the transistor threshold voltage, and V_{DS} is the drain-to-source voltage. As shown in Figure 2-4(a), the leakage current of the NMOS transistor (operating in subthreshold region) with source terminal connected to 0V (ground), that is $V_{\text{GS}} = 0$. The leakage current can be expressed as:

$$I_{\text{leakage}} = \mu_n C_{\text{ox}} \frac{W}{L} (m - 1) V_T^2 \exp\left(\frac{-V_{\text{TH}}}{m V_T}\right) [1 - \exp\left(\frac{-V_{\text{DS}}}{V_T}\right)] \quad (2-8)$$

From (2-8), the leakage current is a robust temperature-dependent equation. For a fixed transistor size, the V_{DS} is the only parameter that can reduce the leakage current. Based on this idea, a low leakage current design is proposed as shown in Figure 2-4(b),

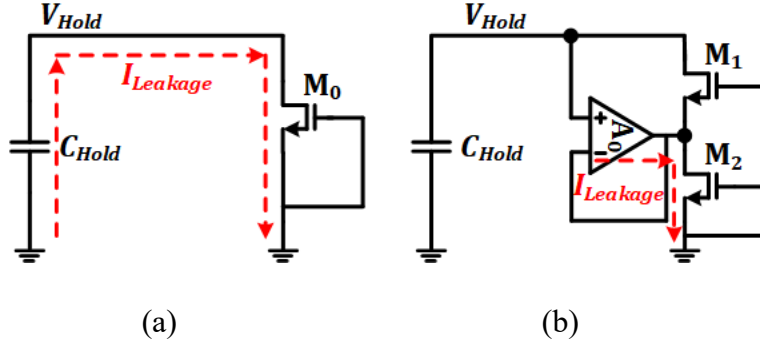


Figure 2-4 The leakage current of NMOS transistor (a) without low leakage design and (b) with low leakage design.

which can significantly reduce the leakage current. As the V_{DS} of transistor M_1 is the offset voltage of an amplifier A_0 only, the leakage through M_1 is minimal. Instead, the leakage current flows to the ground will be provided by the amplifier A_0 , which effect on the charge storing in the holding capacitor C_{Hold} is negligible.

Due to the long holding time of the conventional SCN-BGR in [2.2], the current leakage problem will be significant. By applying the proposed low leakage design as shown in Figure 2-4(b), the current leakage problem due to long holding time can be significantly reduced.

2.3 PROPOSED BGR ARCHITECTURE

2.3.1 CIRCUIT ARCHITECTURE

Figure 2-5 shows the architecture of the proposed BGR circuit. A 2x CP is employed to double the supply voltage to $2V_{DD}$. A clock bootstrap circuit is employed to boost the 20kHz external clock voltage level to $2V_{DD}$. A PTAT current source is used to generate the V_{EB} which is sampled by a voltage divider SCN and a series-parallel

SCN to generate the coefficients α and γ for the corresponding CTAT and PTAT voltages. This work mainly targets on solving the operating range and TC limitations in [2.2], by proposing a second-order effect analysis to relax the SCN non-ideal effect while reducing the leakage problem. Specifically, the contributions of this work are: 1) the SCN non-ideality can contribute to more than 50% of the overall TC in V_{REF} and

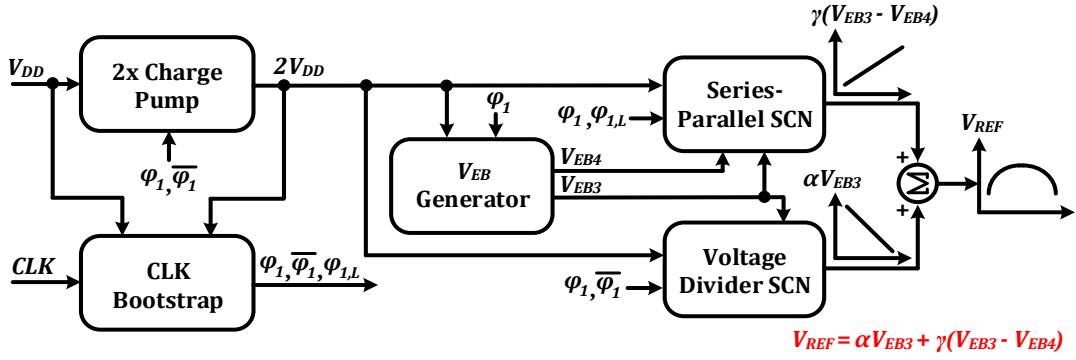


Figure 2-5 Architecture of proposed BGR circuit.

dominates over the curvature of V_{EB} . Consequently, this work focuses on the SCN non-ideality optimization, resulting in a 44% improvement in the overall TC when compared with [2.2]; 2) The proposed work employs a low-leakage design in the voltage divider SCN in order to extend the operating temperature range, leading to a $2\times$ improvement when compared with [2.2]; 3) Instead of biasing using the supply voltage as in [2.2], this work utilizes a PTAT current source to bias the PNP BJT in order to reduce the V_{EB} nonlinearity, and hence improves the achievable TC and temperature range simultaneously. The circuit details of each block of Figure 2-5 will be introduced as follows.

2.3.2 2X CHARGE PUMP AND CLOCK BOOTSTRAP CIRCUIT

Figure 2-6 shows the 2x CP and clock bootstrap circuit with its start-up circuit. The 2x CP is employed to double the supply voltage from V_{DD} to $2V_{DD}$. Two buffers (formed by M_{P4} , M_{P5} , M_{N2} , and M_{N3}) are employed to double the voltage swing of the 20kHz external clock signal CLK to $2V_{DD}$ and hence generated a pair of clock signals ϕ_1 and $\bar{\phi}_1$. Noticed that the external clock signal has passed through 3 buffers to obtain a sharp

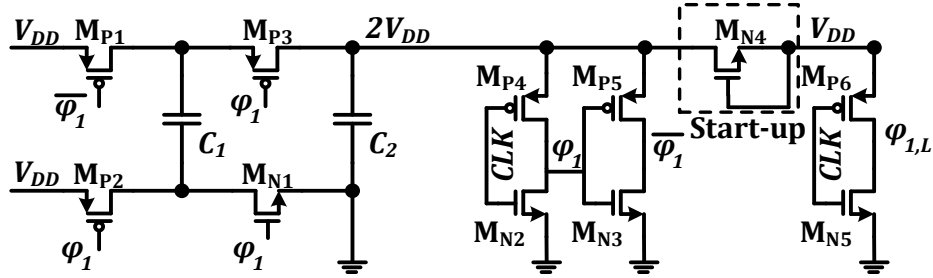


Figure 2-6 2x CP and clock bootstrap circuit with its start-up circuit.

rising edge and falling edge of the clock signal. The M_{N4} acts as the circuit start-up for the clock bootstrap, which guarantees the clock signal level starting from V_{DD} to ensure the operation of the 2x CP. Assume that M_{N4} is removed, since the clock bootstrapping circuit (formed by M_{P4} , M_{P5} , M_{N2} , and M_{N3}) is supplied by the voltage V_{C2} of the capacitor C_2 , we cannot guarantee the voltage level of ϕ_1 and $\bar{\phi}_1$. Thus, this can result in an incorrect operation of M_{P1-3} and M_{N1} . When M_{N4} is added, it forms a diode connection to pre-charge C_2 first, to guarantee the minimum voltage swing of ϕ_1 and $\bar{\phi}_1$. Once the reference circuit starts to operate properly (i.e., when $V_{C2} > V_{DD}$), M_{N4} will be turned off.

As two SCNs are employed to generate the coefficients of both the CTAT and PTAT voltages, the V_{EB} of the PNP BJT is required to be active during the sampling state only. Figure 2-7 shows the simulated output ripple of the 2x CP as a function of the capacitor ratio $C_2/(C_1 + C_2)$. In this work, the capacitor $C_1 = 12\text{pF}$ and $C_2 = 20\text{pF}$ are chosen to optimize the output ripple level of the 2x CP output.

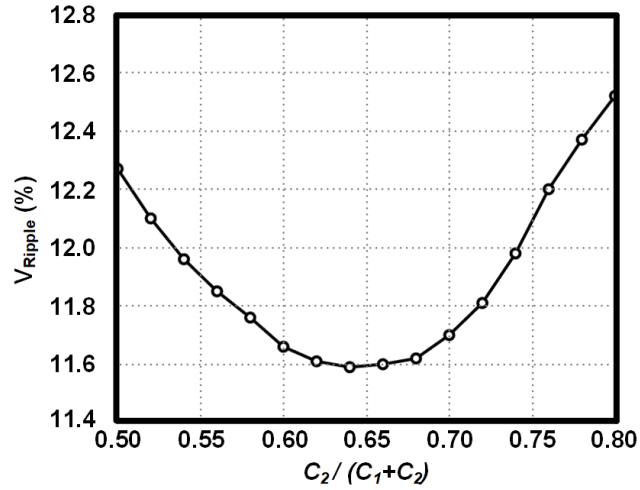


Figure 2-7 Simulated output ripple of the 2x CP as a function of capacitor ratio of $C_2/(C_1 + C_2)$.

2.3.3 V_{EB} GENERATOR

Figure 2-8 shows the proposed V_{EB} generator with PTAT current biasing. The structure of the amplifier A_1 is a basic single-stage differential amplifier similar to [2.4] and is optimized according to our system requirement. This amplifier ensures its low power consumption and does not encounter the stability problem due to its single-pole characteristic. However, the limited DC gain of $\sim 30\text{dB}$ can result in an offset voltage (V_{OS}) of $\sim 1.5\text{mV}$, which is temperature and process dependent. The temperature dependence of V_{OS} in this design is $\sim 1.2\text{mV}$ over the target temperature range, in which it can be calibrated out at one temperature point. The residual error due to process and

mismatch is $\sim 200\mu\text{V}$ through simulation and is therefore negligible in this design. A_1 ensures $V_X = V_Y$ and hence the voltage across the resistor R_1 is $V_{EB1} - V_{EB2}$. The PTAT biasing current can be expressed as:

$$I_{PTAT} = \frac{V_{EB1} - V_{EB2}}{R_1} = \frac{kT}{qR_1} \ln \left(\frac{A_{Q2}}{A_{Q1}} \right) \quad (2-9)$$

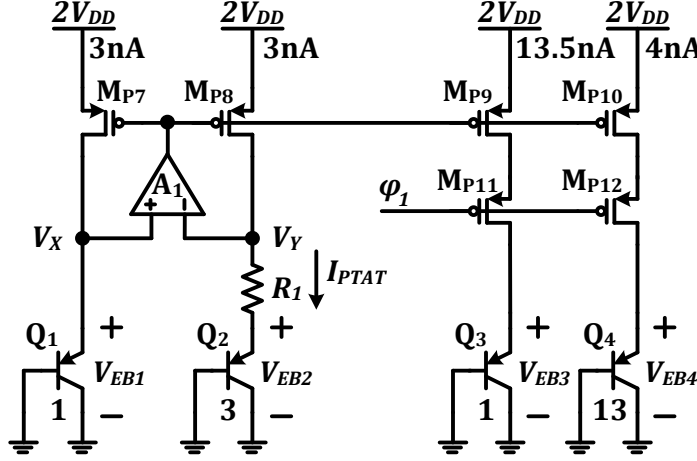


Figure 2-8 V_{EB} generator with PTAT current biasing.

where A_{Q1} and A_{Q2} is the area of the BJT Q_1 and Q_2 , respectively. A large resistor R_1 of $10\text{M}\Omega$ is employed in this work to reduce the current consumption into several nA levels. Although a PTAT current I_{PTAT} can improve the TC of the V_{EB} , if the TC of the PTAT current is too large, the charging current will be too small at low temperatures to cause the voltage divider SCN to settle, which will be further discussed in Section 2.3.4.

In this work, the area ratio of Q_1 and Q_2 is chosen to be 3. From Figure 2-8, the PTAT current I_{PTAT} is copied by the current mirror M_{P9} and M_{P10} to bias the BJT Q_3 and Q_4 , which are employed to generate the BGR reference voltage V_{REF} as:

$$V_{REF} = \alpha V_{EB3} + \gamma(V_{EB3} - V_{EB4})$$

$$= \alpha V_{EB3} + \gamma \frac{kT}{q} \ln \left(\frac{I_{C,Q3} A_{Q4}}{I_{C,Q4} A_{Q3}} \right) = \alpha V_{EB3} + \gamma \frac{kT}{q} \ln(N) \quad (2-10)$$

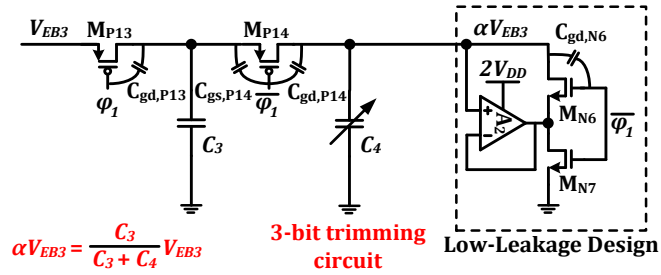
where $I_{C,Q3}$ and $I_{C,Q4}$ are the collector currents of the BJT Q_3 and Q_4 , respectively, N is the effective BJT area ratio, α is the coefficient of the CTAT voltage while γ is the coefficient of the PTAT voltage. V_{EB3} and V_{EB4} in Figure 2-8 represent the V_{EB} generator output in Figure 2-5, which are then sampled by the voltage divider SCN and the series-parallel SCN to form the coefficient α and γ , as well as the CTAT and PTAT voltages, respectively, in which the details will be discussed in the following Section 2.3.4 and 2.3.5.

2.3.4 VOLTAGE DIVIDER SCN

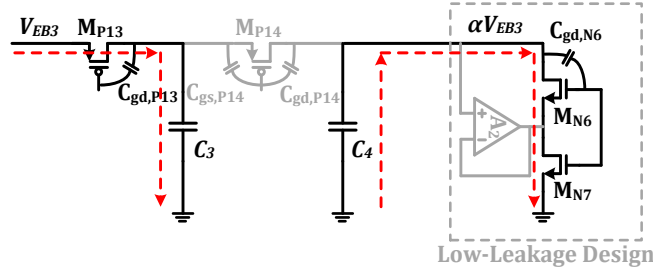
Figure 2-9(a) shows the proposed voltage divider SCN to generate a constant CTAT coefficient α for scaling the CTAT voltage. By charging C_3 to V_{EB3} and fully discharging C_4 to ground in the sampling state and then redistributing their charge in the holding state, we can obtain the coefficient α as described in (2-11). Due to the requirement of a discharging switch for C_4 . As described in Section 2.3.2, the conventional design in Figure 2-4(a) can result in the subthreshold leakage problem. This can affect the coefficient α and increase the TC of V_{REF} . In this work, we alleviated this issue by implementing a low-leakage design, formed by $M_{N6,7}$ and A_2 as shown in Figure 2-9(a). In sampling state illustrated in Figure 2-9(b), M_{P13} , M_{N6} , and M_{N7} are on, M_{P14} is off. C_3 is charged to V_{EB3} while C_4 is discharged to ground. As shown in Figure 2-9(c), in holding state, M_{P13} , M_{N6} , and M_{N7} are off, M_{P14} is on. The charge stored in C_3 will be redistributed as:

$$\alpha V_{EB3} = \frac{C_3}{C_3 + C_4} V_{EB3} \quad (2-11)$$

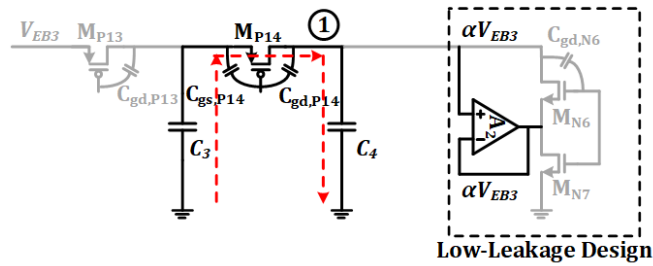
where α is 0.32 in this work which will be discussed in detail in Section 2.3.5. In general, there are mainly four factors that will affect the accuracy of (2-11), which are current leakage, charge injection, clock feedthrough, and capacitor parasitic. Since α is 0.32 and V_{REF} is 500mV in this work, the TC contributed by the V_{EB} curvature is $0.32 \times 2.5\text{mV}$



(a)



(b)



(c)

Figure 2-9 (a) Voltage divider SCN with low leakage design. (b) Sampling state. (c)

Holding state.

$=0.8\text{mV}$ or $10\text{ppm}/^\circ\text{C}$. In simulation, the TC of the SCN-based first-order compensation BGR is $24\text{ ppm}/^\circ\text{C}$, which means that extra non-ideal effects contribute to the remaining TC of $14\text{ ppm}/^\circ\text{C}$ and can be significant (58% of the total TC).

2.3.4.1 CURRENT LEAKAGE

The first factor is the current leakage through NMOS transistor M_{N6} to the ground in holding state (if the unity gain buffer with amplifier A_2 is not added) as discussed in Section 2.3.2. Different from PMOS, the minimum gate-to-source voltage V_{GS} of M_{N6} is zero. Thus, there will be leakage current as indicated in (2-7) and (2-8). The leakage current nonlinearly increases as the temperature rises due to the long holding time, which results in the coefficient α nonlinearly decreases as the temperature rises. Without the low-leakage design, the voltage divider SCN in the holding state is equivalent to the circuit presented in Figure 2-4(a), which results in a leakage current following (2-8) through M_0 . Since the transistor threshold voltage V_{TH} is CTAT voltage and is linked to the leakage current by an exponential function, its current-temperature characteristic is nonlinear. This leakage current can be in the order of several hundreds of pA at high temperature. Assume that the leakage current is 200pA at 120°C , with a holding time of 250ns (20kHz clock signal) and $C_{\text{HOLD}} = C_3 + C_4 = 854\text{fF}$, the voltage variation $\Delta V_{C,\text{HOLD}}$ due to the leakage current is about 6mV , which corresponds to a TC of $75\text{ppm}/^\circ\text{C}$. In this work, a unity gain buffer, which consumes 1.27nW power, is employed to copy the voltage αV_{EB3} to the source terminal of M_{N6} as shown in Figure 2-9(c). The amplifier A_2 is a basic differential pair using a PMOS input pair and unity

feedback, with the bias current mirrored from the PTAT current generator. Also, A_2 is turned off in sampling state to reduce power consumption. By adding the unity gain buffer for low leakage design consideration, although the drain-to-source voltage V_{DS} of M_{N6} is not zero due to the offset of the amplifier A_2 , the leakage current flowing through M_{N6} is negligible, which will not affect the coefficient α .

Figure 2-10 shows the simulated CTAT voltage coefficient α with and without low-leakage design. Without the low-leakage design, α would drop nonlinearly at high

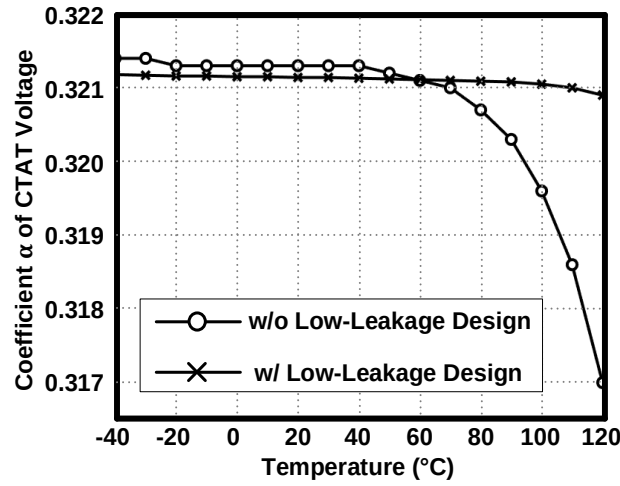


Figure 2-10 Simulated CTAT voltage coefficient α with and without low-leakage design.

temperate and the simulated TC can be up to 80 ppm/°C. The achievable TC can be reduced to 8 ppm/°C if the low-leakage design is employed. The TC contributed by the leakage current is about 57% of the 14 ppm/°C. Notice that the DC offset of A_2 can result in a drain/source voltage difference $V_{DS,N6}$. The introduced residual leakage current can affect the TC at high temperature, ultimately limiting the operating range to 120°C in this work.

2.3.4.2 CHARGE INJECTION

For the charge injection, M_{P13} and M_{P14} are considered as the two main error sources, while the effect of M_{N6} can be neglected. For the charge injection of M_{N6} , as its source terminal is connected to the output node of the buffer A_2 as shown in Figure 2-9(a), which is a low impedance node, the charge injects to the capacitor node 1 as shown in Figure 2-9(c) is negligible. For the charge injection of M_{P13} , since the bias current of Q_3 is not large as shown in Figure 2-8, the impedance at the emitter of Q_3 is large. As a result, the charge injection will flow to both node 1 and the emitter of Q_3 . In this work, we assume k ($0 < k < 1$) times of the charge of M_{P13} will flow to node 1 during each holding cycle. The charge injection due to M_{P13} and M_{P14} can be expressed:

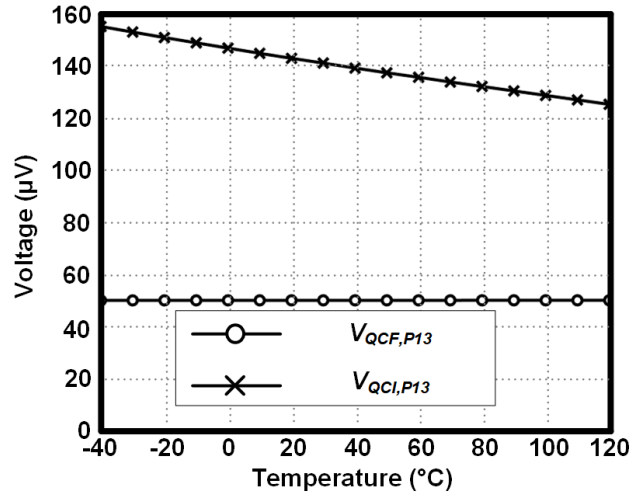
$$Q_{ci,P13} = kW_{P13}L_{P13}C_{ox}(|0 - V_{EB3}| - |V_{THP}|) \quad (2-12)$$

$$Q_{ci,P14} = -W_{P14}L_{P14}C_{ox}(|V_{DD} - \alpha V_{EB3}| - |V_{THP}|) \quad (2-13)$$

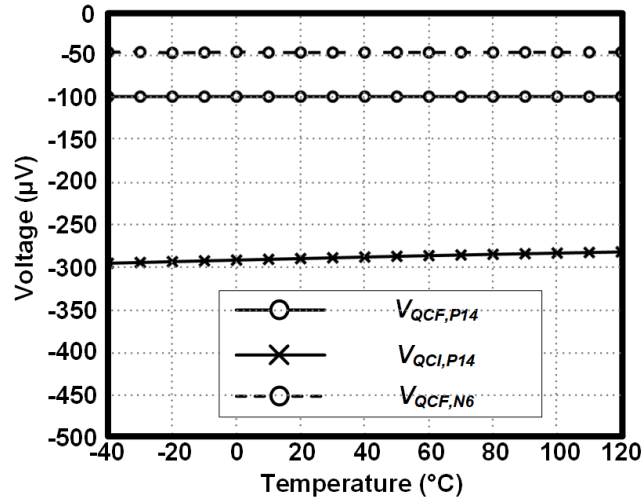
where W_{P13} , L_{P13} , W_{P14} , and L_{P14} are the width and length of the PMOS transistor M_{P13} and M_{P14} , V_{THP} is the threshold voltage of the PMOS transistor, $0 - V_{EB3}$ is the V_{GS} of M_{P13} in sampling state, $0 - \alpha V_{EB3}$ is the V_{GS} of M_{P14} in holding state. Following [2.10], we assumed half of the charge would inject to the capacitor node, that is $k = 0.5$. Since an inversion layer consisting of holes is formed at the oxide-silicon interface [2.10], it attracted the holes from the capacitors C_3 and C_4 , thus there is a negative sign in (2-13). By dividing the charge due to charge injection in (2-12) and (2-13) by the effective capacitance at node 1, the voltage variation due to charge injection can be obtained. Since the TC of V_{EB3} is larger than $|V_{THP}|$, the voltage variation due to charge injection

of M_{P13} in (2-12) is a CTAT voltage; Also, since the TC of αV_{EB3} is larger than $|V_{THP}|$ and there is a negative sign, the voltage variation due to charge injection of M_{P14} in (2-13) is a PTAT voltage.

Figure 2-11(a) and (b) show the simulated result of the voltage variation due to charge injection of M_{P13} and M_{P14} , which verifies (2-12) and (2-13). From (2-12) we can observe that the TC of $Q_{ci,P13}$ depends on $|V_{EB3}|$ and $|V_{THP}|$, which exhibit a TC of $-2.5\text{mV}/^\circ\text{C}$ and $-0.6\text{ mV}/^\circ\text{C}$, respectively. This will result in a $Q_{ci,P13}$ TC of about $-1.9\text{ mV}/^\circ\text{C}$. In contrast, from (2-13), the TC of $Q_{ci,P14}$ depends on $|\alpha V_{EB3}|$ and $|V_{THP}|$. Since α is about 0.32 in this work, the TC of $Q_{ci,P14}$ can be calculated as roughly $-0.2\text{mV}/^\circ\text{C}$. Finally, since k is assumed to be about 0.5, the TC of $V_{QCI,P13}$ should be about 5 times (i.e., $-1.9*0.5/-0.2$) larger than $V_{QCI,P14}$. As shown in Figure 2-11(a) and (b), the TC of $V_{QCI,P13}$ is about $-1,530\text{ ppm}/^\circ\text{C}$, while that of $V_{QCI,P14}$ is about $320\text{ppm}/^\circ\text{C}$. Also, as the voltage variation due to charge injection of M_{P13} and M_{P14} is linear as shown in Figure 2-11, we can compensate the TC of voltage variation due to charge injection during the design process, which result in a DC offset that cannot affect the TC of α . To further explain this idea, we use the charge injection of M_{P13} as an example. Following [2.6] and [2.13], the V_{EB} -temperature and V_{TH} -temperature relationship can be expressed as a constant, first-order and second-order terms. Since the effect of the second-order term is very small, it is $kW_{P13}L_{P13}C_{ox}(a_1 + c_1)$ to cancel the effect of the charge injection (γ is the number of the series-parallel capacitor, a_1 is the first-order term of the V_{EB3} and c_1 is the first-order term of the V_{TH}). However,



(a)



(b)

Figure 2-11 Simulated voltage variation due to (a) the charge injection and clock feedthrough of M_{P13} , and (b) charge injection and clock feedthrough of M_{P14} and clock feedthrough of M_{N6} .

as a_I and c_I are process dependent, the charge injection error would degrade the 3σ accuracy, and a simulated maximum TC (at worst corner) after compensation is about $10\text{ppm}/^\circ\text{C}$ due to the charge injection, which is not included in the remaining TC of $14\text{ppm}/^\circ\text{C}$.

2.3.4.3 CLOCK FEEDTHROUGH

For the clock feedthrough, during the transition process from sampling state to holding state, M_{P13} , M_{P14} , and M_{N6} will couple the voltage to the capacitors C_3 and C_4 through their parasitic capacitances.

However, as the M_{P14} is not connected to the nodes other than the SCN, the clock feedthrough due to M_{P14} is a DC offset, which does not affect the TC of α . In contrast, as the drain terminal of M_{P13} is connected to a CTAT voltage V_{EB3} while the source terminal of M_{N6} is connected to the ground in sampling state as shown in Figure 2-9 (b), as well as M_{P13} conducts current before its gate signal reaches $V_{EB3} - V_{THP}$, and M_{N6} conducts current before its gate signal reaches V_{THN} , the clock feedthrough effect due to M_{P13} and M_{N6} is temperature-dependent. The clock feedthrough due to M_{P13} , M_{P14} , and M_{N6} can be expressed,

$$Q_{cf,P13} = C_{ov,P13} W_{P13} (2V_{DD} - V_{EB3} - |V_{THP}|) \quad (2-14)$$

$$Q_{cf,P14} = 2C_{ov,P14} W_{P14} (0 - 2V_{DD}) \quad (2-15)$$

$$Q_{cf,N6} = C_{ov,N6} W_{N6} (V_{THN}) \quad (2-16)$$

where W is widths of M_{P13} , M_{P14} , and M_{N4} , and V_{THN} is the threshold voltage of the NMOS transistor. By dividing the charge due to clock feedthrough in (2-14) – (2-16) by the effective capacitance at node where $C_{ov,P13}$, $C_{ov,P14}$ and $C_{ov,N6}$ are the overlap capacitance per unit width of M_{P13} , M_{P14} , and M_{N4} , W_{P13} , W_{P14} and W_{N6} are the 1, the voltage variation due to clock feedthrough can be obtained. Since both $-V_{EB3}$ and $-|V_{THP}|$ are PTAT voltages, the voltage variation due to clock feedthrough of M_{P13} in

(2-14) is PTAT voltage; Also, as mentioned before, the voltage variation due to clock feedthrough in (2-15) of M_{P14} is an offset voltage independent of the temperature; Besides, since V_{THN} is a CTAT voltage, the voltage variation due to clock feedthrough of M_{N6} in (2-16) is a CTAT voltage. To relax the temperature dependence of the clock feedthrough, it is suggested to employ a clock signal with a sharp rising and falling edge. In this case, although a larger clock feedthrough will be obtained due to a sharp edge, the TC of the clock feedthrough effect on α is negligible.

As shown in Figure 2-11(a) and (b), the voltage variation due to the clock feedthrough of M_{P13} , M_{P14} , and M_{N6} are almost constant. In this work, the clock signal has passed through 3 buffers to obtain a clock signal with sharp rising and falling edge, thus the clock feedthrough effect on α can be neglected.

2.3.4.4 CAPACITOR PARASITIC

For the capacitor parasitic, as the MIM capacitor is employed for C_3 and C_4 , the parasitic capacitor at the top plate of C_3 and C_4 is tiny, which can be simply neglected. Therefore, the α variation due to the top plate parasitic is negligible.

2.3.4.5 SECOND-ORDER EFFECT OF VOLTAGE DIVIDER SCN

By dividing the second-order effect in (2-12) – (2-16) by the effective capacitance at node 1, the αV_{EB3} in (2-11) becomes:

$$\alpha V_{EB3} = \frac{C_3}{C_3 + C_4} V_{EB3} + \frac{Q_{ci,P13} + Q_{ci,P14} + Q_{cj,P13} + Q_{cj,P14} + Q_{cj,N6}}{C_3 + C_4} \quad (2-17)$$

where $C_3 + C_4$ is the effective capacitance during holding state as shown in Figure

2-9(c). To relax the second-order effect of the voltage divider SCN due to the charge injection and clock feedthrough, it is suggested that the size of M_{P13} and M_{P14} should be minimized while a larger capacitor for C_3 and C_4 is preferable. However, a larger capacitor either requires a longer settling time or a larger charging current, which leads to increase 2x CP output ripple due to a lower operating frequency or increase the power consumption of the BGR as a consequence. Therefore, we should appropriate design the value of C_3 and C_4 . As C_4 would not suffer from the settling problem, Figure 2-12 shows the simulated settling error versus different C_3 values at -40°C only (settling

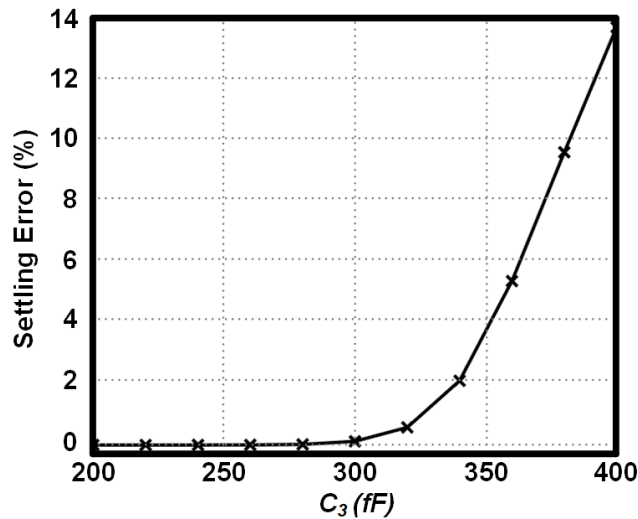


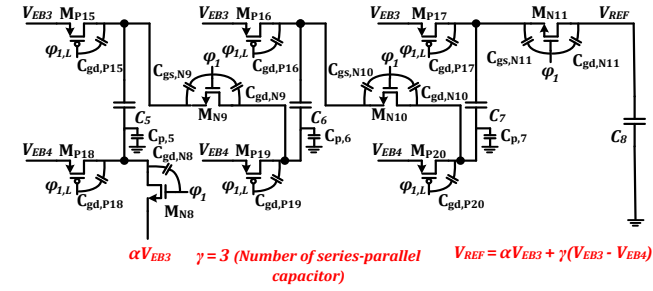
Figure 2-12 Simulated settling error versus different C_3 value in sampling state at -40°C .

error limits the lower limit of the operating temperature range). From Figure 2-12, after the C_3 is larger than 300fF, the settling error starts to increase nonlinearly. In this work, the capacitor $C_3=300\text{fF}$ and $C_4=545\text{fF}$ are designed. From Figure 2-9(a), a 3bit

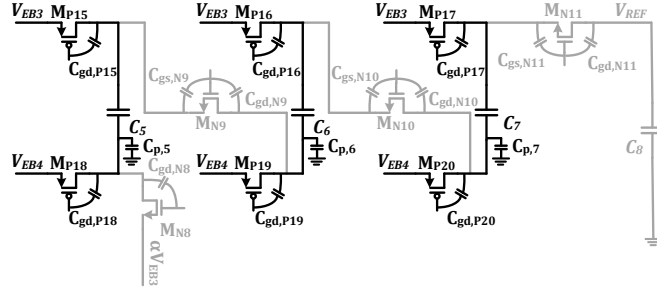
trimming circuit with a unit capacitor equal to 10fF is implemented in parallel with the 545fF capacitor to compensate the process drift.

2.3.5 SERIES-PARALLEL SCN

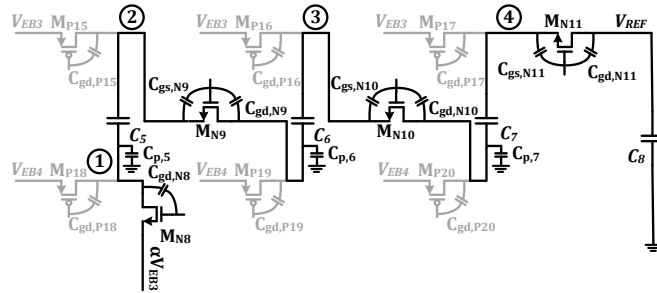
Figure 2-13(a) shows the series-parallel SCN for the PTAT voltage coefficient γ generation and the final reference voltage V_{REF} generation.



(a)



(b)



(c)

Figure 2-13 (a) Series-parallel SCN for PTAT voltage coefficient γ generation and V_{REF} generation. (b) Sampling state. (c) Holding state.

As shown in Figure 2-13(b), in sampling state, $M_{P15} - M_{P20}$ are on while $M_{N8} - M_{N11}$ are off, V_{EB3} is sampled on the top plate of the capacitors $C_5 - C_7$, while V_{EB4} is sampled on the bottom plate. As shown in Figure 2-13(c), in holding state, $M_{P15} - M_{P20}$ are off while $M_{N8} - M_{N11}$ are on, the capacitors $C_5 - C_7$ connect in series together with the voltage divider SCN through M_{N8} .

Therefore, the reference voltage V_{REF} in (2-10) becomes:

$$\begin{aligned} V_{REF} &= \alpha V_{EB3} + \gamma(V_{EB3} - V_{EB4}) \\ &= \alpha V_{EB3} + 3 \frac{kT}{q} \ln \left(\frac{I_{C,Q3} A_{Q4}}{I_{C,Q4} A_{Q3}} \right) = \alpha V_{EB3} + 3 \frac{kT}{q} \ln(N) \end{aligned} \quad (2-18)$$

where γ is the number of the series-parallel capacitor. (2-18) shows the first-order

Table 2-1 Area, Current Consumption, and TC Optimization

γ (No. of Series-Parallel Cap.)	Current Consumption	TC of the V_{REF}
1	2,590 nA	2.1 ppm/°C
2	93.40 nA	4.4 ppm/°C
3	17.50 nA	7.9 ppm/°C
4	10.25 nA	13.2 ppm/°C
5	6.97 nA	21.5 ppm/°C

equation of the V_{REF} . In general, for a BJT-based BGR, V_{EB} is a CTAT voltage, and can be expressed as a constant, first-order and second-order terms a_0 , $a_1 T$, and $a_2 T^2$ respectively. On the whole, as a_1 is much larger than $k \ln(N)/q$, we may employ the multiply factor α and γ such that $\alpha a_1 = \gamma k \ln(N)/q$ to cancel the first-order term of the CTAT voltage. In the work, as $a_1 T$ is about -2.5mV/°C which is mainly determined by the BJT size, we can adjust α , N , and γ to ensure $\alpha a_1 = \gamma k \ln(N)/q$. We determine γ first, and then choose α and N correspondingly. Referring to Figure 2-8, assuming the area ratio of Q_3 and Q_4 is 1:13 using unit element matching, we can vary the current ratio

of M_{P9} and M_{P10} to change N . Table 2-1 shows the simulated trade-offs among the area, current consumption, and TC. For a smaller γ , the TC of V_{REF} decreases as less parasitic capacitance is introduced in the series-parallel SCN. [i.e., the TC due to the second term on the right-hand side of (2-31)] However, a much larger current is required to increase the PTAT term in (2-10) due to its logarithmic relationship, thus increasing the power consumption. In contrast, a larger γ can reduce the power requirement, but it will also lead to an increased TC in V_{REF} due to the extra SCN parasitic capacitance. In this work, we choose $\gamma = 3$ to optimize the trade-off among area, current consumption, and TC. When both a_1 and γ are designed, there is only one combination of α and N can achieve the target reference voltage level of 500mV. For the design of α , we first fix the size of C_3 , then vary C_4 in order to prevent the settling problem during V_{EB3} sampling. For the design of $k \ln(N)/q$, we vary N in (2-18) through the sizing of $Q_{3,4}$, while keeping the transistor width ratio between M_{P9} and M_{P10} to be less than 4 to ensure good current matching. A detailed analysis of the second-order effect on γ will be discussed in the following section 2.3.5.1 – 2.3.5.5.

2.3.5.1 CURRENT LEAKAGE

From Figure 2-8, the bias current of Q_3 and Q_4 is turned off by M_{P11} and M_{P12} in holding state and hence V_{EB3} and V_{EB4} are very small. In this case, although the V_{SD} of the $M_{P15} - M_{P20}$ in holding state [Figure 2-13(c)] is a negative number, the V_{SG} of the $M_{P15} - M_{P20}$ is a negative number as well. The current of the PMOS transistor operating in subthreshold region is deduced as in (2-19) from (2-7). From (2-19), if the V_{SD} is a

negative number, $\exp(V_{SD}/V_T) \approx 0$, the leakage current depends on the V_{SG} only.

Therefore, as the V_{SG} and V_{THP} are negative numbers, $\exp[(V_{SG} + V_{THP})/mV_T] \approx 0$

and hence the leakage current of the series-parallel SCN is negligible.

$$I_{SUB} = \mu C_{ox} \frac{W}{L} (m - 1) V_T^2 \exp\left(\frac{V_{SG} + V_{THP}}{mV_T}\right) [1 - \exp\left(\frac{V_{SD}}{V_T}\right)] \quad (2-19)$$

2.3.5.2 CHARGE INJECTION

For the charge injection, $M_{P15} - M_{P20}$ and $M_{N8} - M_{N11}$ are considered as the error source. As the bias current of Q_3 and Q_4 is not large as shown in Figure 2-8, the impedance at the emitter of the Q_3 and Q_4 is large. As a result, the charge injection due to $M_{P15} - M_{P20}$ will flow to both the node 1-4 and the emitter of Q_3 and Q_4 . In this work, we assume k ($0 < k < 1$) times of the charge of $M_{P15} - M_{P20}$ will flow to the node 1-4 during each holding cycle. The charge injection due to $M_{P15} - M_{P20}$ and $M_{N8} - M_{N11}$ can be expressed as:

$$Q_{ci,P15-P17} = \sum_{i=15}^{17} kW_{Pi}L_{Pi}C_{ox}(|V_{EB3}| - |V_{THP}|) \quad (2-20)$$

$$Q_{ci,P18-P20} = \sum_{i=20}^{18} kW_{Pi}L_{Pi}C_{ox}(|V_{EB4}| - |V_{THP}|) \quad (2-21)$$

$$Q_{ci,N8-N11} = \sum_{i=8}^{11} W_{Ni}L_{Ni}C_{ox} [2V_{DD} - \alpha V_{EB3} \quad (2-22)$$

where W_{Pi} , L_{Pi} , W_{Ni} , and L_{Ni} are the width and length of the PMOS transistor $M_{P15} - M_{P20}$ and NMOS transistor $M_{N8} - M_{N11}$, V_{THP} is the threshold voltage of the PMOS transistor, while V_{THN} is the threshold voltage of the NMOS transistor, $\Delta V_{EB} = V_{EB3} - V_{EB4}$, $2V_{DD}$ is the gate voltage of $M_{N8} - M_{N11}$ in holding state, αV_{EB3} , $\alpha V_{EB3} + \Delta V_{EB}$, $\alpha V_{EB3} + 2\Delta V_{EB}$, and $\alpha V_{EB3} + 3\Delta V_{EB}$ is the node voltage of the node 1-4 respectively as shown in Figure 2-13(c). Since $M_{P15} - M_{P20}$ inject holes to the node 1-4, there is a

positive sign in (2-20) and (2-21). Also, since an inversion layer of charge carriers is formed under the gate oxide of $M_{N8} - M_{N11}$ [2.10], it attracts the charge from the node 1-4, hence there is also a positive sign in (2-22). By dividing the charge due to charge injection in (2-20) – (2-22) by the effective capacitance at node 1-4, the voltage variation due to charge injection can be obtained. Since the TC of V_{EB3} and V_{EB4} is larger than $|V_{THP}|$, the voltage variation due to charge injection of $M_{P15} - M_{P20}$ in (2-20) – (2-21) is a CTAT voltage; Also, as the TC of the PTAT voltage $-aV_{EB3} - V_{THN}$ is larger than the TC of CTAT voltage $-3\Delta V_{EB}$, the voltage variation due to charge injection of $M_{N8} - M_{N11}$ in (2-22) is a PTAT voltage.

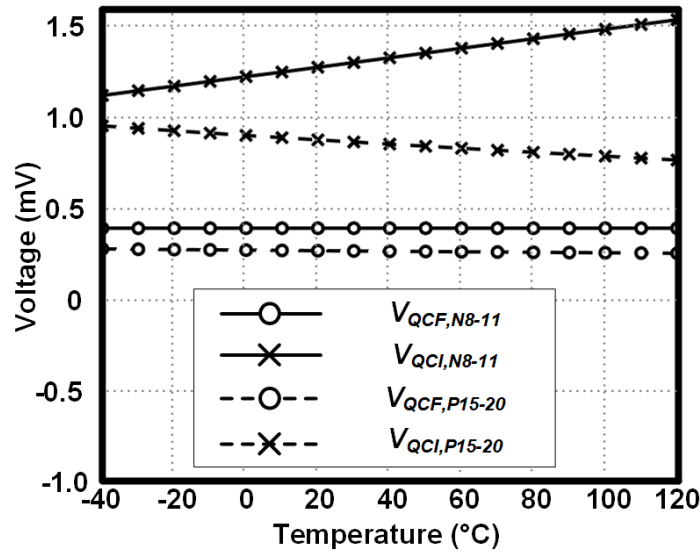


Figure 2-14 Simulated voltage variation due to the charge injection and clock feedthrough of $M_{P15} - M_{P20}$ and $M_{N8} - M_{N11}$.

Figure 2-14 shows the simulated result of the voltage variation due to charge injection of $M_{P15} - M_{P20}$ and $M_{N8} - M_{N11}$, which verifies (2-20) – (2-21). Also, as the voltage variation due to charge injection of $M_{P15} - M_{P20}$ and $M_{N8} - M_{N11}$ is linear as shown in Figure 2-14, we can compensate the TC of voltage variation due to charge

injection during the design process, which result in a DC offset that cannot affect the TC of γ .

2.3.5.3 CLOCK FEEDTHROUGH

For the clock feedthrough, $M_{P15} - M_{P20}$ and $M_{N8} - M_{N11}$ will couple the voltage to the capacitors $C_5 - C_8$ through their parasitic capacitance. During the transition process from sampling state to holding state, $M_{P15} - M_{P20}$ transit from on to off state while $M_{N8} - M_{N11}$ transit from off to on state, and hence all their gate signals transit from 0 to high. As shown in Figure 2-13(c), as the $M_{N8} - M_{N11}$ are not connected to the nodes other than the SCN, the clock feedthrough due to $M_{N8} - M_{N11}$ is a DC offset that does not affect the TC of γ . For $M_{P15} - M_{P20}$, as their source terminals are connected to V_{EB3} and V_{EB4} , the clock feedthrough due to $M_{P15} - M_{P20}$ is temperature-dependent. Similar as Section 2.3.4.3, the clock feedthrough due to $M_{P15} - M_{P20}$, and $M_{N8} - M_{N11}$ can be expressed:

$$Q_{cf,P15-17} = \sum_{i=15}^{17} C_{ov,Pi} W_{Pi} (V_{DD} - V_{EB3} - |V_{THP}|) \quad (2-23)$$

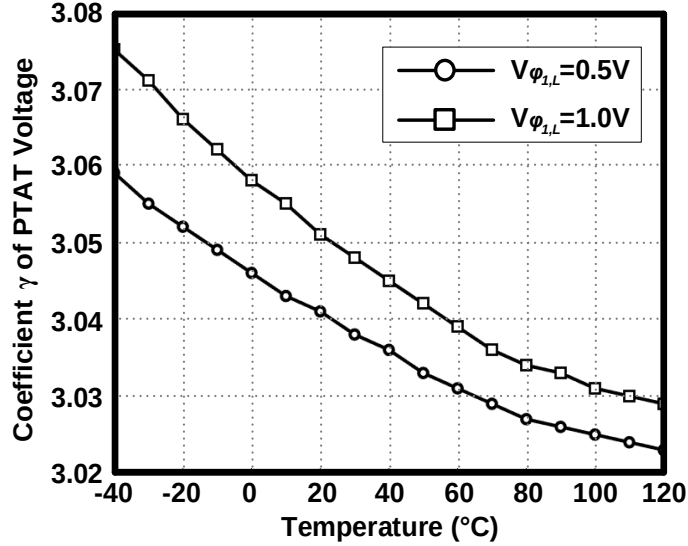
$$Q_{cf,P18-20} = \sum_{i=18}^{20} C_{ov,Pi} W_{Pi} (V_{DD} - V_{EB4} - |V_{THP}|) \quad (2-24)$$

$$Q_{cf,N8-11} = \sum_{i=8}^{11} C_{ov,Ni} W_{Ni} (2V_{DD}) \quad (2-25)$$

where $C_{ov,Pi}$ and $C_{ov,Ni}$ are the overlap capacitance per unit width of $M_{P15} - M_{P20}$ and $M_{N8} - M_{N11}$, V_{DD} is the gate voltage of $M_{P15} - M_{P20}$ in holding state, $2V_{DD}$ is the gate voltage of $M_{N8} - M_{N11}$ in holding state, V_{EB3} is the source voltage of $M_{P15} - M_{P17}$, V_{EB4} is the source voltage of $M_{P18} - M_{P20}$. By dividing the charge due to clock feedthrough in (2-23) – (2-25) by the effective capacitance at node 1-4, the voltage variation due to

clock feedthrough can be obtained. Since $-V_{EB3}$, $-V_{EB4}$ and $-|V_{THP}|$ are PTAT voltage, the voltage variation due to clock feedthrough in (2-23) – (2-24) of $M_{P15} - M_{P20}$ is PTAT voltage; Also, the voltage variation due to clock feedthrough in (2-25) of $M_{N8} - M_{N11}$ is an offset voltage that is temperature independent. Also, since a sharp rising and falling edge is employed as we mentioned in Section 2.3.4.3, the TC of the clock feedthrough effect on γ is negligible.

Figure 2-14 shows the simulated voltage variation due to clock feedthrough of $M_{P15} - M_{P20}$ and $M_{N8} - M_{N11}$, which are almost constant within the temperature range.



(c)

Figure 2-15 Simulated PTAT voltage coefficient γ with 0.5V and 1V clock voltage swing.

Besides, Figure 2-15 shows the simulated PTAT voltage coefficient γ with different clock voltage swing. It shows that the accuracy of γ can be improved by about 20% if a lower clock voltage swing is employed.

2.3.5.4 CAPACITOR PARASITIC

As the MIM capacitor is selected for $C_5 - C_7$, the parasitic at the top plate is negligible, hence only the parasitic capacitors $C_{P,5} - C_{P,7}$ at the bottom plate of the capacitor as shown in Figure 2-13(a) will affect γ . In the sampling state, V_{EB4} is sampled by $C_{P,5} - C_{P,7}$ and redistributed in the holding state. The error charge due to the parasitic capacitor $C_{P,5} - C_{P,7}$ can be expressed as:

$$Q_{p,C5-C7} = \sum_{i=5}^7 C_{P,i} [V_{EB4} - \alpha V_{EB3} - (i - 5)\Delta V_{EB}] \quad (2-26)$$

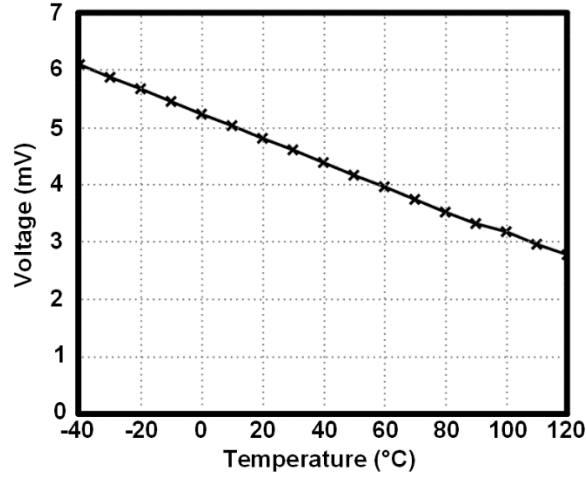


Figure 2-16 Simulated voltage variation due to the bottom parasitic $C_{P,5} - C_{P,7}$.

where $\Delta V_{EB} = V_{EB3} - V_{EB4}$, V_{EB4} is the bottom plate voltage of the capacitor $C_5 - C_7$ in sampling state as shown in Figure 2-13(b), αV_{EB3} , $\alpha V_{EB3} - \Delta V_{EB}$, and $\alpha V_{EB3} - 2\Delta V_{EB}$ is the bottom plate voltage of the capacitor $C_5 - C_7$ in holding state as shown in Figure 2-17. Since the TC of CTAT voltage $V_{EB4} - \Delta V_{EB}$ is larger than αV_{EB3} , the voltage variation due to the bottom plate parasitic in (2-26) is a CTAT voltage. Figure 2-16 shows the simulated voltage variation due to the bottom parasitic $C_{P,5} - C_{P,7}$, which verifies (2-26). Typically, the bottom-plate parasitic capacitance of the mimcap is

about 6.4% [2.14]. As $C_5 - C_7$ are 1pF, the parasitic capacitance is about 64fF. Here, we use the bottom-plate parasitic $C_{P,5}$ as an example to illustrate the parasitic effect. By (2-26) and (2-27), error voltage due to the parasitic capacitance $C_{P,5}$ can be expressed as $V_{PC,C5}=C_{P,5}(V_{EB4} - \alpha V_{EB3})/C_{eff,N1}$, where $C_{P,5}/C_{eff,N1}$ is about 0.06. Following [2.6], the V_{EB} -temperature relationship can be express as the constant, first-order and second-order term. Since the first-order term can be eliminated, only the second-order term can affect the TC of the V_{REF} . With the same emitter size for the PNP BJTs, the second-order term $a_{2,VEB4}$ of V_{EB4} , can be approximated by the second-order term $a_{2,VEB3}$ of V_{EB3} . Therefore, only the term $(1 - \alpha)a_{2,VEB3}$ would affect the TC of V_{REF} . As α is about 0.32 and the non-linear curvature of V_{EB} is about 2.5mV, the overall voltage variation due to $C_{P,5-7}$ is about 6 ppm/°C (or 43% of the remaining TC of 14 ppm/°C).

2.3.5.5 SECOND-ORDER EFFECT OF SERIES-PARALLEL SCN

To study the second-order effect on $\gamma(V_{EB3} - V_{EB4})$ in (2-18), it is necessary to

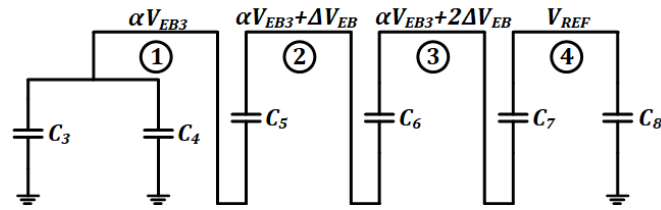


Figure 2-17 Equivalent capacitor of the two SCNs and V_{REF} in holding state.

find out the effective capacitance at each node. By ignoring the parasitic capacitance and the on resistances of all the MOS transistors in Figure 2-9(c) and Figure 2-13(c), the equivalent capacitor of the two SCNs and V_{REF} in holding state is shown in Figure

2-17. By considering the second-order effect in (2-20) – (2-26) and the effective capacitance at node 1-4, the voltage variation due to the second-order effect at node 1-4 can be expressed as:

$$V_{SE,N1} = \frac{Q_{ci,N10} + Q_{ci,P18} + Q_{cf,P18} + Q_{cf,N10} + Q_{pc,C5}}{C_{eff,N1}} \quad (2-27)$$

$$V_{SE,N2} = \frac{Q_{ci,N8} + Q_{ci,P19} + Q_{ci,P19} + Q_{cf,P15} + Q_{cf,P19} + Q_{cf,N8} + Q_{pc,C6}}{C_{eff,N2}} \quad (2-28)$$

$$V_{SE,N3} = \frac{Q_{ci,N9} + Q_{ci,P16} + Q_{ci,P20} + Q_{cf,P16} + Q_{cf,P20} + Q_{cf,N9} + Q_{pc,C7}}{C_{eff,N3}} \quad (2-29)$$

$$V_{SE,N4} = \frac{Q_{cj,N11} + Q_{ci,P17} + Q_{cf,P17} + Q_{cf,N11}}{C_{eff,N3}} \quad (2-30)$$

where $C_{eff, N1}$, $C_{eff, N2}$, $C_{eff, N3}$, and $C_{eff, N4}$ are the effective capacitance at node 1-4 respectively as shown in Figure 2-17. By considering the second-order effect in (2-27) – (2-30), $\gamma(V_{EB3} - V_{EB4})$ in (2-18) becomes:

$$\gamma(V_{EB3} - V_{EB4}) = 3 \frac{kT}{q} \ln(N) + \sum_{i=1}^4 V_{SE, Ni} \quad (2-31)$$

Also, as the series-parallel SCN connect in series together with the voltage divider SCN through M_{N8} , the αV_{EB3} in (2-17) becomes:

$$\alpha V_{EB3} = \frac{C_3}{C_{eff,N1}} V_{EB3} + \frac{Q_{ci,P13} + Q_{ci,P14} + Q_{cj,P13} + Q_{cj,P14} + Q_{cj,N6}}{C_{eff,N1}} \quad (2-32)$$

From (2-31), we can notice that a larger capacitor $C_5 - C_7$ can relax the voltage variation due to the second-order effect. However, as the bottom plate parasitic depends on the size of the capacitor, it will contribute small improvement only when the capacitor reaches picofarads (pF) level. In this work, a 1pF MIM capacitor is chosen for $C_5 - C_8$.

2.4 SIMULATION AND MEASUREMENT RESULTS

The proposed SCN-based BGR was implemented in a 65nm process and occupied a chip area of $0.18 \times 0.29 \text{ mm}^2$. The die photo is shown in Figure 2-18. Figure 2-19 shows 500 times Monte Carlo simulation results of the BGR V_{REF} at 27°C and $V_{DD}=0.5\text{V}$. The average V_{REF} is 498.1 mV with the standard deviation σ of 2.89mV and a 3σ variation of 1.74%. Table 2-2 shows the simulated power consumption of each block of the proposed BGR. As there is no static current in the series-parallel SCN with operating frequency of 20kHz, the SCN consumes very low power.

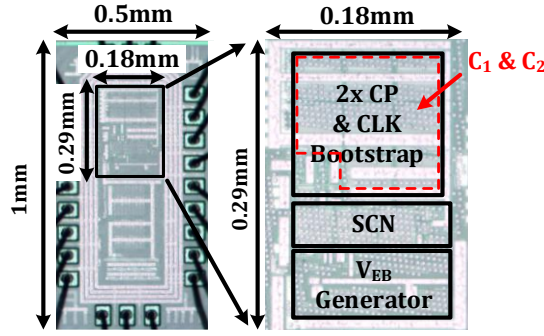


Figure 2-18 The die photo of the proposed BGR

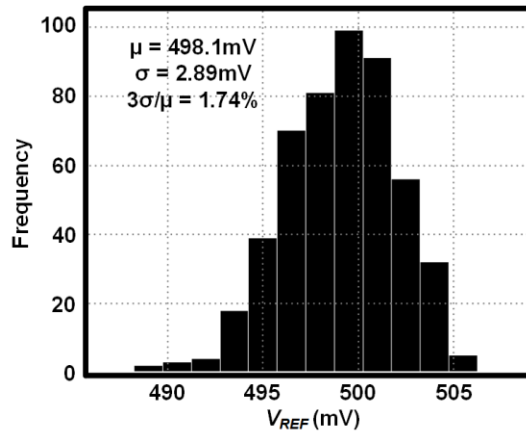
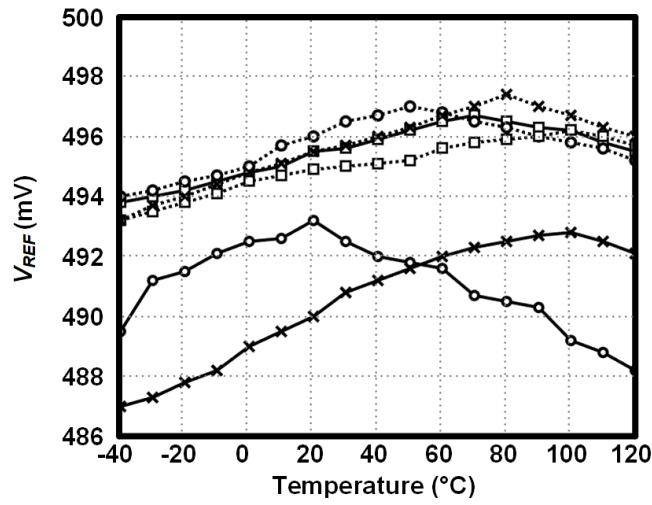


Figure 2-19 500 times process only Monte Carlo simulation results of the BGR

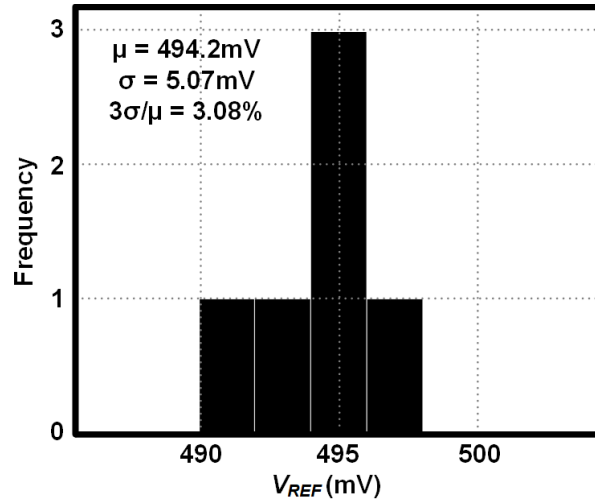
V_{REF} at 20°C and $V_{DD}=0.5\text{V}$.

TABLE 2-2 SIMULATED POWER CONSUMPTION OF EACH BLOCK OF THE PROPOSED
BGR AT 27°C AND $V_{DD}=0.5V$

Each Block	Power Consumption
2x CP	1.41nW
CLK Bootstrap	2.63nW
V_{EB} Generator	29.04nW
Series-Parallel SCN	0.01nW
Voltage Divider SCN	1.27nW



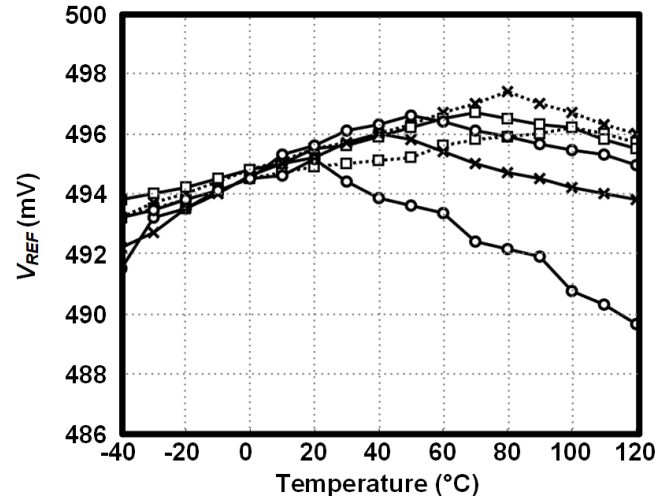
(a)



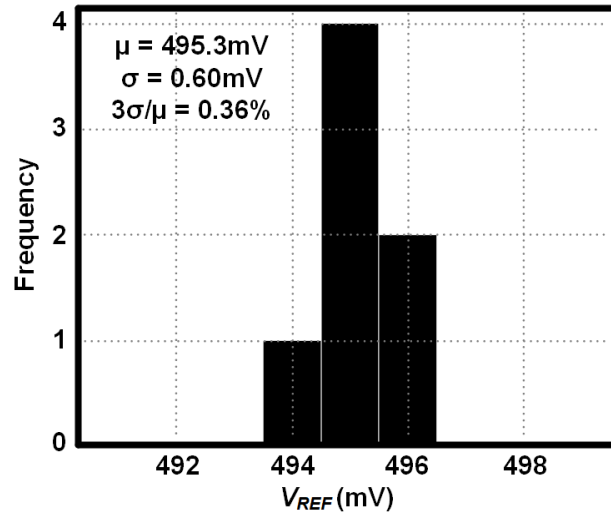
(b)

Figure 2-20 The measured untrimmed (a) BGR V_{REF} from -40 to 120 °C and (b) distribution of the BGR V_{REF} across 6 chips at 20°C and $V_{DD}=0.5V$.

The power consumption in voltage divider SCN is mainly contributed by the low leakage buffer. The simulated total power consumption is 34.4nW at 27°C and $V_{DD}=0.5V$.



(a)



(b)

Figure 2-21 The measured trimmed (a) BGR V_{REF} from -40 to 120 °C and (b) distribution of the BGR V_{REF} across 6 chips at 20°C and $V_{DD}=0.5V$.

Figure 2-20 shows the BGR V_{REF} measurement result of 6 chips before trimming under $V_{DD}=0.5V$. As shown in Figure 2-20(a), the TC of the BGR varies from 38ppm/°C to 74ppm/°C and obtains an average TC value of 48ppm/°C. Figure 2-20(b) shows the BGR V_{REF} variation of 6 chips at 20°C, which shows the average V_{REF} is 494.2mV with a standard deviation of 5.07mV as well as a $3\sigma/\mu$ variation of 3.08%. Compared with the Monte Carlo simulation result as shown in Figure 2-19, the measurement result of the $3\sigma/\mu$ increases about two times, which is due to the mismatch issue.

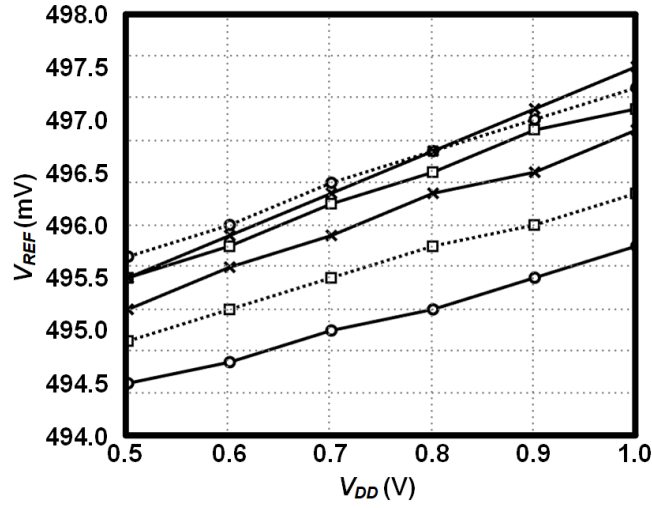


Figure 2-22 Measured LNR of 6 trimmed chips at 20°C.

Figure 2-21 shows the BGR V_{REF} measurement result of 6 chips after trimming under the same testing condition. After applying one-point trimming at 20°C by changing 3bit calibration of C_4 as shown in Figure 2-9(a), the TC of the BGR varies from 38ppm/°C to 70ppm/°C and obtains an average value of 42ppm/°C, as shown in Figure 2-21(a). Figure 2-21(b) shows the average V_{REF} is 495.3mV with a standard deviation of 0.60 mV as well as a $3\sigma/\mu$ variation of 0.36%. Figure 2-22 shows the measured LNR results of 6 trimmed chips at 20°C, which shows that the V_{REF} variation is about 1.3mV to 2mV as V_{DD} varies from 0.5V to 1.0V. Therefore, the LNR of the

proposed BGR varies from 2.6mV/V to 4mV/V and obtains an average value of 3.2mV/V.

Table 2-3 compares the proposed BGR with the previously reported state-of-the-art low power and low V_{DD} BGR works. For fair comparison, the power consumption and area overhead of the on-chip oscillator is included in Table 2-3. It results in a power consumption of 2nW at room temperature, which is close to that of [2.2] (2nW and 2.7nW respectively), as well as an area overhead of 0.001 μm^2 .

For the SCN-based BGR works in [2.2], [2.4] – [2.6], the proposed BGR obtains a better TC and wider temperature range as well as a comparable power consumption compared with [2.2] and [2.5]; while the proposed BGR consumes much less power and obtains a comparable TC compared with [2.4] and [2.6], and a wider temperature range than [2.4]. The area of the proposed BGR is the smallest compared with the SCN-based BGR works except [2.2], which is mainly due to large capacitor of the 2x CP, but this work obtains a better line regulation than [2.2]. A clock signal of different frequencies for the 2x CP and SCNs is an alternative solution for the proposed BGR to reduce the area. Compared with [2.11] – [2.13], the proposed BGR obtains a better TC; while it obtains a wider temperature range than [2.11] and a lower power consumption than [2.12] and [2.13].

Table 2-3 Comparison of BGR Metrics with Other Low Power and Low Supply Voltage designs

Model	This work	[2.2]	[2.4]	[2.5]	[2.6]	[2.11]	[2.12]	[2.13]
Year	2020	ISSCC 2015	JSSC 2012	TCAS-II 2017	TCAS-I 2018	TCAS-I 2016	JSSC 2013	TCAS-II 2017
Process	65nm	0.13 μ m	0.13 μ m	0.18 μ m	0.18 μ m	0.18 μ m	0.18 μ m	0.18 μ m
Reference (mV)	495	500	256	240	460	118	548	212
Power Consumption (nW)	$\frac{36}{38}^*$	32	170	40	83	16	52.5	192
Minimum Supply Voltage (V)	0.5	0.5	0.75	0.5	0.55	0.45	0.7	0.4
TC (ppm/ $^{\circ}$ C)	42	75	40	58	28	59	114	84.5
Temperature Range ($^{\circ}$ C)	-40 – 120	0 – 80	-20 – 85	-25 – 85	-45 – 120	-40 – 85	-40 – 120	-40 – 130
Untrimmed error (3 σ / μ)	$\frac{1.74\%^S}{3.08\%^M}$	2% ^M	3% ^M	3% ^M	3.2% ^M	13.5% ^M	4.8% ^M	2% ^S
LNR	0.64%/V	2%/V	0.019%/V	0.46%/V	0.59%/V	0.34%/V	0.29%/V	0.957%/V
PSRR (dB)	$\frac{-50}{@DC}$	$\frac{-40}{@DC}$	$\frac{-93}{@DC}$	$\frac{-62}{@100Hz}$	$\frac{-62}{@100Hz}$	$\frac{-50}{@100Hz}$	$\frac{-62}{@100Hz}$	$\frac{-40}{@100Hz}$
Ripple	200 μ V ^S	50 μ V ^M	20mV ^M	N.A.	N.A.	N.A.	N.A.	N.A.
Area (mm ²)	$\frac{0.0522}{0.0532}^*$	0.0264	0.07	0.058	0.061	0.013	0.246	0.09

^S Simulation Result, ^M Measurement Result, *Included the oscillator

2.5 SUMMARY

This work presents an SCN-based BGR circuit with a wide temperature range, high precision, low supply voltage and low power consumption characteristics, which is suitable for IoT device application. The proposed BGR employs a 2x CP with an adjusted capacitor ratio to minimize the ripple, which is used to supply the V_{EB} generator, in order to relax V_{DD} from 0.9V to 0.5V with 36nW power consumption. A PTAT current source is proposed to bias the PNP BJT in order to reduce the nonlinearity of V_{EB} , thus improving the TC and extending the temperature range of the BGR. Two SCNs with one implemented the low leakage design and the other one operates with adjusted clock swing are proposed to form the PTAT and CTAT voltages, in order to reduce CTAT voltage coefficient nonlinearity as well as improve the LNR of the BGR. Also, four kinds of second-order effect that affect the TC of the coefficient of CTAT and PTAT voltage are studied. According to the analysis and simulation result, it can be eliminated during the design process. Finally, the measurement result verifies the designed BGR, which obtains a TC of 42 ppm/°C at 0.5V supply within -40 °C to 120 °C. The LNR is 3.2mV/V or 0.64%/V from 0.5V to 1V. Also, it shows a $3\sigma/\mu$ variation of 3.08% before trimming. With a 3-bit trimming, it shows a $3\sigma/\mu$ variation of 0.36%.

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Chapter 3 A 0.5-V SUPPLY, 24nW BANDGAP REFERENCE WITH 32ppm/°C AVERAGE TEMPERATURE COEFFICIENT WITHIN -40°C to 120°C

3.1 INTRODUCTION

Low-power smart devices have received much more attention in the field of the IoT. In the future, an estimation is that there will be more than 20 billion IoT devices connected to the internet in 2025 [3.1]. In certain application scenarios, IoT devices will appear in the remote locations where access to reliable power sources is difficult. Consequently, low energy consumption is always the most important constraint in IoT applications as frequent battery replacement will significantly increase the maintenance cost [3.2], [3.3].

Most IoT systems are heavily duty-cycled, and typically operate in sleep mode to reduce the total power consumption [3.4]. However, even in the standby mode, as the reference voltage circuit is an important and fundamental part of the SoC, it is mostly in operation and becomes an important contributor to the power budget in the standby mode [3.4]. In addition, the minimum voltage at which the BGR starts operation [3.5] limits the start-up voltage of an ULP system. Hence, we can extend the battery runtime of the ULP system by lowering the supply voltage of the BGR, such that a harvested voltage as low as 0.5V [3.6] can supply the BGR.

The SCN based BGR [3.7], [3.5], and [3.8] – [3.10] is suitable for IoT applications due to its low power consumption. To reduce the power consumption into several tens

of nW levels, the conventional resistor-based BGR [3.11] requires the resistance of around hundred $M\Omega$ while the SCN based BGR requires the capacitance of just several pF only, which saves a lot of area. The work in [3.7] employed the reverse bandgap concept to reduce the supply voltage to 0.75V; However, this technique required a large current ratio to generate the PTAT voltage, resulting in large power consumption. The studies in [3.5], [3.8] – [3.10] employed a $2\times$ CP to reduce the supply voltage by half. However, due to the nonlinearity of the BJT biasing [3.5] and the leakage current in the SCN [3.5], [3.8], the designs in [3.5], [3.8] typically suffered from a large TC and a narrow operating temperature range. The study in [3.9] utilized a curvature compensation scheme to extend the temperature range, but at the cost of increased power consumption and lowered untrimmed accuracy due to the match between the compensation and BGR circuits. In our previous work [3.10], the PTAT current biasing improved the TC and extended the temperature range, simultaneously. However, a large settling error occurs at both the voltage divider SCN and the series-parallel SCN especially during low temperature, inevitably degrading the TC as well as increasing the power consumption. Also, due to the PTAT current biasing, the capacitor of $2\times$ CP becomes very large unavoidably in order to reduce the ripple at high temperature, resulting in a large capacitor size.

In this work, we propose an SCN based BGR using a dual PTAT clock topology. The fast PTAT clock drives a dual-branch interleaved $2\times$ CP to reduce the output voltage ripple of $2\times$ CP while relaxing the capacitor size. The slow PTAT clock

controls the voltage divider SCN to reduce its settling error at low temperature while lowering the leakage-induced error at high temperature, resulting in lower power consumption and smaller TC. We also propose a replica V_{EB} generation branch to further alleviate the settling-induced error in the series-parallel SCN during switching. The organization of this work is the following. Section 3.2 presents the principle of the performance optimization of the proposed BGR. Section 3.3 introduces the proposed BGR architecture and its analysis. Section 3.4 exhibits the measurement results. Section 3.5 draws the Conclusions.

3.2 PRINCIPLE OF PERFORMANCE OPTIMIZATION

Figure 3-1 shows the architecture of our previously proposed SCN based BGR circuit [3.10]. A $2\times$ CP is employed to double the supply voltage to $2V_{DD}$. A CLK bootstrap circuit is employed to boost the voltage level of the 20kHz external clock from V_{DD} to $2V_{DD}$. A PTAT current source (referring to I_{R2} in Figure 3-14) generating by the V_{EB} generator is used to generate the V_{EB} which is sampled by a voltage divider SCN and a series-parallel SCN to generate the coefficients α and γ for the corresponding CTAT and PTAT voltages. The reference voltage V_{REF} can be expressed as:

$$V_{REF} = \alpha V_{EB3} + \gamma \Delta V_{EB} \quad (3-1)$$

The BGR design in [3.10] mainly suffers from two problems: 1) As the PTAT current (referring to I_{R2} in Figure 3-14) was used to bias the BJT for reducing the nonlinearity of V_{EB} for improving the TC and extending the temperature range, the

reduction in the bias current at low temperature can induce settling errors in both

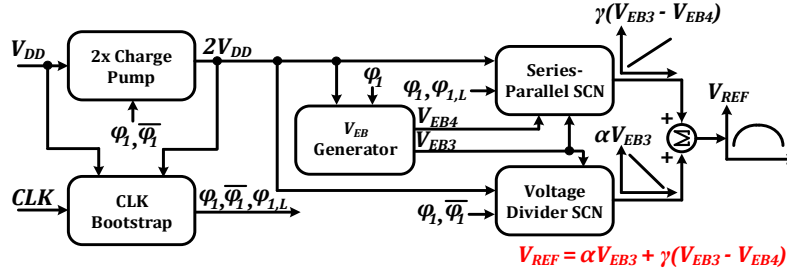


Figure 3-1 Architecture of a SCN based BGR [3.10].

voltage divider SCN and series-parallel SCN, effectively degrading the achievable TC.

This inevitably set a lower bound for the bias current for extending the lower temperature range, leading to a larger power consumption; 2) Large capacitors were employed to reduce the output ripple of the $2\times$ CP, which occupied a large chip area. For the rest of this section, the settling errors of the two SCNs, the large area problem of the CP capacitor, and their corresponding solutions will be discussed in detail.

3.2.1 SETTLING ERROR REDUCTION IN VOLTAGE DIVIDER SCN

Figure 3-2(a) shows the voltage divider SCN in [3.10]. Conventionally, we can employ the conventional sample and hold circuit with an ideal voltage source [3.12] to model the left part of the voltage divider SCN, as shown in Figure 3-2(b). The settling error in this model can be estimated as:

$$V_{err} = V_S - V_{C1} = V_S \cdot e^{-\frac{t}{(R_S + R_{P30})C_1}} \quad (3-2)$$

where V_S is the sampling voltage source signal, V_{C1} is the voltage of the sampling capacitor, C_1 is the sampling capacitor, R_S is the internal resistance of voltage source

V_S , R_{P30} is the equivalent resistance of the ideal switch S_{P30} in order to model the transistor M_{P30} , and t is the sampling time. Typically, as $R_S \ll R_{P30}$, the resistance in the sample and hold circuit is dominated by the switch on-resistance R_{P30} . However, this is true only if the sampling voltage source can provide a sufficient charging current [i.e., R_S is negligible]. In practice, the voltage divider would sample V_{EB3} of Q_3 onto C_1 as shown in Figure 3-2(c). In this case, V_S is formed by the BJT Q_3 and the BJT biasing current source I_{BIAS} . Consequently, the charging current I_{C1} can be expressed as:

$$I_{C1} = I_{BIAS} - I_{Q3} = I_{BIAS} - I_{S,Q3} e^{\frac{V_{EB3}}{V_T}} \quad (3-3)$$

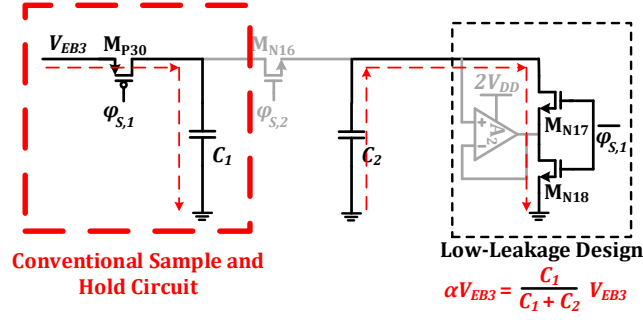
where $I_{S,Q3}$ is the saturation current of the BJT, and V_T is the thermal voltage kT/q .

From (3-1), we can notice that I_{C1} depends on both V_{EB3} and I_{BIAS} . For a large I_{BIAS} , the resistance in the sample and hold circuit is dominated by the switch on-resistance R_{P30} as in the ideal case. In contrast, for a small I_{BIAS} , the internal resistance dominates. Therefore, we employ R_{BIAS} (equivalent to R_S in Figure 3-2 (b)) as shown in Figure 3-2 (c) to model this characteristic.

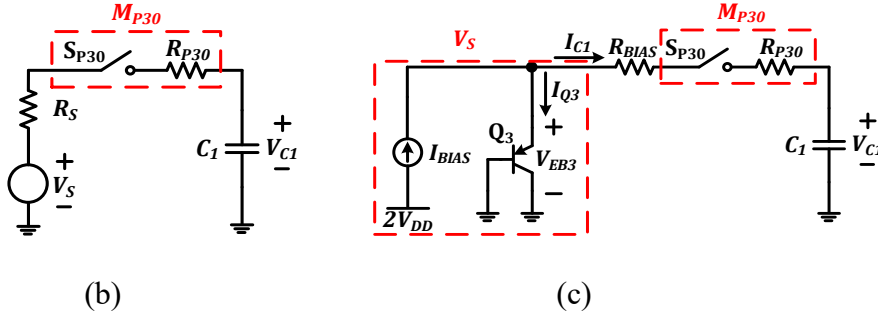
For a low-power BGR, as the biasing current is in the order of several nA, $R_{BIAS} \gg R_{P30}$. In this case, $R_{P30} + R_{BIAS}$ is dominated by R_{BIAS} and hence it strongly depends on I_{BIAS} . Typically, I_{BIAS} is a PTAT current to reduce the nonlinearity of the BJT [3.13]. The reduction in I_{BIAS} at low temperature can introduce significant settling error of V_{C1} . Assume $R_{BIAS} \gg R_{P30}$ to simplify the study, with the help of (3-2), the settling error of V_{C1} with a constant sampling time can be expressed as:

$$V_{err} = V_{EB3} - V_{C1} = V_{EB3} \cdot e^{-\frac{1}{C_1} \frac{t_{SMP}}{R_{BIAS}}} \quad (3-4)$$

$$R_{BIAS} = R_{BIAS,0} [1 + TC_{R,BIAS}(T - T_0)] \quad (3-5)$$



(a)



(b)

(c)

Figure 3-2 Circuit configuration of our previous proposed voltage divider SCN in sampling state [3.10], (b) equivalent model of the sample and hold circuit with ideal voltage source, and (c) equivalent model of the sample and hold circuit with V_S formed by the BJT Q_3 and the BJT biasing current source I_{BIAS} .

where t_{SMP} is the sampling time, $R_{BIAS,0}$ is the R_{BIAS} at the reference temperature T_0 , $TC_{R,BIAS}$ is the first-order TC of R_{BIAS} which will be further discussed in Section 3.3.5.

Figure 3-3 shows the settling error of V_{C1} with I_{BIAS} adjusted at 27°C . Since the settling error increases nonlinearly as temperature decreases, it cannot be directly compensated by simply adjusting the PTAT voltage. The effect of V_{err} contributing to

V_{REF} can be expressed as:

$$V_{REF} = \alpha V_{EB3} \left(1 - e^{-\frac{1}{C_1 R_{BIAS,0} [1 + T C_{R,BIAS} (T - T_0)]} \frac{t_{SMP}}{}} \right) + \gamma \Delta V_{EB} \quad (3-6)$$

It shows that even a 0.1% settling error in the voltage divider SCN would result in

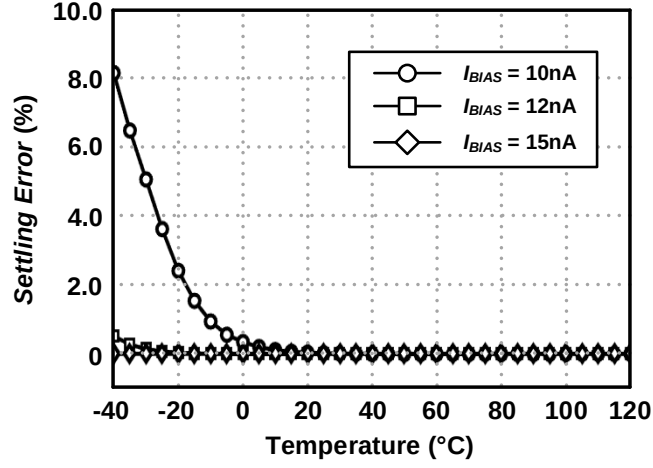


Figure 3-3. Settling error of V_{CI} under different biasing current.

a $\sim 250\mu\text{V}$ variation in the BGR reference voltage V_{REF} or $\sim 3\text{ppm}/^\circ\text{C}$ extra TC of the BGR circuit. To relax this error, one of the solutions is to increase I_{BIAS} such that the settling error is negligible even at the lowest temperature point. However, this requires a large biasing current for the BJT, dominating the BGR power consumption. As shown in Figure 3-3, I_{BIAS} should increase from 10 to 15nA to ensure a $\sim 0.01\%$ settling error from -40 to 120°C, corresponding to a 50% increase in BJT biasing power to relax this settling error problem. To resolve the settling error problem without increasing I_{BIAS} , we propose to drive the voltage divider SCN using a PTAT clock signal which will be discussed in detail in Section 3.3.5.

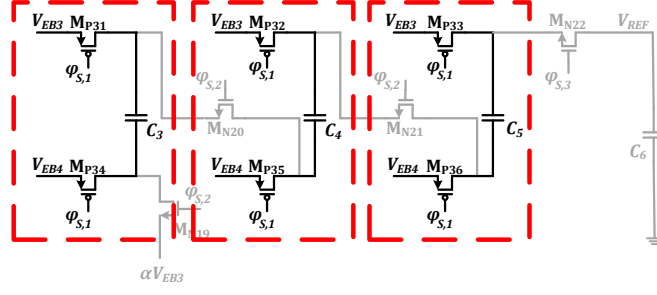
3.2.2 SETTling ERROR REDUCTION IN SERIES-PARALLEL SCN

As shown in Figure 3-1, the PTAT voltage is generated by the difference between two V_{EB} voltages V_{EB3} and V_{EB4} , which can be expressed as:

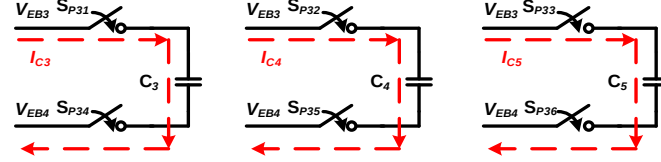
$$\Delta V_{EB} = V_{EB3} - V_{EB4} = \frac{kT}{q} \ln \left(\frac{I_{Q3} A_{Q4}}{I_{Q4} A_{Q3}} \right) \quad (3-7)$$

where k is the Boltzmann constant, T is the absolute temperature in Kelvin, q is the electron charge. I_{Q3} and I_{Q4} are the biasing currents of the two BJTs Q_3 and Q_4 to generate V_{EB3} and V_{EB4} respectively as shown in Figure 3-14. A_{Q3} and A_{Q4} are the area of the two BJTs Q_3 and Q_4 . Following (3-7), in order to generate a large PTAT voltage ΔV_{EB} , either the biasing current of Q_3 (I_{Q3}) or the area of Q_4 (A_{Q4}) should be increased. Usually, we amplify the PTAT voltage ΔV_{EB} by a coefficient γ [3.7]. In the proposed design, γ is generated by the series-parallel SCN, where γ is the number of the series-parallel capacitor. Figure 3-4(a) shows the series-parallel SCN in the sampling state [3.10], with its equivalent circuit model as shown in Figure 3-4(b).

In the sampling state, V_{EB3} is connected to the top plate of C_3 , C_4 , and C_5 , while V_{EB4} is connected to the bottom plate of C_3 , C_4 , and C_5 for storing ΔV_{EB} as shown in Figure 3-4(b). Since V_{EB3} will be sampled by the voltage divider SCN in each period in order to form the CTAT voltage coefficient α as discussed in Section 3.2.1, there exist



(a)



(b)

Figure 3-4 (a) Circuit configuration of our previous proposed series-parallel SCN in sampling state [3.10], and (b) its equivalent circuit model.

charging currents I_{C3} , I_{C4} , and I_{C5} in the series-parallel SCN as shown in Figure 3-4(b). I_{C3} , I_{C4} , and I_{C5} would affect the biasing point of V_{EB4} , resulting in a settling error for ΔV_{EB} sampling. Figure 3-5 shows the settling error of the series-parallel SCN under different numbers of series-parallel capacitors, which is mainly caused by the charging current I_{C3} , I_{C4} , and I_{C5} (a 0.1% settling error in the series-parallel SCN would result in a $\sim 185\mu\text{V}$ variation of the V_{REF} or a $\sim 2.3\text{ppm}/^\circ\text{C}$ extra TC of the BGR circuit).

Increasing the biasing current I_{Q4} can relax this error. However, either I_{Q3} or A_{Q4} should be increased proportionally in order to form the same PTAT voltage, inevitably increasing the consumed power and area. Instead, lowering the sampling speed can also relax this error. Yet, it would increase the error caused by the leakage current, leading to an increase in TC.

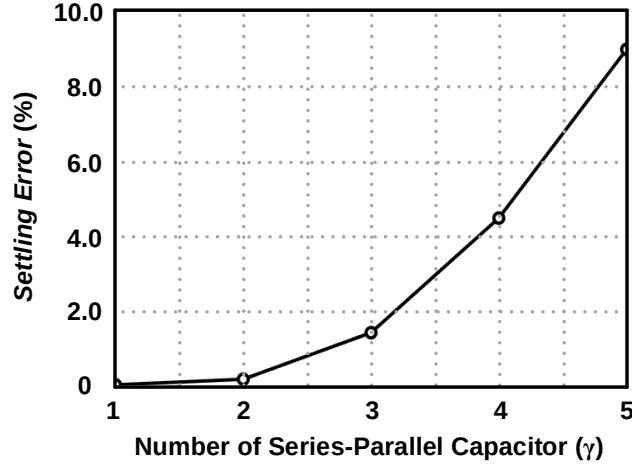


Figure 3-5 Settling error of the series-parallel SCN in sampling state under different γ .

To resolve the above problem, we introduce replicated branches for V_{EB4} generation which will be discussed in Section 3.3.6. By employing the replica V_{EB4} generation branches, the settling error ($\gamma = 3$) can be suppressed to the level when single-branch V_{EB4} generation and $\gamma = 1$ are applied [i.e., the settling error is suppressed from 1.8% to 0.01%]. As the biasing current of V_{EB4} is very small, the extra power consumption of the replicated branches is still negligible. The cost for the extra BJT array is also negligible as the area is typically dominated by the CP capacitors.

3.2.3 AREA OPTIMIZATION

To reduce the minimum supply voltage, the general solution is to employ a conventional $2\times$ CP [3.5] as shown in Figure 3-6 to double the supply voltage from V_{DD} to $2V_{DD}$. However, due to the settling errors of the two SCNs, the operating frequency of the $2\times$ CP is limited [3.5] and [3.10]. To reduce the output ripple of the $2\times$ CP, a large capacitor is usually employed, thus occupying a large chip area.

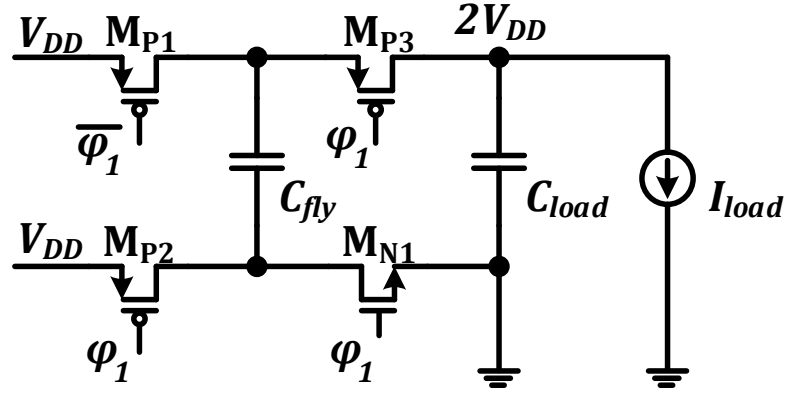


Figure 3-6 Circuit configuration of conventional $2\times$ CP [3.5].

To solve this problem, two approaches are employed in the proposed design: (1) Using a dual-branch interleaved $2\times$ CP; and (2) Employing a dual clock path topology, where the clock signal with faster operating frequency (the fast clock) is employed to control the $2\times$ CP while the clock signal with slower operating frequency (the slow clock) is employed to control the two SCNs. The details will be discussed in Section 3.3.3.

3.3 PROPOSED BGR ARCHITECTURE

3.3.1 CIRCUIT ARCHITECTURE

Figure 3-7 shows the architecture of the proposed BGR circuit. An oscillator with a PTAT current biasing is employed to generate the fast and slow clock signals $CLK_{F/S}$. The bootstrapping circuit boosts the voltage level of the fast and slow clock to $2V_{DD}$. A dual-branch interleaved $2\times$ CP is employed to double the supply voltage of the BGR core to $2V_{DD}$. Biased with a PTAT current source, the BJT generates the required V_{EB} , which is sampled by the voltage divider SCN and the series-parallel SCN to generate

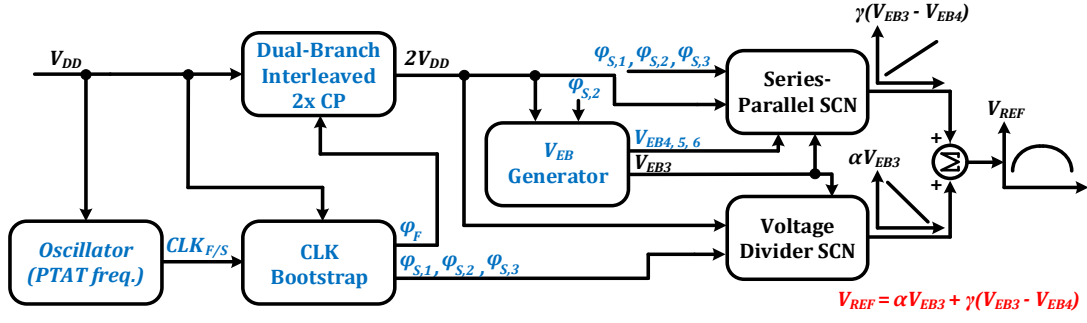


Figure 3-7 Architecture of proposed BGR circuit.

the coefficients α and γ for the corresponding CTAT and PTAT voltages. As discussed in Section 3.2, the PTAT fast clock drives the proposed $2\times$ CP in order to reduce the capacitor area of the $2\times$ CP, while the PTAT slow clock controls the two SCNs in order to relax the settling error and thus obtaining a lower power consumption as well as a smaller TC. The dual-branch interleaved $2\times$ CP is employed to relax the output voltage ripple constrain for a smaller chip area. Also, replica V_{EB} generation branches are explored to relax the settling error in the series-parallel SCN for a smaller TC. The details of each circuit block in Figure 3-7 will be discussed as follows.

3.3.2 OSCILLATOR WITH PTAT FREQUENCY

Figure 3-8 shows the circuit configuration of the current-controlled oscillator as well as its PTAT current biasing circuit. The PTAT current I_{R1} as shown in Figure 3-8(a) can be expressed as:

$$I_{R1} = \frac{\eta V_T \ln(K)}{R_1} \quad (3-8)$$

where η is the process-dependent parameter, K is the W/L ratio of M_{P2} and M_{P3} .

M_{P1} , M_{P4} , M_{N1} and M_{N4} are employed to form an amplifier to improve the PSRR. I_{R1} is

then copied by the current mirrors $M_{P5,7,9}$ and $M_{N6,8,10}$ as shown in Figure 3-8(b) to

bias the oscillator. The oscillating frequency can be expressed as:

$$f = \frac{3I_{R1}}{C_{OSC}V_{DD}} \quad (3-9)$$

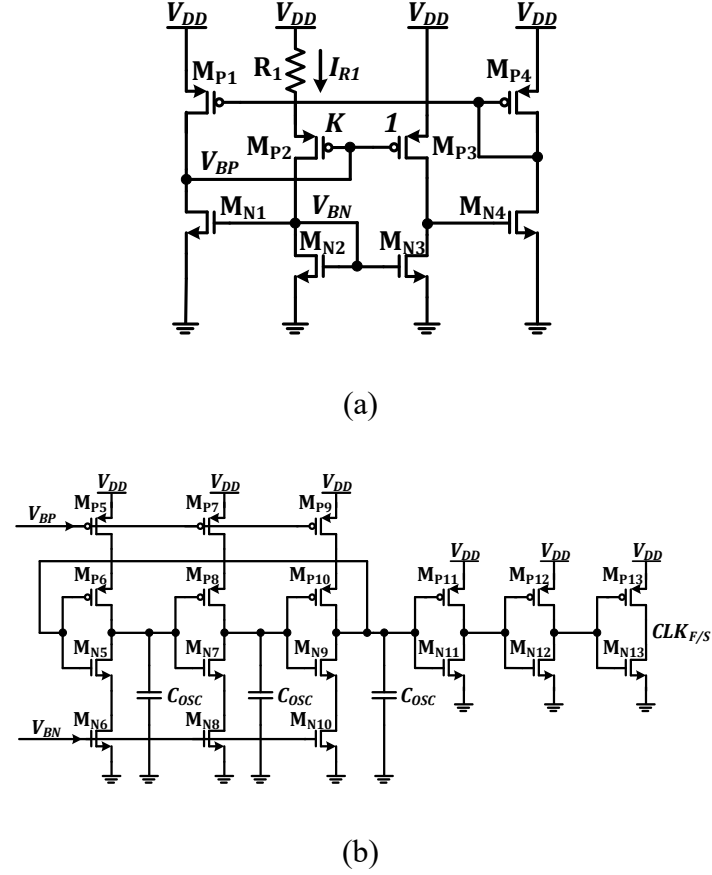


Figure 3-8(a) PTAT current biasing circuit of the oscillator [3.8], (b) current-controlled oscillator for the fast or slow clock generation.

By adjusting the TC of I_{R1} , the TC of the oscillating frequency can be adjusted correspondingly. To optimize the settling error as discussed in Section 3.2.1, the TC of I_{R1} should follow (3-19) which will be discussed in Section 3.3.5. In this work, $R_1 = 10\text{M}\Omega$ and $K = 3$. Considering a $\pm 15\%$ variation in R_1 , the biasing current of V_{EB3} and the capacitor size of $2 \times \text{CP}$ are chosen accordingly. Also, we set $C_{OSC,F} = 100\text{fF}$ to

generate the fast clock, with $f_{CLK,F} = 40\text{kHz}$. Furthermore, $C_{OSC,S} = 250\text{fF}$, resulting in $f_{CLK,S} = 5\text{kHz}$. The details of the selection of $f_{CLK,F}$ and $f_{CLK,S}$ will be discussed in Section 3.3.3 and Section 3.3.5 respectively.

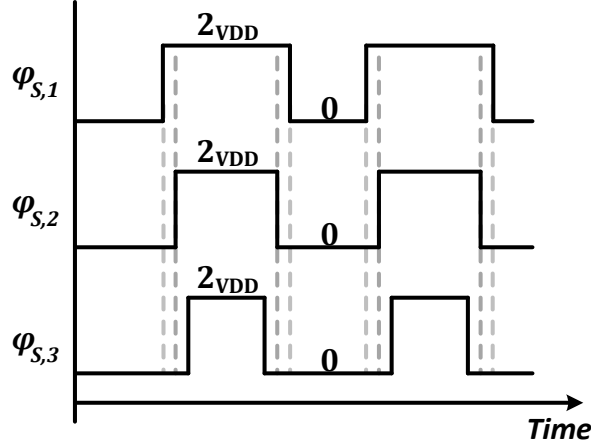


Figure 3-9 Non-overlap slow clock waveforms.

Besides, three buffers are employed to ensure fast clock transitions. $CLK_{F/S}$ would pass through the bootstrapping circuit [3.10] for boosting the voltage swing of the fast clock φ_F and the slow clock φ_S to $2V_{DD}$. φ_F is employed to control the dual-branch interleaved $2\times$ CP as shown in Figure 3-9. Meanwhile, φ_S would pass through a non-overlap circuit to generate the non-overlap slow clock signals as shown in Figure 3-9 for controlling the V_{EB} generator and the two SCNs. The simulated total power consumption of the oscillators and the bootstrapping circuit for the fast and slow clocks is 9nW .

3.3.3 DUAL-BRANCH INTERLEAVED $2\times$ CP

Figure 3-10 shows the circuit configuration of the dual-branch interleaved $2\times$ CP.

When φ_F is low and $\bar{\varphi}_F$ is high, C_{fly1} is charged to V_{DD} while the C_{fly2} is connected to

C_{load} to supply $2V_{DD}$ to the core circuit; when φ_F is high and $\bar{\varphi}_F$ is low, C_{fly2} is charged to V_{DD} while C_{fly1} is connected to C_{load} to supply $2V_{DD}$ to the core circuit.

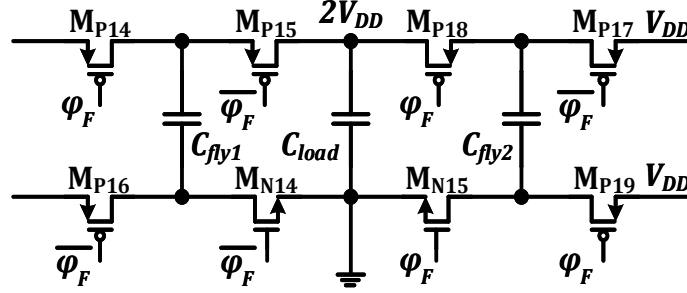


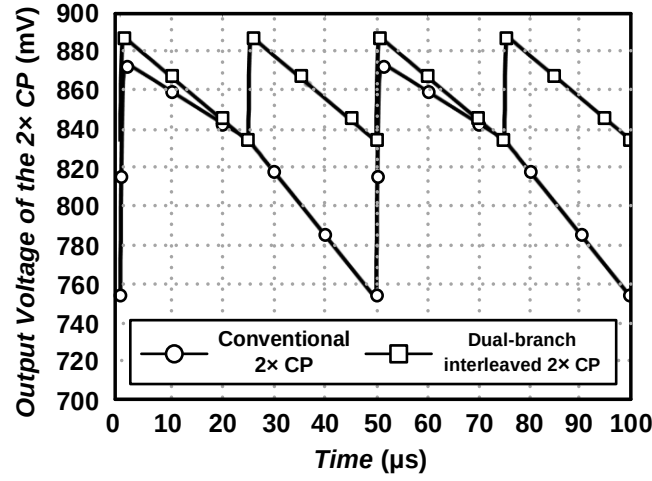
Figure 3-10 Circuit configuration of the dual-branch interleaved $2\times$ CP with fast clock control.

Compared with the conventional $2\times$ CP as shown in Figure 3-6, due to its interleaved charging and discharging process, the effective frequency is doubled and hence the output voltage ripple can also be halved. The output voltage ripple of the conventional $2\times$ CP and the dual-branch interleaved $2\times$ CP can be expressed as:

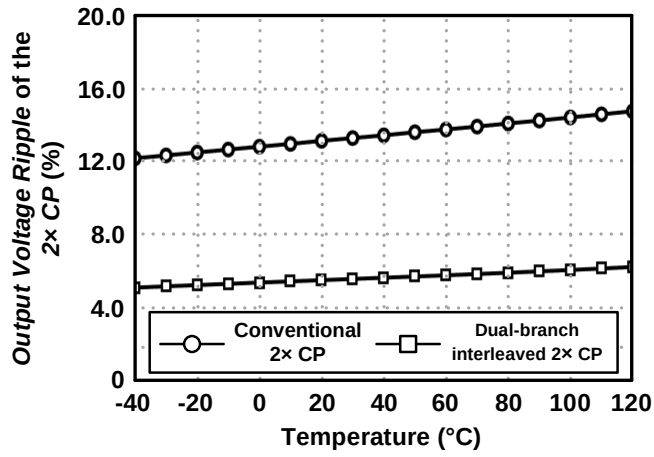
$$V_{Ripple_Con} = \frac{I_{load}}{2fC_{load}} + \frac{I_{load}}{2f(C_{fly}+C_{load})} \quad (3-10)$$

$$V_{Ripple_DB} = \frac{I_{load}}{2f(C_{fly1}+C_{load})} \quad (3-11)$$

where V_{Ripple_Con} is the output voltage ripple of the conventional $2\times$ CP, V_{Ripple_DB} is the output voltage ripple of the dual-branch interleaved $2\times$ CP, C_{fly} is the flying capacitor of the conventional $2\times$ CP as shown in Figure 3-10, and C_{fly1} is the flying capacitor of the dual-branch interleaved $2\times$ CP as shown in Figure 3-10. By choosing $C_{fly1}=C_{fly2}$, if the total capacitance of the conventional $2\times$ CP is the same as the proposed dual-branch interleaved $2\times$ CP, the output voltage ripple of the proposed one is about half of the conventional one, that is $V_{Ripple_DB} \approx 0.5V_{Ripple_Con}$ as validated by the simulation



(a)

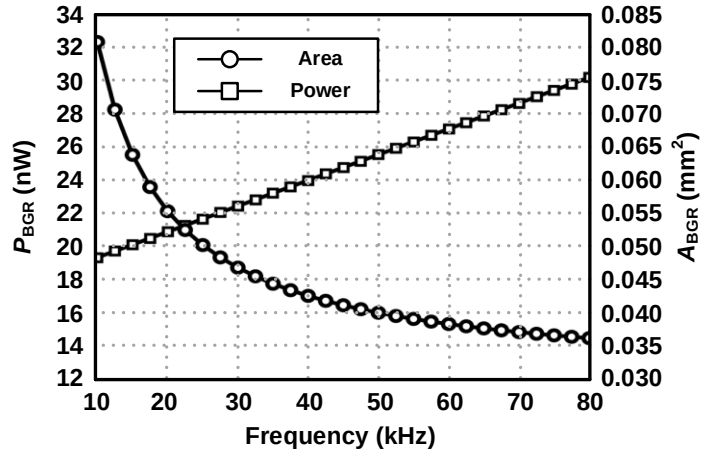


(b)

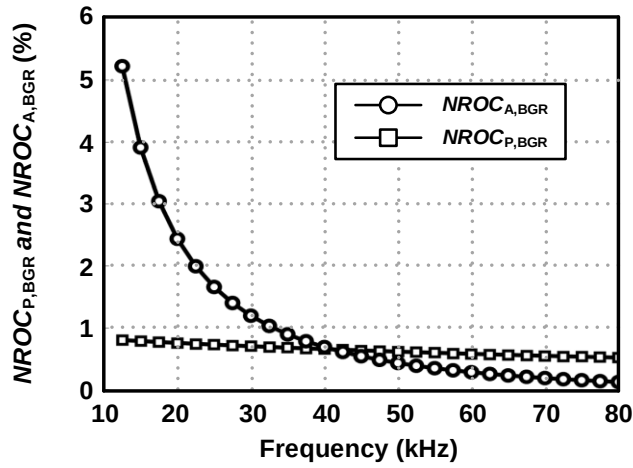
Figure 3-11(a) Transient response of the $2V_{DD}$ of the conventional $2\times$ CP and the proposed dual-branch interleaved $2\times$ CP with 20kHz clock frequency, 0.5V supply voltage, 30nA PTAT loading current, and 20pF total capacitance, and (b) output voltage ripple of the conventional $2\times$ CP (V_{Ripple_Con}) and the proposed dual-branch interleaved $2\times$ CP (V_{Ripple_DB}) versus temperature

result in Figure 3-11. Figure 3-11(a) shows the transient response of $2V_{DD}$ of the conventional $2\times$ CP and the proposed dual-branch interleaved $2\times$ CP. It clearly illustrates that the effective frequency of the dual-branch interleaved $2\times$ CP is doubled

when compared with that of the conventional $2\times$ CP, and hence the output voltage ripple can be halved. Figure 3-11(b) shows the $V_{\text{Ripple_Con}}$ and the $V_{\text{Ripple_DB}}$ versus temperature. As the BGR is biased by a PTAT current, the I_{load} is large at high temperature and hence results in a large $V_{\text{Ripple_DB}}$. Therefore, a PTAT clock is employed to control the dual-branch interleaved $2\times$ CP in order to relax the $V_{\text{Ripple_DB}}$.



(a)



(b)

Figure 3-12(a) A_{BGR} and P_{BGR} versus the frequency of the fast clock, and (b) $NROC_{\text{A,BGR}}$ and $NROC_{\text{P,BGR}}$ versus the frequency of fast clock.

Although the dual-branch interleaved $2\times$ CP can reduce the output voltage ripple, the operating frequency should be low due to the settling error constraints. However, this essentially contradicts with the small output voltage ripple requirement. To solve this problem, this work proposes a dual clock path topology, where a fast clock drives the $2\times$ CP and a slow clock controls the two SCNs. One major design consideration is that the fast clock results in a large power consumption in the oscillator. Also, as the operating frequency of the $2\times$ CP increases, the bandwidth of the amplifier should be enhanced, raising further the power consumption requirement.

In conclusion, doubling the switching frequency can reduce the area of the $2\times$ CP by half, but the power consumption of both the oscillator and the amplifier will be doubled. Figure 3-12(a) shows the area of the overall BGR (A_{BGR}) and power of the overall BGR (P_{BGR}) versus the operating frequency of the fast clock. To determine the optimized point, it is necessary to figure out their normalized rate of change (NROC) values in order to show the trade-off relationship between A_{BGR} and P_{BGR} . The NROC of P_{BGR} ($NROC_{P,BGR}$) and the NROC of A_{BGR} ($NROC_{A,BGR}$) can be expressed as:

$$NROC_{P,BGR} = \frac{1}{P_{BGR}} \left| \frac{dP_{BGR}}{df} \right| \quad (3-12)$$

$$NROC_{A,BGR} = \frac{1}{A_{BGR}} \left| \frac{dA_{BGR}}{df} \right| \quad (3-13)$$

Figure 3-12(b) shows the $NROC_{P,BGR}$ and the $NROC_{A,BGR}$ versus the frequency of the fast clock. The intersection point locates at $f=40.4\text{kHz}$. In this work, the operating frequency of the $2\times$ CP is chosen to be 40kHz .

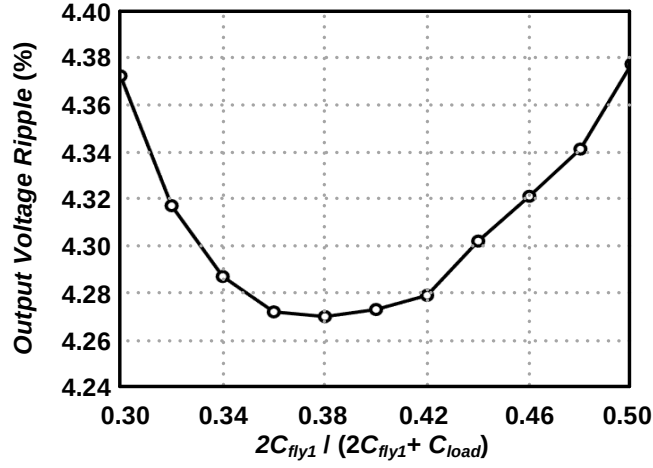


Figure 3-13 Simulated output voltage ripple of the dual-branch interleaved $2\times$ CP as a function of capacitor ratio $2C_{fly1}/(2C_{fly1} + C_{load})$.

Figure 3-13 shows the simulated output voltage ripple of the dual-branch interleaved $2\times$ CP as a function of the capacitor ratio $2C_{fly1}/(2C_{fly1}+C_{load})$, where $C_{fly1}=C_{fly2}$. We can observe that the output voltage ripple can be optimized when $2C_{fly1}/(2C_{fly1}+C_{load})=0.38$. In this work, we set $C_{fly1}=C_{fly2}=3.8\text{pF}$ and $C_{load}=12.4\text{pF}$ to optimize the output voltage ripple level. Because of the application of the proposed dual-branch interleaved $2\times$ CP with a PTAT fast clock, the output ripple of the $2\times$ CP is small within the whole temperature range. As a result, there is no voltage drop in the V_{REF} at the high temperature as shown in Figure 3-24(a), when compared with that of the design in [3.10] and hence the TC is further improved by $\sim 4\text{ppm}/^\circ\text{C}$.

3.3.4 V_{EB} GENERATOR

Figure 3-14 shows the circuit configuration of the V_{EB} generator. A basic single-ended output differential pair amplifier A_1 with a power consumption of 3nW is employed to generate the biasing current I_{R2} , which can be expressed as:

$$I_{R2} = \frac{V_{EB1} - V_{EB2}}{R_2} = \frac{V_T \ln(N)}{R_2} \quad (3-14)$$

where N is the area ratio of the Q_2 and Q_1 . We can ensure good BJT matching by using an integer ratio of $N=3$ [3.14] and arranging them in a 2×2 array. This can ensure that each BJT unit is matched in a compact area.

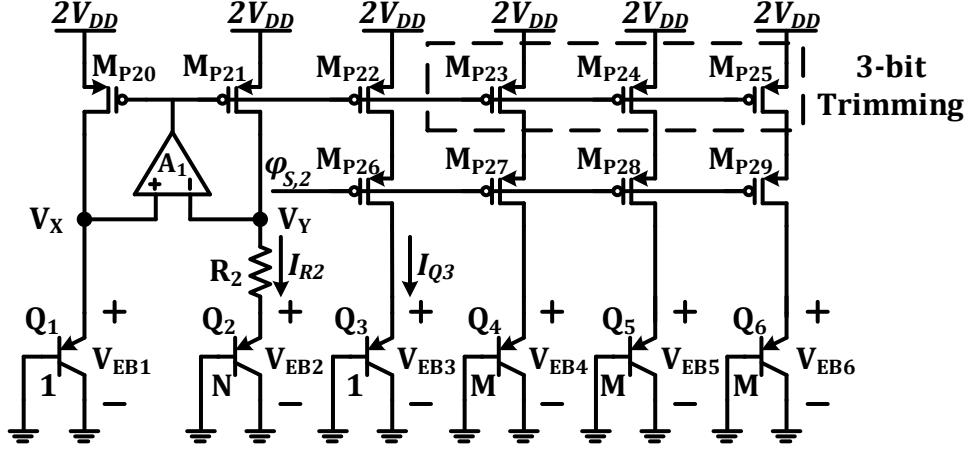


Figure 3-14 Circuit configuration of the V_{EB} generator.

In this work, $R_2=10\text{M}\Omega$ to generate nA level current. I_{R2} is copied by the current mirror $M_{P22-P25}$ to bias the BJTs Q_{3-6} to generate $V_{EB3-EB6}$, which will then be sampled by the voltage divider SCN and the series-parallel SCN to yield BGR output V_{REF} as shown in Figure 3-7. The BGR reference voltage V_{REF} can be expressed as:

$$V_{REF} = \alpha V_{EB3} + \gamma \Delta V_{EB} = \alpha V_T \ln\left(\frac{I_{Q3}}{I_{S,Q3}}\right) + \gamma V_T \ln\left(M \frac{W_{P22}}{W_{P23}}\right) \quad (3-15)$$

where α is the CTAT voltage coefficient generated by the voltage divider SCN and γ is the PTAT voltage coefficient generated by the series-parallel SCN (the details will be discussed in Section 3.3.5 and 3.3.6, respectively). The difference in V_{EB} is defined as $\Delta V_{EB} = V_{EB3} - V_{EB4} = V_T \ln(M \cdot W_{P22}/W_{P23})$, where M is the area ratio of Q_3 and Q_4 , W_{P22} and W_{P23} are the width of the transistors M_{P22} and M_{P23} , and $I_{S,Q3}$ is the saturation

current of the BJT Q_3 . Also, as shown in Figure 3-14, V_{EB5} and V_{EB6} are the replicas of V_{EB4} (i.e., $V_{EB4}=V_{EB5}=V_{EB6}$), aiming to reduce the settling error in the series-parallel SCN which will be discussed in Section 3.3.6. In addition, as V_{EB3} depends on I_{Q3} and $I_{S,Q3}$, while ΔV_{EB} depends mainly on device matching, the variation of V_{REF} is mainly caused by the V_{EB3} (about $\pm 9\text{mV}$ at 27°C). Since α is about 0.35 in the proposed design, the variation of V_{REF} is about $\pm 3.2\text{mV}$. To overcome this variation, a 3-bit trimming circuit to control the widths $W_{P23-P25}$ of transistors $M_{P23-P25}$, where the least significant bit (LSB) $\approx 1\text{mV}$.

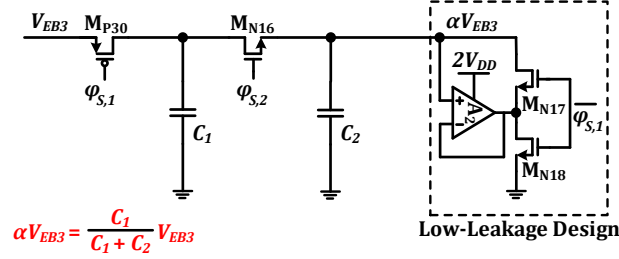
3.3.5 VOLTAGE DIVIDER SCN

Figure 3-15(a) shows the circuit configuration of the voltage divider SCN to generate the CTAT voltage coefficient α in Figure 3-15. In the sampling state as shown in Figure 3-15 (b), M_{P30} , M_{N17} , and M_{N18} are on while M_{N16} is off. C_1 is charged to V_{EB3} and C_2 is fully discharged. In the holding state as shown in Figure 3-15(c), M_{P30} , M_{N17} , and M_{N18} are off while M_{N16} is on. The charge stored in C_1 is redistributed as:

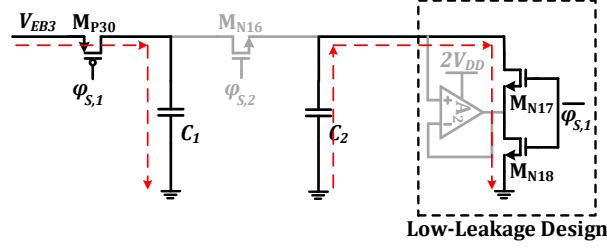
$$\alpha V_{EB3} = \frac{C_1}{C_1+C_2} V_{EB3} \quad (3-16)$$

Hence, M_{P30} , M_{N17} , and M_{N18} would be turned off first to prevent the charge leakage. To resolve the settling error problem as discussed in Section 3.2.1, we propose to drive the voltage divider SCN using a PTAT clock signal. First, the first-order TC of R_{BIAS} ($TC_{R,BIAS}$) in Figure 3-16 can be expressed as:

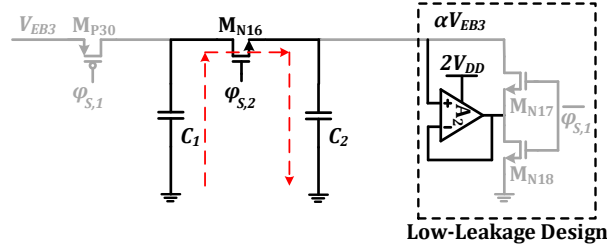
$$TC_{R,BIAS} = \frac{1}{R_{BIAS,0}} \frac{dR_{BIAS}}{dT} \quad (3-17)$$



(a)



(b)



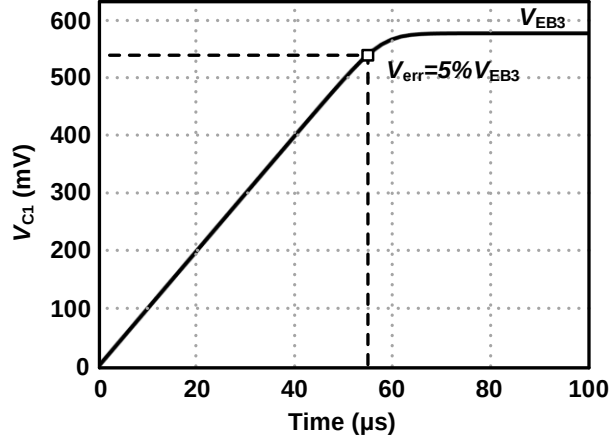
(c)

Figure 3-15 (a) Circuit configuration of the voltage divider SCN with low-leakage design [3.10], (b) in the sampling state, and (c) in the holding state.

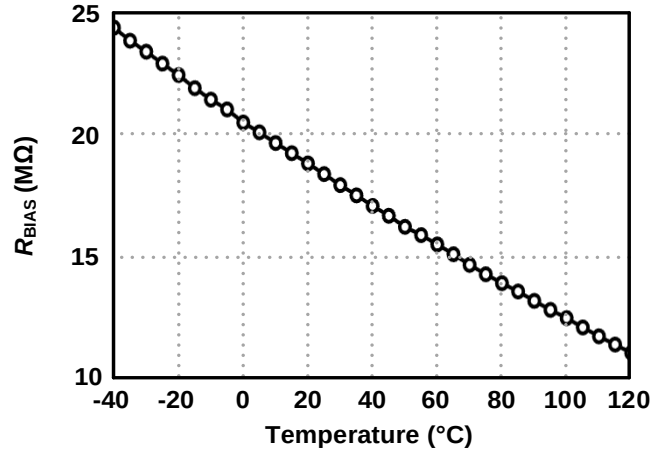
Noticed that the non-overlap clock is employed to control the voltage divider SCN.

As the high-order TC of R_{BIAS} is small, they are neglected to simplify the study.

As $TC_{R,BIAS}$ should be normalized to $R_{BIAS,0}$, the absolute value of R_{BIAS} would not affect $TC_{R,BIAS}$. Therefore, no matter $V_{err} = 0.05V_{EB3}$ or $V_{err} = xV_{EB3}$, this coefficient x will not affect $TC_{R,BIAS}$. In this work, we choose $V_{err} = 0.05V_{EB3}$. Substitute



(a)



(b)

Figure 3-16 (a) Transient response of V_{C1} under $R_{P30}=0$, $I_{BIAS}=10\text{nA}$, $C_I=1\text{pF}$, and $T=20^\circ\text{C}$ and, (b) estimated R_{BIAS} under $V_{err} = 0.05V_{EB}$, $R_{P30}=0$, $I_{BIAS}=10\text{nA}$ and $C=1\text{pF}$.

$V_{err} = 0.05V_{EB3}$ into (3-4), it becomes:

$$R_{BIAS} = -\frac{1}{\ln(0.05)} \frac{t_{5\%}}{C_1} \quad (3-18)$$

where $t_{5\%}$ is the time at which $V_{err} = 0.05V_{EB3}$. To obtain $t_{5\%}$, a transient simulation with the circuit configuration at 20°C as shown in Figure 3-2(c) is plotted in Figure

3-16 (a). Let R_{P30} , I_{BIAS} and C_1 equal to 0, 10nA and 1pF, respectively. Figure 3-16(a) shows the transient response of V_{C1} where $t_{5\%}=55.9\mu s$ when $V_{err}=0.05V_{EB3}$. Substitute $t_{5\%}=55.9\mu s$ and $C_1=1pF$ into (3-18), we get $R_{BIAS,20^\circ C}=18.7M\Omega$. Repeat this step at different temperatures, we can obtain the R_{BIAS} curve as shown in Figure 3-16(b).

To eliminate the settling error without sacrificing power consumption, we employed a clock signal with PTAT frequency to compensate the $TC_{R,BIAS}$. The settling error in (3-4) [considering $TC_{R,BIAS}$] with a proposed PTAT-frequency clock ($f_{CLK,S}$) for compensation can be expressed as:

$$V_{err} = V_{EB3} \cdot e^{-\frac{1}{C_1} \frac{t_{SMP,0} [1+TC_{t,SMP}(T-T_0)]}{R_{BIAS,0} [1+TC_{R,BIAS}(T-T_0)]}} \quad (3-19)$$

where $t_{SMP,0}$ is the sampling time at the reference temperature T_0 , $TC_{t,SMP}$ is the first-order TC of t_{SMP} ($t_{SMP}=\frac{1}{2f_{CLK,S}}$). If $TC_{R,BIAS}=TC_{t,SMP}$, the TC of the exponential function can be eliminated [i.e., the nonlinear curve as shown in Figure 3-3 becomes a linear curve]. Therefore, (3-19) can be simplified as:

$$V_{err} = V_{EB3} \cdot e^{-\frac{1}{C_1} \frac{t_{SMP,0}}{R_{BIAS,0}}} \quad (3-20)$$

As C_1 , $t_{SMP,0}$, and $R_{BIAS,0}$ are constant, the exponential function is temperature-independent. The $TC_{t,SMP}$ can be controlled by the W/L ratio K in (3-8). Figure 3-17 shows the settling error under different $TC_{t,SMP}$. It clearly shows that the settling error can be eliminated if $TC_{R,BIAS} = TC_{t,SMP}$.

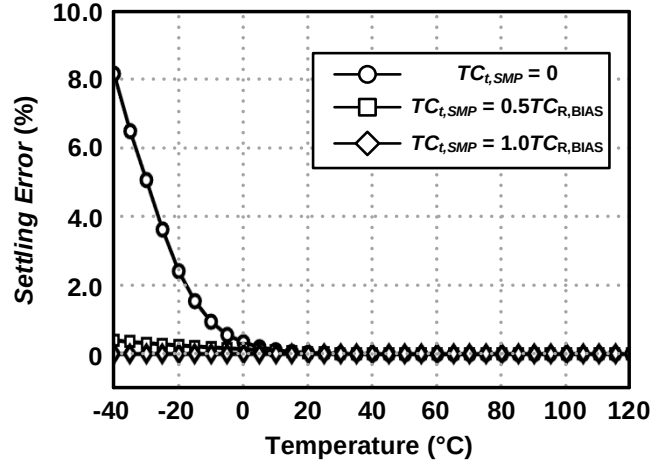


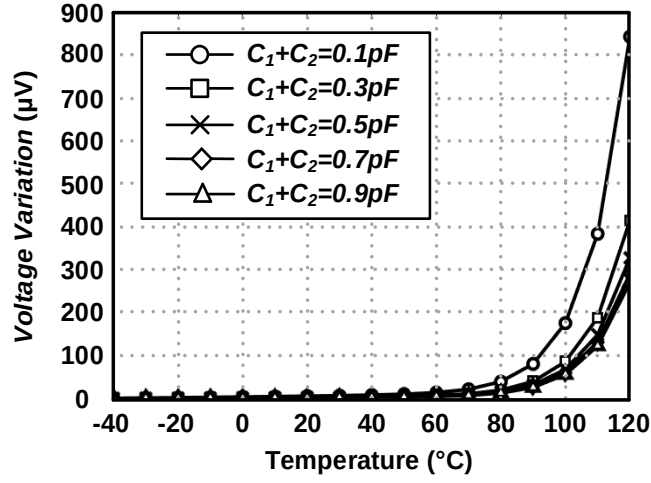
Figure 3-17 Settling error under different $TC_{t,SMP}$.

The other key parameter of the voltage divider SCN is the size of C_1 and C_2 , which mainly depends on the settling error and the leakage current. As the TC of the settling error of C_1 can be directly compensated by the PTAT frequency clock, the size of C_1 can be determined by the leakage current first. After that, we can design the corresponding operating frequency and the biasing current of V_{EB3} . The voltage variation caused by the leakage current can be expressed as:

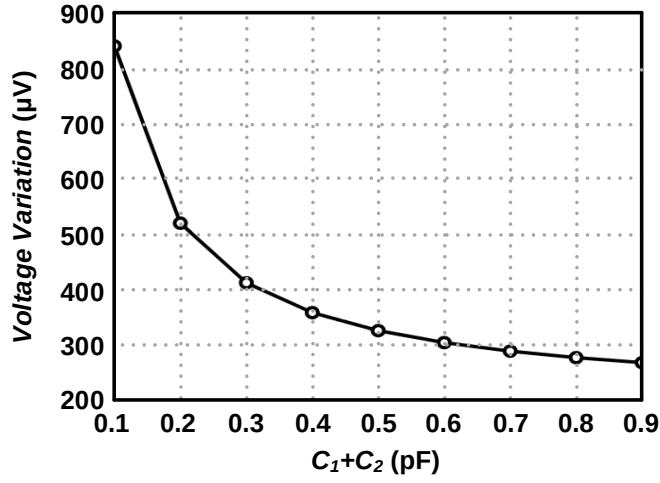
$$V_{Leakage} = \frac{I_{SUB}}{2f(C_1 + C_2)} \quad (3-21)$$

$$I_{SUB} = \mu C_{ox} \frac{W}{L} (m - 1) V_T^2 \exp\left(\frac{V_{GS} - V_{TH}}{mV_T}\right) (1 - \exp\left(\frac{-V_{DS}}{V_T}\right)) \quad (3-22)$$

where f is the slow clock frequency $f_{CLK,S}$, μ is the carrier mobility, C_{ox} is oxide capacitance, W is transistor width, L is transistor length, m is subthreshold slope factor ($m = 1 + C_d / C_{ox}$, where C_d is the depletion capacitance), V_{GS} is gate-to-source voltage, V_{TH} is the transistor threshold voltage, and V_{DS} is the drain-to-source voltage. Figure 3-18(a) shows the voltage variation caused by the leakage current of M_{N17} (referring to Figure 3-15) at different temperatures according to (3-21).



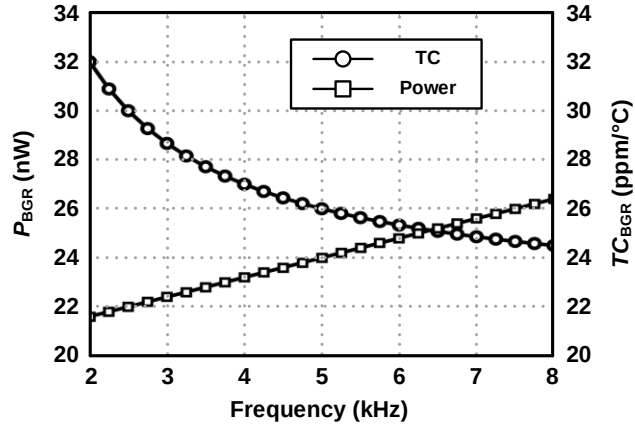
(a)



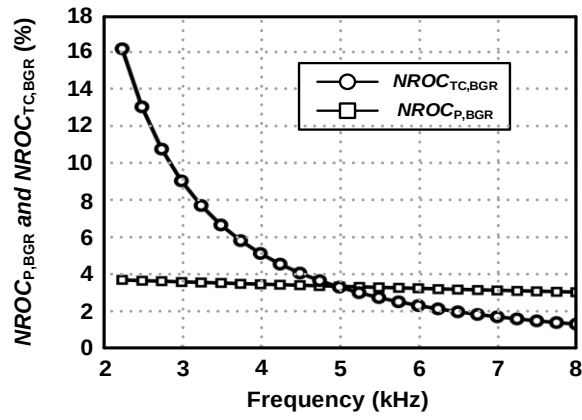
(b)

Figure 3-18 Voltage variation due to the leakage current versus (a) temperature and (b) capacitance C_1+C_2 at 5kHz operating frequency.

Figure 3-18(b) shows the relationship between V_{Leakage} and (C_1+C_2) . The variation will be saturated at around 0.8pF because of the effective capacitance of the SCNs which is proposed in [3.10], with $V_{\text{Leakage}} \approx 300\mu\text{V}$ or 4ppm/°C under 5kHz operating frequency. Therefore, $C_1+C_2 \approx 0.8\text{pF}$ is chosen in the proposed design. To form the CTAT voltage coefficient α , we set $C_1=250\text{fF}$ and $C_2=510\text{fF}$. For a fixed capacitance, V_{Leakage} depends on the operating frequency only.



(a)



(b)

Figure 3-19(a) TC_{BGR} and P_{BGR} versus the frequency of the slow clock, and (b) $NROC_{TC,BGR}$ and $NROC_{P,BGR}$ versus the frequency of slow clock.

By doubling the operating frequency of the slow clock, the voltage variation caused by the leakage current can be reduced by half, however, the biasing current I_{Q3} as shown in Figure 3-14 will also be doubled in order to relax the settling error. Figure 3-19(a) shows the TC of the overall BGR (TC_{BGR}) and the power of the overall BGR (P_{BGR}) versus the frequency of the slow clock. To determine the optimized point, it is necessary to figure out their NROC values in order to show the trade-off relationship between TC_{BGR} and P_{BGR} . The NROC of P_{BGR} ($NROC_{P,BGR}$) is defined in (3-12) and

the NROC of TC_{BGR} ($NROC_{TC,BGR}$) can be expressed as:

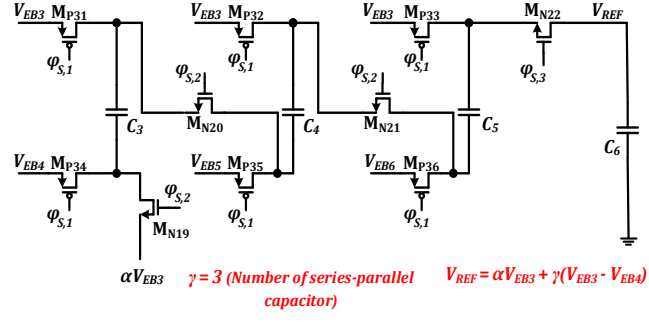
$$NROC_{TC,BGR} = \frac{1}{TC_{BGR}} \left| \frac{dTC_{BGR}}{df} \right| \quad (3-23)$$

Figure 3-19(b) shows $NROC_{P,BGR}$ and $NROC_{TC,BGR}$ versus the frequency of the slow clock. The intersection point locates at $f = 4.95\text{kHz}$. In this work, the frequency of the slow clock is chosen to be 5kHz .

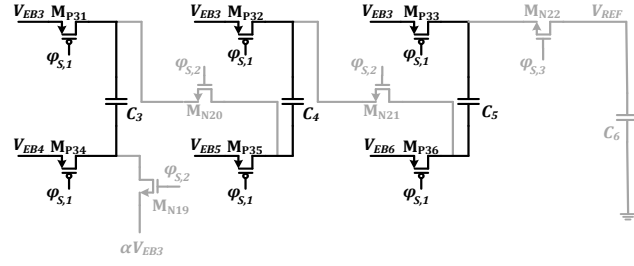
3.3.6 SERIES-PARALLEL SCN

Figure 3-20(a) shows the circuit configuration of the series-parallel SCN to generate the PTAT voltage coefficient γ as well as V_{REF} . In the sampling state as shown in Figure 3-20(b), M_{P31-36} are on while the M_{N19-22} are off, the difference between V_{EB3} and V_{EB4} , V_{EB3} and V_{EB5} , and V_{EB3} and V_{EB6} are stored by C_3 , C_4 and C_5 , respectively. In the holding state as shown in Figure 3-20(c), M_{P31-36} are off while M_{N19-22} are on. C_{3-5} are connected in series together with the node αV_{EB3} as shown in Figure 3-20. The non-overlap clock signals $\varphi_{S,1}$, $\varphi_{S,2}$ and $\varphi_{S,3}$ as shown in Figure 3-9 are employed to control the series-parallel SCN in order to prevent the charge leakage. During the transition process from sampling state to holding state, M_{P31-36} will be turned off first to prevent the charge stored in capacitors C_{3-5} to flow back to the V_{EB} node. Then, M_{N19-21} are turned on to form γ . Finally, M_{N22} is turned on to sample V_{REF} to C_6 to prevent large ripple at the V_{REF} node. The V_{REF} generated by the series-parallel SCN in

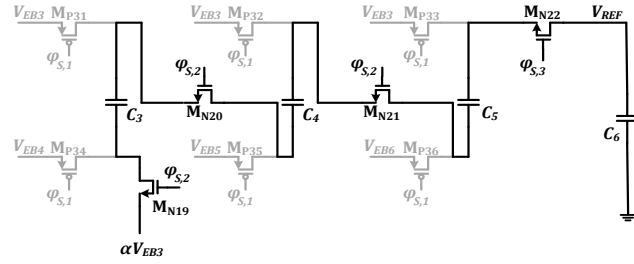
(3-15)



(a)



(b)



(c)

Figure 3-20 (a) Schematic of the series-parallel SCN, (b) sampling state, and (c) holding state.

becomes:

$$\begin{aligned}
 V_{REF} &= \alpha V_T \ln \left(\frac{I_{Q3}}{I_{S,Q3}} \right) + \gamma V_T \ln \left(M \frac{W_{P22}}{W_{P23}} \right) \\
 &= \frac{C_1}{C_1 + C_2} V_T \ln \left(\frac{I_{Q3}}{I_{S,Q3}} \right) + 3 V_T \ln \left(M \frac{W_{P22}}{W_{P23}} \right)
 \end{aligned} \tag{3-24}$$

where γ is the number of the series-parallel capacitor. Since V_{EB3} will be sampled by the

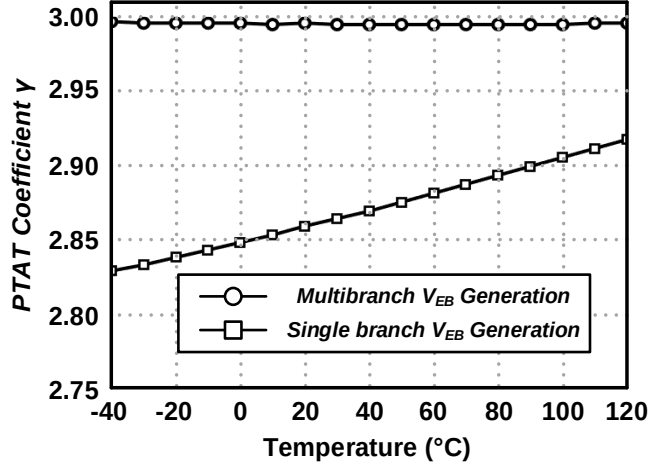


Figure 3-21 PTAT coefficient γ versus temperature.

voltage divider SCN in each cycle, the top plate voltage of C_3 varies from αV_{EB3} to V_{EB3} in each sampling state. This variation will be coupled by C_3 to the V_{EB4} node. As the biasing current of V_{EB4} is relatively small, this coupling charge would create a significant error and hence affect the accuracy of γ . In this case, we implement the replica V_{EB} generation branches as shown in Figure 3-14 and Figure 3-20(b). By employing the replica V_{EB} generation branches, the coupling charge to the V_{EB4} node is one-third compare with the case as shown in Figure 3-14. Figure 3-21 shows the simulated γ versus temperature. By applying the replica V_{EB} generation branches for V_{EB4} (i.e., V_{EB5} and V_{EB6}), the settling error can be suppressed to a negligible level.

3.4 SIMULATION AND MEASUREMENT RESULTS

The proposed SCN based BGR was implemented in a 65nm process and occupied a chip area of $0.17 \times 0.25 \text{ mm}^2$. The die photo together with the corresponding layout is shown in Figure 3-22. Since the die is covered by the AP layer, the circuit cannot be seen under the microscope. Table 3-1 shows the simulated power consumption of each

TABLE 3-1 SIMULATED POWER CONSUMPTION OF EACH BLOCK OF THE PROPOSED BGR AT 27°C AND $V_{DD}=0.5V$

Each Block	Power Consumption
2× CP	2.7nW
CLK oscillator and Bootstrap	9.0nW
V_{EB} Generator	16.5nW
Two SCNs	0.5nW

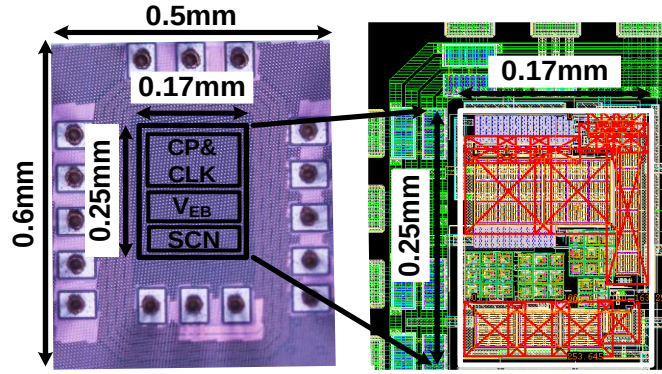
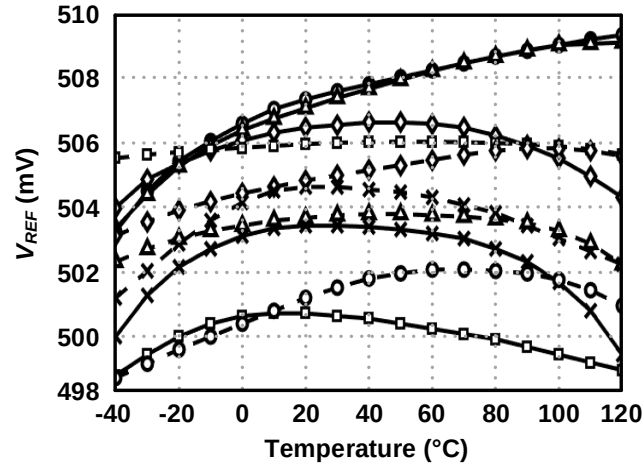
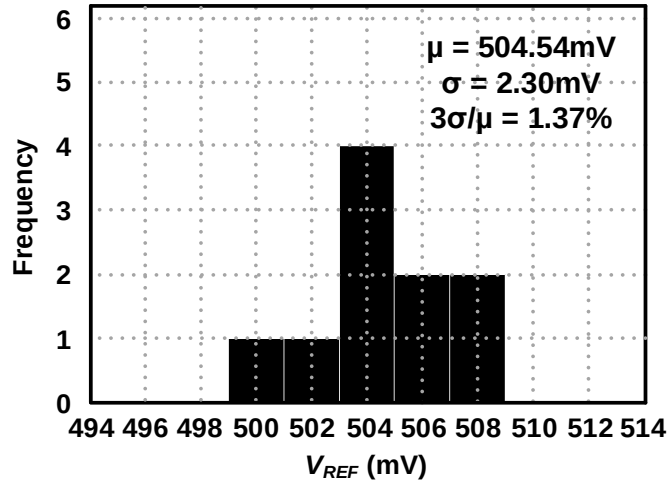


Figure 3-22 Die photo of the proposed BGR.

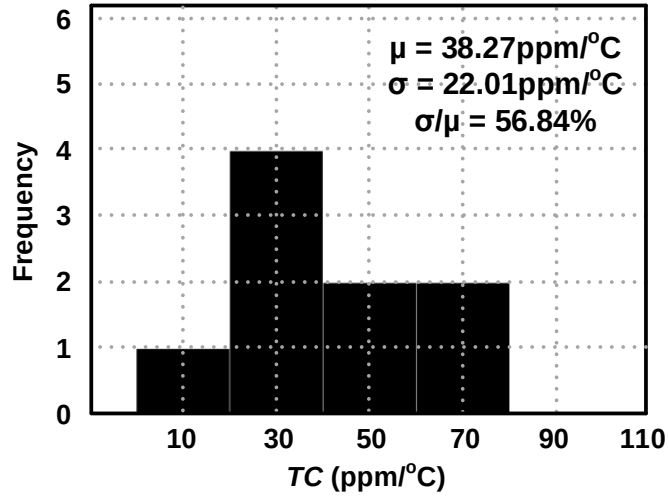
block of the proposed BGR. As the power of the SCNs is mainly contributed by the low leakage buffer without static current consumption, they consume very low power. The simulated total power consumption is 28.7nW at 27°C and $V_{DD}=0.5V$. The testing environment of the proposed BGR is similar to the testing environment in [3.9]. Figure 3-23 shows the measured V_{REF} from 10 chips before trimming under $V_{DD}=0.5V$. As shown in Figure 3-23(a), the TC of the BGR varies from 20ppm/°C to 70ppm/°C and obtains an average TC value of 38ppm/°C. Figure 3-23 (b) shows the V_{REF} variation at 20°C, demonstrating an average V_{REF} is 504.54mV with a standard deviation of 2.30mV, corresponding to a $3\sigma/\mu$ variation of 1.37%. Figure 3-23(c) shows the distribution of the BGR TC before trimming, which demonstrates an average TC of



(a)

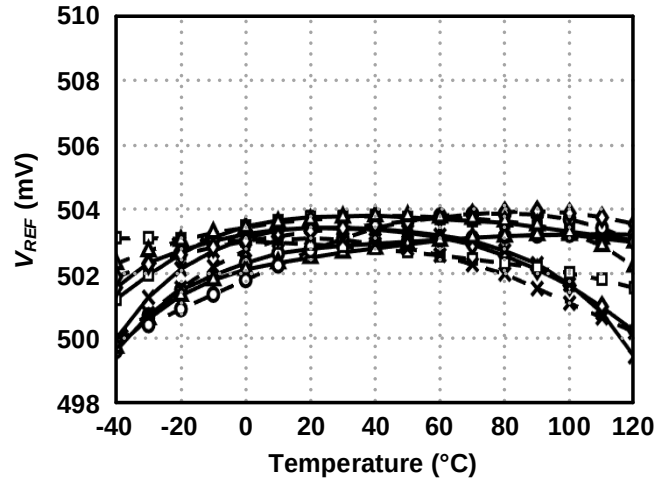


(b)

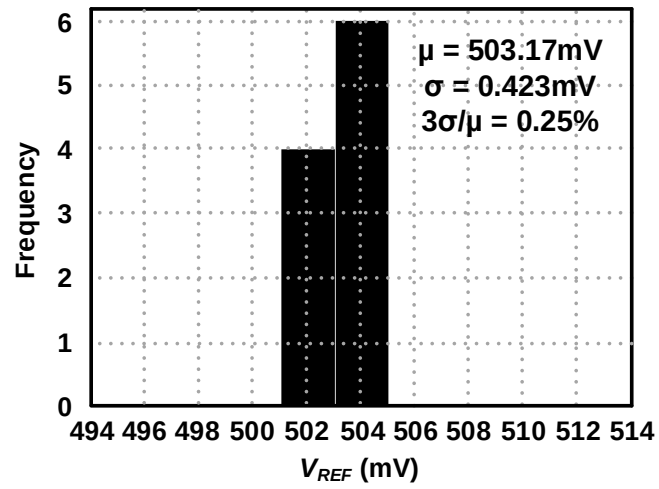


(c)

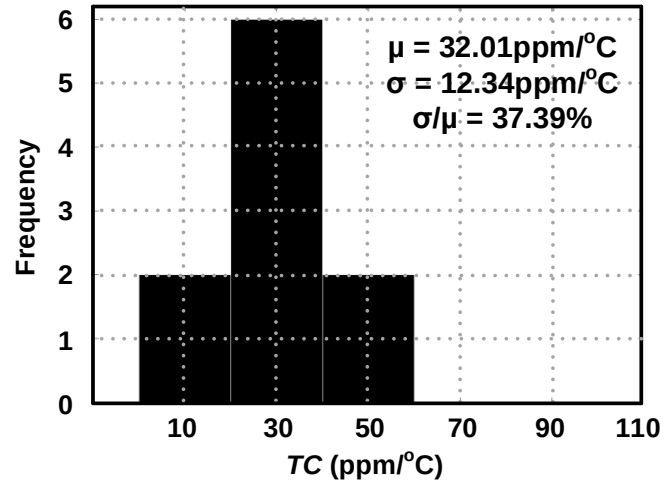
Figure 3-23 Measured untrimmed (a) BGR V_{REF} from -40 to 120 $^{\circ}\text{C}$, (b) distribution of the BGR V_{REF} across 10 chips at 20 $^{\circ}\text{C}$ and $V_{DD}=0.5\text{V}$, and (c) distribution of the BGR TC across 10 chips at $V_{DD}=0.5\text{V}$



(a)



(b)



(c)

Figure 3-24 Measured trimmed (a) BGR V_{REF} from -40 to 120 $^{\circ}\text{C}$ and (b) distribution of the BGR V_{REF} across 10 chips at 20 $^{\circ}\text{C}$ and $V_{DD}=0.5\text{V}$, and (c) distribution of the BGR TC across 10 chips at $V_{DD}=0.5\text{V}$

38.27ppm/°C, and a standard deviation of 22.01ppm/°C, corresponding to a σ/μ variation of 56.84%. Figure 3-24 shows the measured V_{REF} after trimming under the same testing condition. After applying one-point trimming at 20°C by changing 3bit calibration of $M_{P23} - M_{P25}$ as shown in Figure 3-14, the TC of the BGR varies from 20ppm/°C to 50ppm/°C and obtains an average value of 32ppm/°C, as shown in Figure 3-24(a). Figure 3-24(b) shows the average V_{REF} is 503.17mV with a standard deviation of 0.423mV as well as a $3\sigma/\mu$ variation of 0.25%. Figure 3-24(c) shows the distribution of the BGR TC after trimming, which demonstrates an average TC of 32.01ppm/°C, and a standard deviation of 12.34ppm/°C, corresponding to a σ/μ variation of 37.39%. Figure 3-25 shows the measured LNR results of 10 trimmed chips at 20°C, which shows that the V_{REF} has a deviation from 1.3mV to 2.1mV as V_{DD} varies from 0.5V to 1V. The corresponding LNR is from 2.6mV/V to 4.2mV/V or 0.52%/V to 0.84%/V, with an average value of 3.3mV/V or 0.66%/V.

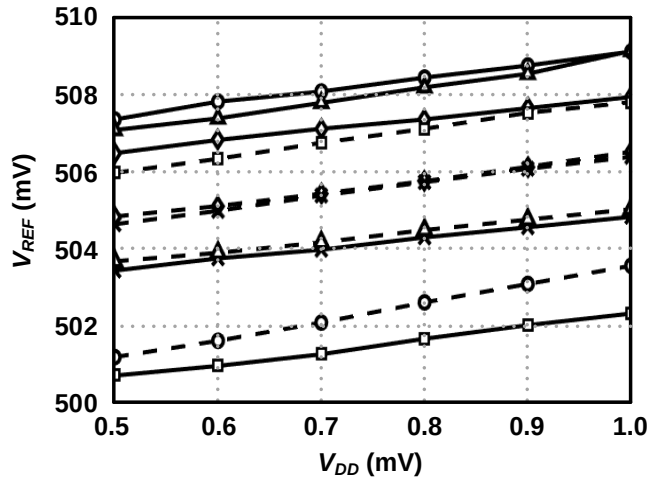


Figure 3-25 Measured V_{REF} of 10 trimmed chips versus V_{DD} at 20°C.

Table 3-2 Comparison of BGR Metrics with Other Low Power and Low Supply Voltage designs

Model	This work	[3.10]	[3.5]	[3.7]	[3.8]	[3.9]	[3.15]	[3.16]	[3.17]	[3.18]
Year	2021	2020	ISSCC 2015	JSSC 2012	TCAS-II 2017	TCAS-I 2018	TCAS-I 2019	TVLSI 2018	TCAS-II 2018	TCAS-II 2017
Process	65nm	65nm	0.13 μ m	0.13 μ m	0.18 μ m	0.18 μ m	0.18 μ m	0.18 μ m	0.18 μ m	0.18 μ m
Reference (mV)	503	495	500	256	240	460	368	412	328	212
Power Consumption (nW)	24	36	32	170	40	83	28	85	79	192
Minimum Supply Voltage (V)	0.5	0.5	0.5	0.75	0.5	0.55	0.7	0.9	0.8	0.4
TC (ppm/ $^{\circ}$ C)	32	42	75	40	58	28	43.1	33.7	33.8	84.5
Temperature Range ($^{\circ}$ C)	-40 – 120	-40 – 120	0 – 80	-20 – 85	-25 – 85	-45 – 120	-40 – 125	-40 – 125	-10 – 100	-40 – 130
Untrimmed error (3 σ / μ)	1.37%	3.08% ^M	2% ^M	3% ^M	3% ^M	3.2% ^M	1.05%	1.17%	N.A.	2% ^S
LNR	0.66%/V	0.64%/V	2%/V	0.019%/V	0.46%/V	0.59%/V	0.027%/V	0.060%/V	0.21%/V	0.957%/V
PSRR (dB)	-50 @DC	-50 @DC	-40 @DC	-93 @DC	-62 @100Hz	-62 @100Hz	-55 @100Hz	-40 @100Hz	-50 @100Hz	-40 @100Hz
Ripple	200 μ V ^S	200 μ V ^S	50 μ V ^M	20mV ^M	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
Area (mm ²)	0.0425	0.0522 0.0532*	0.0264	0.07	0.058	0.061	0.055	0.11	0.014	0.09

^S Simulation Result, ^M Measurement Result

Table 3-2 compares the proposed BGR with the previously reported state-of-the-art low power and low V_{DD} BGRs. For the SCN based BGRs in [3.5], [3.7] – [3.9], the proposed BGR obtains a better TC, a smaller power consumption and a wider temperature range compared with [3.5], [3.7] and [3.8]. Moreover, the proposed BGR consumes much less power and obtains a comparable TC compared with [3.9]. The area of the proposed BGR is the smallest among the existing SCN based BGRs, except from [3.5] mainly due to the large capacitors of the $2\times$ CP as well as the BJT array. Compared with [3.15] – [3.18], the proposed BGR obtains a better TC and smaller power consumption, while achieving a wider temperature range than [3.17] and a smaller area than [3.15], [3.16] and [3.18]. Compared with our previous work [3.10], the power consumption is 33% smaller which is mainly contributed by the power reduction in the V_{EB} generator and offsets by the power increment in the oscillator, the TC is 24% better, and the area is 19% smaller which is mainly contributed by the area reduction in the $2\times$ CP and offsets by the area increment in the oscillator and the V_{EB} generator while maintaining other performance similarly. Noticed that the ripple of V_{REF} is similar to the ripple of V_{REF} in [3.10] as it is mainly depending on the clock feedthrough and charge injection of M_{N22} during the transition process from holding state to sampling state referring to Figure 3-20.

3.5 SUMMARY

This work presented an SCN based BGR circuit with a wide temperature range, high precision, low supply voltage and low power consumption characteristics,

suitable for IoT applications. The proposed BGR employed a dual-branch interleaved $2\times$ CP to minimize the CP output ripple, used to supply the V_{EB} generator in order to relax V_{DD} from 0.9V to 0.5V. A PTAT fast clock controls the $2\times$ CP in order to reduce its capacitor area, and a PTAT slow clock controls the SCNs in order to relax the settling error with lower power consumption as well as to reduce TC. We also explored a replica V_{EB} generation branch to relax the settling error in the series-parallel SCN to further improve the TC. Finally, the measurement results verify the designed BGR, which obtains a TC of 32 ppm/ $^{\circ}\text{C}$ at 0.5V supply within $-40\text{ }^{\circ}\text{C}$ to $120\text{ }^{\circ}\text{C}$. The LNR is 3.3mV/V or 0.66%/V from 0.5V to 1V. Besides, the 10-chip measurement results showed a $3\sigma/\mu$ variation of 1.37% before trimming. We further improved the $3\sigma/\mu$ variation to 0.25% with a 3-bit trimming.

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Chapter 4 SUB- μ W AUTO-CALIBRATION BANDGAP VOLTAGE REFERENCE WITH 1σ INACCURACY OF $\pm 0.12\%$ WITHIN -40 TO 120 $^{\circ}\text{C}$

4.1 INTRODUCTION

High-precision ICs such as sensors received much attention in the field of the IoT and electric vehicles. The global industrial sensors market growth expectations vary from 20.6 billion USD in 2021 to 31.9 billion USD in 2026 [4.1]. The precision of high-performance ICs highly depends on the accuracy of the on-chip references [4.2]. For this reason, with such high-precision ICs mass produced, an integrated high-precision BGR circuit is indispensable. To provide a stable voltage reference against PVT variations, numerous studies have systematically analyzed voltage and temperature variations in BGRs [4.3]-[4.8]. Furthermore, the analysis of temperature-related and voltage-related errors in the current-based BGR have been reported in [4.7] and [4.8], respectively. Despite the extensive research works on voltage and temperature variations in BGRs, a systematic analysis of process-related errors is still lacking in the existing literatures [4.3]-[4.7] and [4.9]-[4.17], in which this work aims to address and contribute on this issue. Specifically, the generated reference voltage (V_{REF}) of the BGR will drift due to mismatch and process variation, significantly affecting its accuracy. To solve this problem, the studies in [4.3]-[4.6] and [4.9]-[4.11] employed one-point trimming techniques to improve the BGR accuracy, but they either required a complex structure [4.3]-[4.4], or cannot trim both V_{REF} and its TC error [4.5]- [4.6],

and [4.9]-[4.11]. Besides, the performance of the aforementioned designs can deteriorate as the BJT common-emitter current gain (β_F) becomes significant in advanced processes, and two-point trimming techniques [4.12] are typically required to improve the BGR accuracy, inevitably requiring a time-consuming calibration process after wafer-level fabrication and hence an increased labor cost. For instance, calibration alone adds ~60% production costs for a sensor IC in [4.13]. To avoid the costly calibration process, the studies in [4.7] and [4.14]-[4.17] presented different voltage reference circuits that do not require trimming. The studies in [4.14]-[4.15] can achieve a 3σ inaccuracy of less than 1% but were only validated in dated CMOS technologies, while those in [4.7] and [4.16]-[4.17] can suffer from a large 3σ inaccuracy due to a small β_F or the sub- μ W designs.

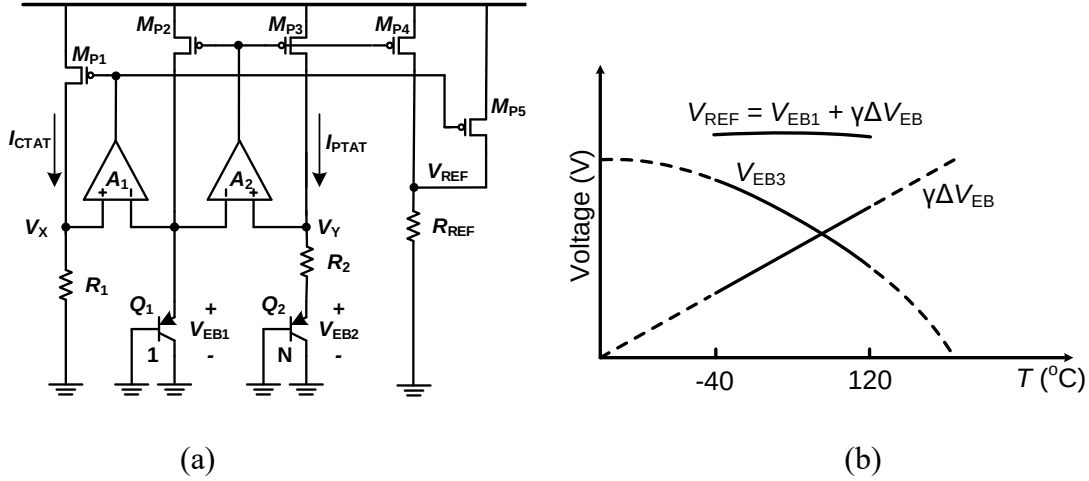


Figure 4-1(a) Circuit configuration of a conventional current-based BGR, and (b) relationship between voltage and temperature of the reference circuit.

Figure 4-1(a) presents the conventional current-based BGR which can suffer from several error sources, including amplifier offset as well as the process drifts of the

BJT's saturation current (I_S), forward emission coefficient (η_F), and common-emitter current gain (β_F). Typically, the PTAT error as introduced by the I_S process drift can be greatly reduced by the conventional one-point trimming method, with the amplifier offset tackled by dynamic techniques (e.g., chopping) [4.4]. However, it is still a challenging problem to systematically alleviate the η_F and β_F induced error without extra calibration efforts, especially in advanced CMOS processes.

This work, an extended version of [4.18], presents an auto-calibration technique for a current-based BGR. Utilizing the relationship of I_S , η_F , and β_F , we theoretically develop a trimming ratio ($\Delta m/\Delta n$) of the PTAT and the CTAT coefficients, which can compensate both the drift in V_{REF} and its TC simultaneously. We also propose an auto-calibrated BGR using a digital-assisted auto-calibration loop, which can automatically alleviate the error induced by η_F and β_F using only one-point trimming, while relaxing simultaneously the error caused by I_S and the amplifier offset. This work first studies the effect of process variation on I_S , η_F , and β_F with reference to V_{EB} and ΔV_{EB} of the BJT, which leads to the drift in V_{REF} and its TC even after applying the conventional one-point trimming. We also propose a IDAC for auto-trimming, demonstrated in a small drift in both V_{REF} and its TC after one-point calibration. Section 4.2 presents the principle of the proposed auto-calibration technique. Section 4.3 highlights the circuit implementation and analysis of the proposed BGR. Section 4.4 exhibits the measurement results. Finally, section 4.5 draws the conclusions.

4.2 THE PRINCIPLE OF AUTO-CALIBRATION

4.2.1 PROPOSED TRIMMING CONCEPT

From the conventional current-based BGR as shown in Figure 4-1(a), we first assume that there is no offset in the amplifier (A_1 and A_2) and no mismatch in the current mirror $M_{P1} - M_{P5}$ for simplicity. With $V_X = V_Y = V_{EB1}$ and considering the reverse Early Effect [4.19], the current flowing through resistor R_2 and R_1 (i.e., I_{PTAT} and I_{CTAT} respectively) can be expressed as,

$$I_{PTAT} = \frac{\Delta V_{EB}}{R_2} = \frac{V_{EB1} - V_{EB2}}{R_2} = \frac{\eta_F V_T}{R_2} \ln(N) \quad (4-1)$$

$$I_{CTAT} = \frac{V_{EB1}}{R_1} \quad (4-2)$$

where $V_T = kT/q$ is the thermal voltage, η_F is the BJT forward emission coefficient, k is the Boltzmann constant, T is the absolute temperature in Kelvin, q is the electron charge, and N is the area ratio of BJTs Q_1 and Q_2 . As Q_1 and Q_2 are biased at the region where β_F is not sensitive to I_C , the error caused by the difference between β_{F1} (β_F of Q_1) and β_{F2} (β_F of Q_2) in (4-1) is negligible. I_{PTAT} and I_{CTAT} are copied by the current mirror to bias R_{REF} to produce the reference voltage V_{REF} , which can be elaborated as,

$$\begin{aligned} V_{REF} &= R_{REF}(I_{CTAT} + I_{PTAT}) = \alpha V_{EB1} + \gamma \Delta V_{EB} \\ &= \frac{R_{REF}}{R_1} V_{EB1} + \frac{R_{REF}}{R_2} \eta_F V_T \ln(N) \end{aligned} \quad (4-3)$$

Assume $R_{REF} = R_1$, (4-3) becomes,

$$V_{REF} = V_{EB1} + \gamma \Delta V_{EB} = V_{EB1} + \frac{R_{REF}}{R_2} \eta_F V_T \ln(N) \quad (4-4)$$

By adjusting the PTAT voltage coefficient $\gamma = R_{REF}/R_2$ and the area ratio N to eliminate

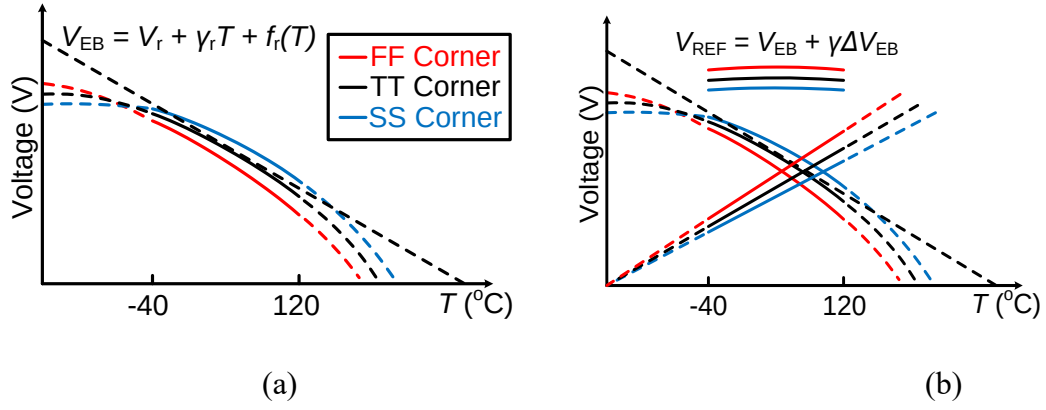


Figure 4-2(a) Temperature curve of V_{EB} , and (b) temperature curve of V_{EB} and corresponding $\gamma \Delta V_{EB}$ under different process corners.

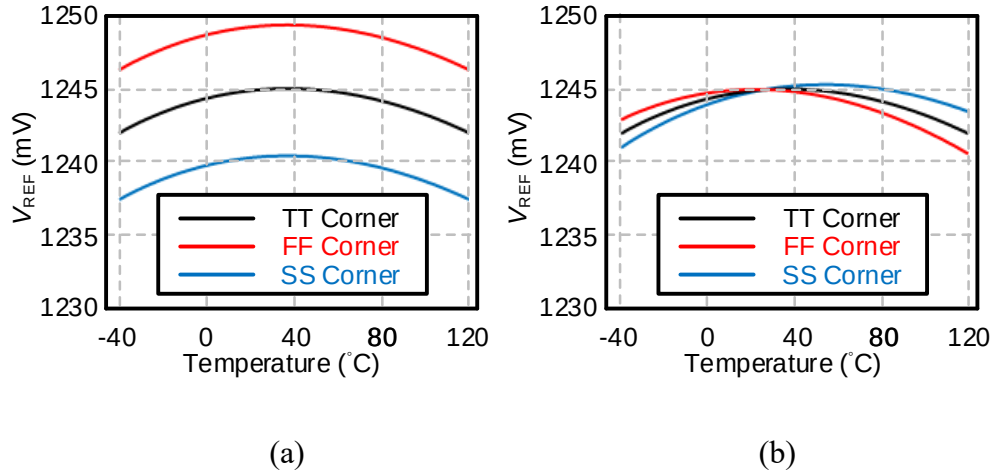


Figure 4-3 Simulated temperature curve of V_{REF} under different process corners

in 65nm standard CMOS process by controlling $\gamma \Delta V_{EB}$ to (a) adjust TC and (b) adjust V_{REF} value.

the first-order component of the CTAT voltage V_{EB1} , the reference voltage V_{REF} can be generated as shown in Figure 4-1(b). Due to process variation, there is a voltage drift in V_{EB} under different process corners, producing a significant error in V_{REF} as shown in Figure 4-2(a). Since ΔV_{EB} is usually considered as a process insensitive parameter, we may adjust γ in (4-4) to perform PTAT trimming by adjusting R_2 in Figure 4-1(a)

in order to eliminate the voltage drift in V_{EB} , as depicted in Figure 4-2(b). Yet, conventional PTAT trimming only eliminates the PTAT voltage drift induced by process variation in V_{EB} but not the non-PTAT part. This results in a residual error in V_{REF} which will be discussed in (4-17). Figure 4-3 shows the simulation result of V_{REF} under different process corners by controlling $\gamma\Delta V_{EB}$, with subfigures (a) and (b) presenting the trimmed V_{REF} with V_{REF} adjusted and its TC adjusted, respectively. As observed, due to the non-PTAT voltage drift in V_{EB} , we can only calibrate either the V_{REF} or its TC by controlling $\gamma\Delta V_{EB}$ but not both using the conventional 1-point trimming methods [4.3]-[4.6] and [4.9]-[4.11]. To better illustrate this issue, we should consider the effect of process variation on V_{EB} and ΔV_{EB} of BJT, which leads to the drift in V_{REF} and its TC. Considering the reverse Early Effect [4.19], the V_{EB} of a BJT can be expressed as,

$$V_{EB} = \eta_F V_T \ln \left(\frac{I_C}{I_S} \right) \quad (4-5)$$

where I_C is the collector current and I_S is the saturation current. As the emitter current I_E of a PNP BJT is set by a current source, the base current should also be considered when determining the resulting V_{EB} [4.19], as defined by

$$V_{EB} = \eta_F V_T \left[\ln(I_E) + \ln \left(\frac{\beta_F}{1+\beta_F} \right) - \ln(I_S) \right] \quad (4-6)$$

where β_F is the BJT's common-emitter current gain. Here, I_S , η_F , and β_F dominate the drift in V_{EB} under process variation. As they all depend on the doping concentration, which increases in the FF corner and vice versa in the SS corner [4.19]. Following the BSIM4 model [4.20], they can be expressed as,

$$I_S = I_{S0}(1 + k_{IS}\sigma) \exp \left[\frac{E_G(T_r)}{V_T(T_r)} - \frac{E_G(T)}{V_T(T)} + X_{ti} \ln \left(\frac{T}{T_r} \right) \right] \quad (4-7)$$

$$\eta_F = \eta_{F0}(1 + k_{\eta F}\sigma)[1 + TC_{\eta F1}(T - T_r)] \quad (4-8)$$

$$\beta_F = \beta_{F0}(1 + k_{\beta F}\sigma)[1 + TC_{\beta F1}(T - T_r)] \quad (4-9)$$

where I_{S0} , η_{F0} , and β_{F0} are the corresponding nominal values, respectively. k_{IS} , $k_{\eta F}$, and $k_{\beta F}$ are the process corner variation coefficient of I_S , η_F , and β_F , respectively. σ is the process-corner-dependent variable. T_r is the reference temperature. $E_G(T)$ and $E_G(T_r)$ are the bandgap energies at T and T_r . $V_T(T)$ and $V_T(T_r)$ are the thermal voltage at T and T_r . X_{ti} is the temperature exponent of I_S . $TC_{\eta F1}$ and $TC_{\beta F1}$ are the first-order TCs of η_F and β_F , respectively. Generally, $TC_{\eta F1}$ is very small as the reverse Early voltage is much larger than the thermal voltage [4.21], and we can assume that η_F is temperature independent and simply neglect its TC. However, η_F is process dependent, so we should consider the corresponding process drift. The voltage drift of V_{REF} due to the process corner variation can be expressed as,

$$\Delta V_{REF} = \frac{\partial V_{REF}}{\partial \sigma} \Big|_{TT} \cdot \Delta \sigma \quad (4-10)$$

where $\Delta \sigma$ is the drift of σ from TT corner (i.e., $\Delta \sigma=0$ at TT corner). Considering (4-4)

and (4-6), $\partial V_{REF}/\partial \sigma$ can be expressed as,

$$\frac{\partial V_{REF}}{\partial \sigma} = \frac{\partial V_{EB}}{\partial \sigma} + \frac{\partial \gamma \Delta V_{EB}}{\partial \sigma} = \frac{\partial V_{EB}}{\partial I_S} \frac{\partial I_S}{\partial \sigma} + \frac{\partial V_{EB}}{\partial \eta_F} \frac{\partial \eta_F}{\partial \sigma} + \frac{\partial V_{EB}}{\partial \beta_F} \frac{\partial \beta_F}{\partial \sigma} + \frac{\partial \gamma \Delta V_{EB}}{\partial \eta_F} \frac{\partial \eta_F}{\partial \sigma} \quad (4-11)$$

Using (4-2), and (4-6)-(4-9), we have,

$$\frac{\partial V_{EB}}{\partial I_S} \frac{\partial I_S}{\partial \sigma} = -\eta_F V_T \frac{k_{IS}}{1+k_{IS}\Delta\sigma} \quad (4-12)$$

$$\frac{\partial V_{EB}}{\partial \eta_F} \frac{\partial \eta_F}{\partial \sigma} = V_{EB} \frac{k_{\eta F}}{1+k_{\eta F}\Delta\sigma} \quad (4-13)$$

$$\frac{\partial V_{EB}}{\partial \beta_F} \frac{\partial \beta_F}{\partial \sigma} = \frac{\eta_F V_T}{1 + \beta_F} \frac{k_{\beta_F}}{1 + k_{\beta_F} \Delta \sigma} \quad (4-14)$$

$$\frac{\partial \gamma \Delta V_{EB}}{\partial \eta_F} \frac{\partial \eta_F}{\partial \sigma} = \gamma \Delta V_{EB} \frac{k_{\eta_F}}{1 + k_{\eta_F} \Delta \sigma} \quad (4-15)$$

With $\Delta \sigma = 0$ at TT corner, by substituting (4-12)-(4-15) into (4-11), we have,

$$\left. \frac{\partial V_{REF}}{\partial \sigma} \right|_{TT} = -\eta_F V_T k_{IS} + V_{EB} k_{\eta_F} + \frac{\eta_F V_T}{1 + \beta_F} k_{\beta_F} + \gamma \Delta V_{EB} k_{\eta_F} \quad (4-16)$$

Using (4-10) and (4-16), the voltage drift of V_{REF} due to the process corner variation can be expressed as,

$$\Delta V_{REF} = \underbrace{(-\eta_F V_T k_{IS})}_{\text{① PTAT Error}} + \underbrace{V_{EB} k_{\eta_F}}_{\text{② CTAT Error}} + \underbrace{\frac{\eta_F V_T}{1 + \beta_F} k_{\beta_F}}_{\text{③ Quasi-PTAT Error}} + \underbrace{\gamma \Delta V_{EB} k_{\eta_F}}_{\text{④ PTAT Error}} \Delta \sigma \quad (4-17)$$

The first term is the I_S -induced V_{EB} drift, which is a PTAT error as its temperature dependence is dominated by V_T . The second term is the η_F -induced V_{EB} drift, which is a CTAT error as a result of V_{EB} . The third term is the β_F -induced V_{EB} drift, which is a quasi-PTAT error due to the dominance of V_T over β_F (i.e., β_F is typically small in the advanced process). Finally, the fourth term is the η_F -induced $\gamma \Delta V_{EB}$ drift, which is a PTAT error owing to ΔV_{EB} . To intuitively understand the effect of each individual term, Figure 4-4 shows the simulated voltage drift breakdown of V_{REF} due to the process corner variation in a standard 65nm CMOS process, with the biasing current I_E set to 100nA and the BJT's emitter area $A_E = 10 \times 10 \mu\text{m}^2$. As observed, the four error terms in (4-17) generate PTAT, CTAT, quasi-PTAT, and PTAT errors, respectively, and the total error would result in the voltage drift of V_{REF} in FF corner. A similar result can be obtained for the SS process corner, and Figure 4-4 only illustrates the FF corner result for simplicity. As discussed in Section 4.2.1, the conventional approach to eliminate the errors in (4-17) is to adjust the PTAT coefficient γ in (4-4) to perform PTAT trimming (i.e., to adjust R_2 for controlling the resistor ratio $\gamma = R_{REF}/R_2$). However, even

though

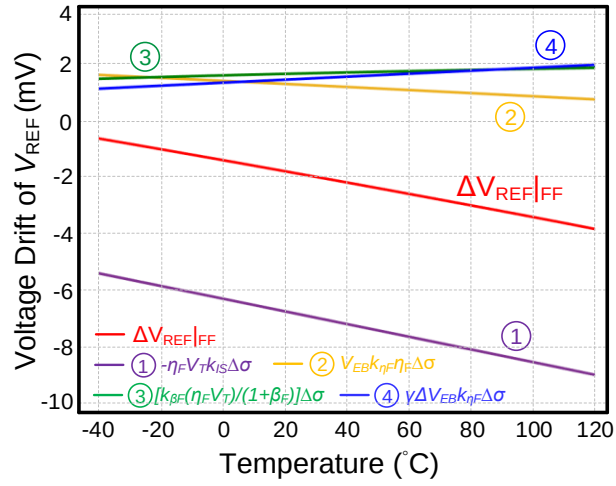


Figure 4-4 Simulated curve of the voltage drift of V_{REF} at FF corner in 65nm standard CMOS process, ① error induced by I_S variation ($-\eta_F V_T k_{IS} \Delta\sigma$), ② and ④ errors induced by η_F variation ($V_{EB} k_{\eta_F} \Delta\sigma$ and $\gamma \Delta V_{EB} k_{\eta_F} \Delta\sigma$), and ③ error induced by β_F variation ($[k_{\beta_F} (\eta_F V_T) / (1 + \beta_F)] \Delta\sigma$).

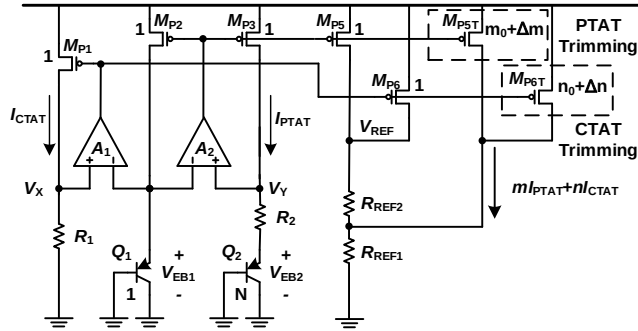


Figure 4-5 Circuit configuration of a conventional current-based BGR with adjustable PTAT and CTAT current mirror.

this approach can relax the PTAT error, the CTAT error still remains, and hence we cannot trim both the error of V_{REF} and its TC by using the conventional PTAT trimming.

To solve this problem, this work proposes a one-point trimming approach in order to adjust both the V_{REF} and its TC error simultaneously. Unlike the conventional PTAT trimming, we propose to adjust both the CTAT and PTAT voltage to eliminate the

corresponding error in (4-17). Figure 4-5 shows the circuit configuration of a conventional current-based BGR with adjustable PTAT and CTAT current mirrors. In this case, V_{REF} can be expressed as,

$$V_{REF} = \left[\frac{R_{REF2} + R_{REF1}}{R_1} + \frac{R_{REF1}}{R_1} (n_0 + \Delta n) \right] V_{EB1} + \left[\gamma + \frac{R_{REF1}}{R_2} (m_0 + \Delta m) \right] \Delta V_{EB} \quad (4-18)$$

$$\Delta V_{REF} = \Delta n \frac{R_{REF1}}{R_1} V_{EB1} + \Delta m \frac{R_{REF1}}{R_2} \Delta V_{EB} \quad (4-19)$$

where $\gamma = (R_{REF2} + R_{REF1})/R_2$ and $R_1 = R_{REF2} + R_{REF1}$. n_0 and m_0 are the nominal values. Here, we can adjust the PTAT and CTAT coefficients Δm and Δn to eliminate the PTAT and CTAT errors in (4-17), respectively. Noticed that the trimming current is injected into the midpoint of the resistors R_{REF1} and R_{REF2} to reduce the transfer ratio of current mirror. Due to the PTAT nature of the first error term caused by I_S in (4-17), we should adjust Δm in (4-19). Hence, we can evaluate the required Δm for cancelling the effect from I_S (i.e., m_{IS}) as,

$$\Delta m_{IS} \frac{R_{REF1}}{R_2} \Delta V_{EB} = -\eta_F V_T k_{IS} \Delta \sigma \Rightarrow \Delta m_{IS} = -\frac{\eta_F V_T k_{IS} \Delta \sigma}{\frac{R_{REF1}}{R_2} \Delta V_{EB}} \quad (4-20)$$

Similarly, we can find the corresponding Δm to eliminate the third and fourth error terms as contributed by β_F (quasi-PTAT in advanced process) and η_F (PTAT), respectively, by ensuring

$$\Delta m_{\beta_F} \frac{R_{REF1}}{R_2} \Delta V_{EB} = \frac{\eta_F V_T}{1 + \beta_F} k_{\beta_F} \Delta \sigma \Rightarrow \Delta m_{\beta_F} = \frac{\eta_F V_T k_{\beta_F} \Delta \sigma}{(1 + \beta_F) \frac{R_{REF1}}{R_2} \Delta V_{EB}} \quad (4-21)$$

$$\Delta m_{\eta_F} \frac{R_{REF1}}{R_2} \Delta V_{EB} = \gamma \Delta V_{EB} k_{\eta_F} \Delta \sigma \Rightarrow \Delta m_{\eta_F} = \frac{R_2}{R_{REF1}} \gamma k_{\eta_F} \Delta \sigma \quad (4-22)$$

Moreover, to eliminate the effect of the second error term in (4-17) caused by η_F ,

we should adjust Δn in (4-19) as it is a CTAT error (with $V_{EB} = V_{EB1}$) and obtain

$$\Delta n_{\eta_F} \frac{R_{REF1}}{R_1} V_{EB} = V_{EB} k_{\eta_F} \Delta \sigma \Rightarrow \Delta n_{\eta_F} = \frac{R_1}{R_{REF1}} k_{\eta_F} \Delta \sigma \quad (4-23)$$

From (4-20)-(4-23), the trimming ratio of the PTAT and CTAT coefficient (i.e., $\Delta m/\Delta n$) can be expressed as,

$$\frac{\Delta m}{\Delta n} = \frac{\Delta m_{IS} + \Delta m_{\eta_F} + \Delta m_{\beta_F}}{\Delta n_{\eta_F}} = \frac{1}{\frac{R_1}{R_2} \ln(N) k_{\eta_F}} \left[-k_{IS} + \frac{k_{\beta_F}}{(1+\beta_F)} \right] + \frac{R_2}{R_1} \gamma \quad (4-24)$$

With $\gamma = (R_{REF2} + R_{REF1})/R_2$ and $R_1 = R_{REF2} + R_{REF1}$, (4-24) becomes,

$$\frac{\Delta m}{\Delta n} = \frac{\Delta m_{IS} + \Delta m_{\eta_F} + \Delta m_{\beta_F}}{\Delta n_{\eta_F}} = \frac{1}{\gamma \ln(N) k_{\eta_F}} \left[-k_{IS} + \frac{k_{\beta_F}}{(1+\beta_F)} \right] + 1 \quad (4-25)$$

As the trimming ratio $\Delta m/\Delta n$ is ratiometric, it is process corner independent (i.e., it does not depend on σ). However, it is sensitive to the adopted CMOS technology (i.e., $\Delta m/\Delta n$ is different for distinct CMOS processes). Furthermore, $\Delta m/\Delta n$ depends on the resistor ratio γ , the BJT area ratio N , the BJT current gain β_F , and the process corner variation coefficients k_{IS} , k_{η_F} , and k_{β_F} . As γ and N depend on device matching while k_{IS} , k_{η_F} , and k_{β_F} are process-specific empirically characterized parameters, they can be considered insensitive to process and temperature. However, as β_F is sensitive to temperature, it still leads to a residual TC after using the proposed trimming inevitably limiting the achievable trimmed accuracy.

Based on (4-25), we can calculate the corresponding trimming ratio in different CMOS processes. For example, in the 65nm standard CMOS process employed, we can first extract the corresponding k_{IS} , k_{η_F} , and k_{β_F} in the model file. Moreover, with the biasing current $I_E = 100\text{nA}$ and the emitter area $A_E = 10 \times 10 \mu\text{m}^2$, we have $\beta_F \sim 0.7$

and $\gamma \ln(N) = V_{EB}/V_T$. We can then obtain the trimming ratio $\Delta m/\Delta n = -2$. Figure 4-6 shows

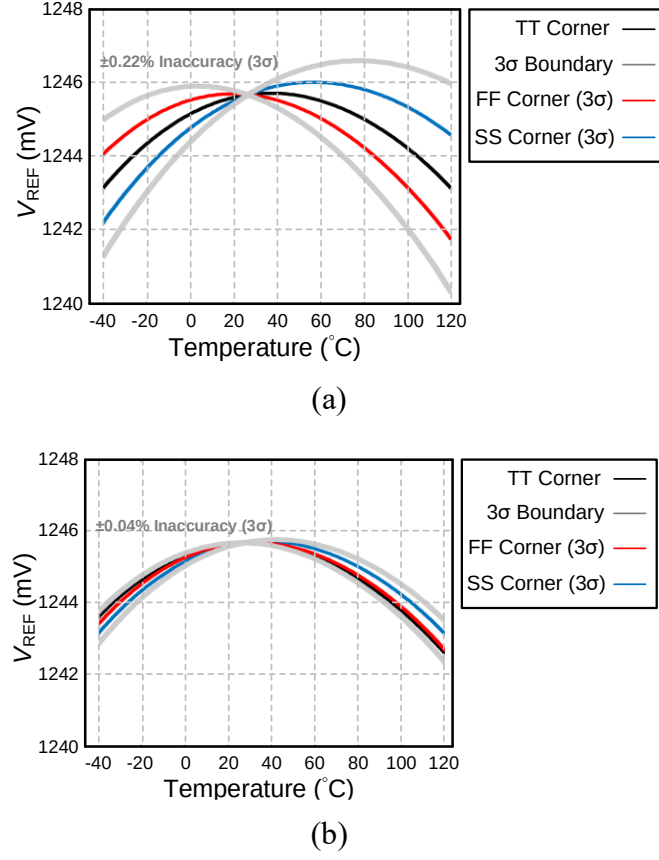
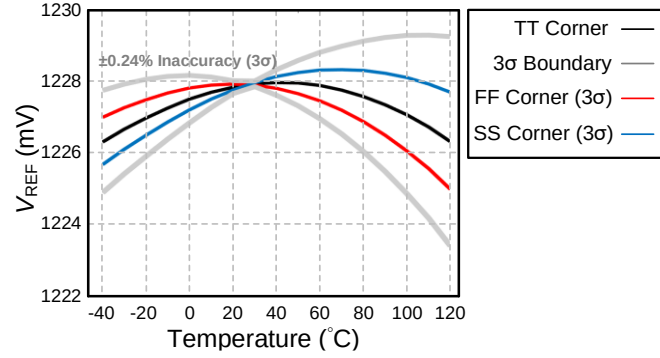


Figure 4-6 Simulated curve of V_{REF} (a) with conventional PTAT trimming, and (b) with proposed trimming ($\Delta m/\Delta n = -2$) in 65nm standard CMOS process with biasing current $I_E = 100\text{nA}$ and emitter area $A_E = 10 \times 10\mu\text{m}^2$.

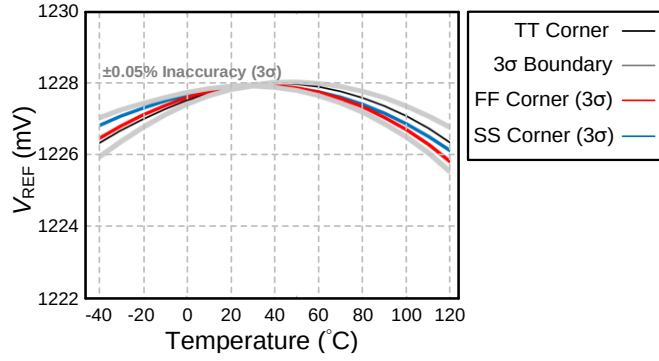
the corresponding simulated V_{REF} .

By applying the conventional PTAT trimming as shown in Figure 4-6(a), due to the η_F -induced CTAT error, the TC is not well adjusted and the 3 σ inaccuracy of V_{REF} is $\pm 0.22\%$ within the whole temperature spectrum. In contrast, by using the proposed trimming approach with the ratio $\Delta m/\Delta n = -2$ and adjusting V_{REF} at 27 $^{\circ}\text{C}$ as shown in Figure 4-6(b), we can greatly reduce both the drift in V_{REF} and its TC, with the 3 σ

inaccuracy decreases from $\pm 0.22\%$ to $\pm 0.04\%$. As expected, there exists a residual TC which is mainly induced by β_F in (4-25).



(a)

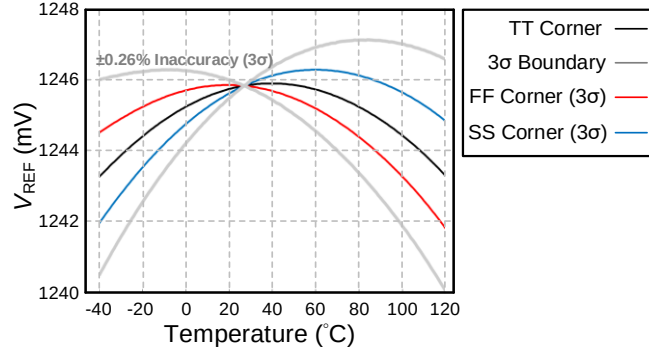


(b)

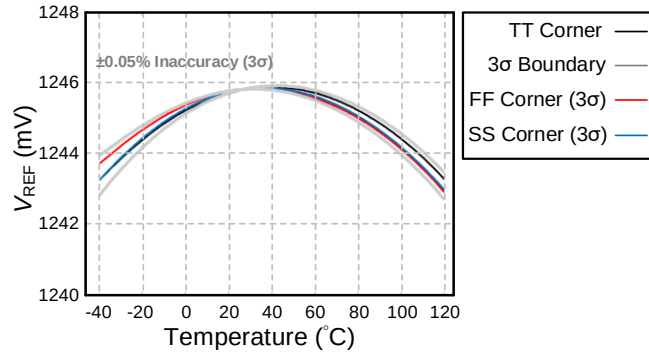
Figure 4-7 Simulated curve of V_{REF} (a) with conventional PTAT trimming, and (b) with proposed trimming ($\Delta m/\Delta n = -2.5$) in 28nm standard CMOS process with biasing current $I_E = 100\text{nA}$ and emitter area $A_E = 10 \times 10\mu\text{m}^2$.

To demonstrate the universality of our proposed auto-calibration technique, we performed simulations in different CMOS processes. Using a 28nm standard CMOS process with the same settings for I_E and A_E , we can obtain the trimming ratio of $\Delta m/\Delta n = -2.5$. Figure 4-7 shows the simulated V_{REF} in a 28nm standard CMOS process. By applying the conventional PTAT trimming as shown in Figure 4-7(a), it only achieves a 3σ inaccuracy of $\pm 0.24\%$ over the targeted temperature range. In contrast, the 3σ

inaccuracy can be suppressed from $\pm 0.24\%$ to $\pm 0.05\%$ by using the proposed trimming method with the ratio $\Delta m/\Delta n = -2.5$ as shown in Figure 4-7(b).



(a)



(b)

Figure 4-8 Simulated curve of V_{REF} (a) with conventional PTAT trimming, and (b) with proposed trimming ($\Delta m/\Delta n = -2.1$) in a 180nm standard CMOS process with biasing current $I_E = 100\text{nA}$ and emitter area $A_E = 10 \times 10\mu\text{m}^2$.

Similarly, in a 180nm standard CMOS process, with the same settings for I_E and A_E , we can obtain the trimming ratio of $\Delta m/\Delta n = -2.1$. Figure 4-8 shows the simulated V_{REF} in a 180nm standard CMOS process. By applying the conventional PTAT trimming as shown in Figure 4-8(a), it only achieves a 3σ inaccuracy of $\pm 0.26\%$ over the targeted temperature range. In contrast, the 3σ inaccuracy can be suppressed from $\pm 0.26\%$ to $\pm 0.05\%$ by using the proposed trimming method with the ratio $\Delta m/\Delta n = -$

2.1 as shown in Figure 4-8(b).

4.2.2 PROPOSED AUTO-CALIBRATION TECHNIQUE

Figure 4-9 displays the implementation of the proposed auto-calibration technique using the I_{PTAT} and I_{CTAT} as generated by the conventional current-based BGR in Figure 4-5. Since $V_{CAL} = (\eta F k T / q) \cdot \ln(N) \cdot (R_{CAL} / R_2)$ (refer to Figure 4-9) is generated by employing I_{PTAT} (refer to Figure 4-5) to bias R_{CAL} , it mainly depends on the physical constants k and q , and the resistor ratio R_{CAL} / R_2 . Considering both mismatch and process variation, V_{CAL} shows a 1σ inaccuracy of $\pm 0.11\%$ from simulation. This error is directly reflected in V_{REF} after applying the proposed auto-calibration. We use a chopped comparator (CMP) to compare V_{REF} and V_{CAL} at room temperature (27°C), and produce the output V_{CMP} to activate the calibration engine. If the temperature drifts from 27°C , then it might result in about 0.5% error in V_{REF} per one degree drift.

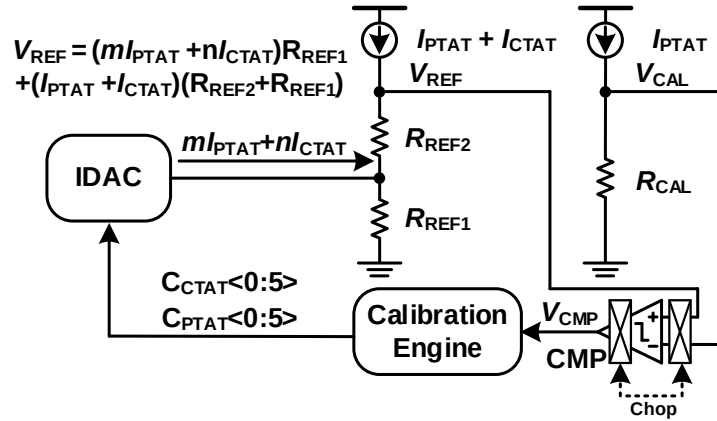


Figure 4-9 Circuit diagram of the proposed auto-calibration technique.

The calibration engine generates two sets of digital signals ($C_{CTAT}<0:5>$ and $C_{PTAT}<0:5>$) to control the IDAC and produces the output current $m \cdot I_{PTAT} + n \cdot I_{CTAT}$ to bias R_{REF1} , so that V_{REF} tracks V_{CAL} iteratively. The resolution of the PTAT and CTAT

trimming currents at 27°C are 1.18nA and 0.56nA, respectively.

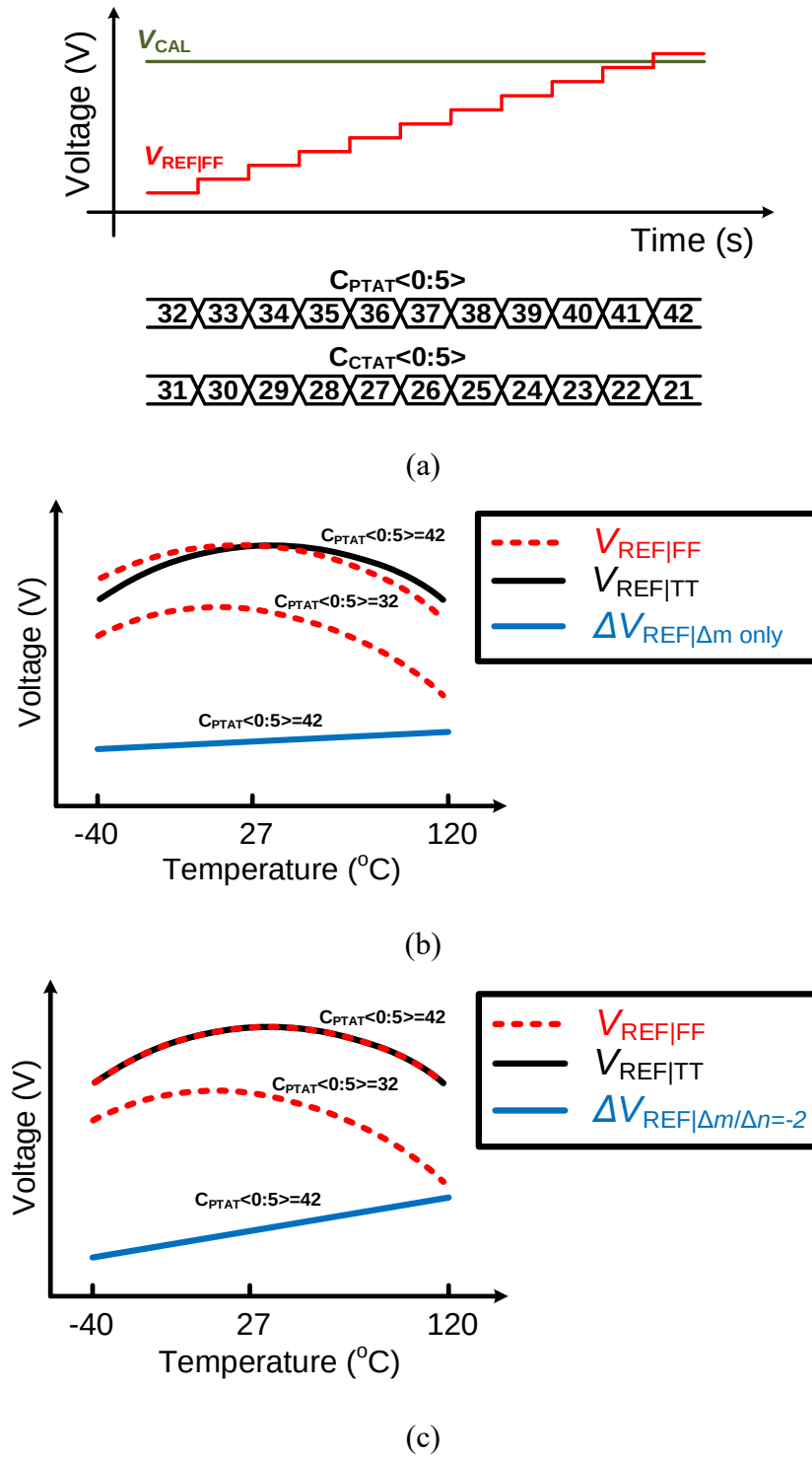


Figure 4-10 Conceptual waveform of V_{REF} at FF corner (a) at the trimming temperature, (b) within the entire temperature range by changing only Δm (conventional PTAT trimming), and (c) within the entire temperature range with

$$\Delta m/\Delta n=-2.$$

Figure 4-10 shows the conceptual waveform of V_{REF} at FF corner. Figure 4-10(a) shows the conceptual waveform of V_{REF} at the FF corner at the trimming temperature (i.e., 27°C). Here, we simultaneously adjust I_{PTAT} and I_{CTAT} to save trimming time and resolution. Firstly, we set $C_{\text{PTAT}}\langle 0:5 \rangle = 32$ and $C_{\text{CTAT}}\langle 0:5 \rangle = 31$ to initialize m_0 and n_0 (i.e., $\Delta m = \Delta n = 0$). When $V_{\text{REF|FF}} < V_{\text{CAL}}$, $C_{\text{PTAT}}\langle 0:5 \rangle$ increases by one and $C_{\text{CTAT}}\langle 0:5 \rangle$ decreases by one in each comparison cycle, i.e., Δm increases while Δn decreases. As a result, $V_{\text{REF|FF}}$ increases to follow V_{CAL} , and the result converges after multiple comparison cycles. Figure 4-10(b) shows the conceptual waveform of V_{REF} at the FF corner within the entire temperature range by changing only Δm (i.e., the conventional PTAT trimming), demonstrated that only V_{REF} or its TC can be adjusted due to the CTAT error in (4-17). In contrast, by using the proposed trimming ratio $\Delta m/\Delta n = -2$ to adjust the slope of ΔV_{REF} according to (4-19) and adding it to the $V_{\text{REF|FF}}$, we can adjust both the V_{REF} and its TC as shown in Figure 4-10(c). The calibration engine operates complementarily when the reference voltage is larger than V_{CAL} .

4.3 CIRCUIT IMPLEMENTATION

Figure 4-11 presents the circuit configuration of the current-based BGR with the proposed auto-calibration technique. The left part is the conventional current-based BGR. The two-stage chopped amplifiers A_1 and A_2 ensure $V_X = V_Y = V_{\text{EB1}}$, and generate $I_{\text{CTAT}} = V_{\text{EB1}}/R_1$ and $I_{\text{PTAT}} = (V_{\text{EB1}} - V_{\text{EB2}})/R_2 = (\eta_F kT/q) \cdot \ln(N)/R_2$. Here, the area ratio of Q_2 and Q_1 is set to $N = 8$. BJTs with a unit A_E of $10 \times 10 \mu\text{m}^2$ are employed to minimize

Table 4-1 Error Source Breakdown in V_{REF} from Monte-Carlo Simulation

Error Source	Typical Value (σ)	Error contribution (σ/μ)
Offset of Chopped Amplifier	$\pm 0.21\text{mV}$	$\pm 0.02\%$
Mismatch of Current Mirror	$\pm 0.49\text{mV}$	$\pm 0.05\%$
Process Variation of I_S	$\pm 2.08\text{mV}$	$\pm 0.21\%$
Process Variation of η_F	$\pm 0.82\text{mV}$	$\pm 0.08\%$
Process Variation of β_F	$\pm 0.58\text{mV}$	$\pm 0.06\%$

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Table 4-1 summarize the main error sources and their contributions to the total error in V_{REF} . Monte Carlo simulations were completed using foundry-supplied statistical device models with the sampling method of Random. It shows that the process variation of I_S , η_F , and β_F dominates the voltage drift in V_{REF} . If a higher accuracy is desired, dynamic matching for current mirror and auto-zeroing for the amplifier can be employed to further suppress mismatch-induced errors. The I_{PTAT} (copied by M_{P4} to bias R_{CAL}) allows the determination of $V_{\text{CAL}} = (\eta_F kT/q) \cdot \ln(N) \cdot (R_{\text{CAL}}/R_2)$, which depends on the physical constants and resistor ratio. Notice that V_{CAL} still suffers from the process variation of η_F . Furthermore, an external enable signal (EN) activates the proposed auto-calibration only when necessary to save power. M_{P5} and M_{P6} copy I_{CTAT} and I_{PTAT} , with the total current biasing R_{REF1} and R_{REF2} . The IDAC injects the trimming current $m \cdot I_{\text{PTAT}}$ and $n \cdot I_{\text{CTAT}}$ through M_{P10} to bias R_{REF1} . Consequently, we can generate a $V_{\text{REF}} = (mI_{\text{PTAT}} + nI_{\text{CTAT}}) \cdot R_{\text{REF1}} + (I_{\text{PTAT}} + I_{\text{CTAT}}) \cdot (R_{\text{REF2}} + R_{\text{REF1}})$. By comparing V_{REF} and V_{CAL} using the chopped comparator CMP [4.22], V_{CMP} can be generated to activate the calibration engine, and the accuracy of the calibration loop is mainly determined by the offset voltage of CMP.

Figure 4-12 shows the block diagram and the key waveforms of the calibration engine. EN activates the oscillator that generates three clock signals (ϕ_{CMP} , ϕ_{T} , ϕ_{P}). When ϕ_{CMP} is high, the register updates V_{DIR} from V_{CMP} . The assertion of ϕ_{P} judges whether V_{DIR} is high or low to control the two accumulators into either plus or minus mode. The update of $C_{\text{CTAT}}\langle 0:5 \rangle$ and $C_{\text{PTAT}}\langle 0:5 \rangle$ happens when ϕ_{T} is high.

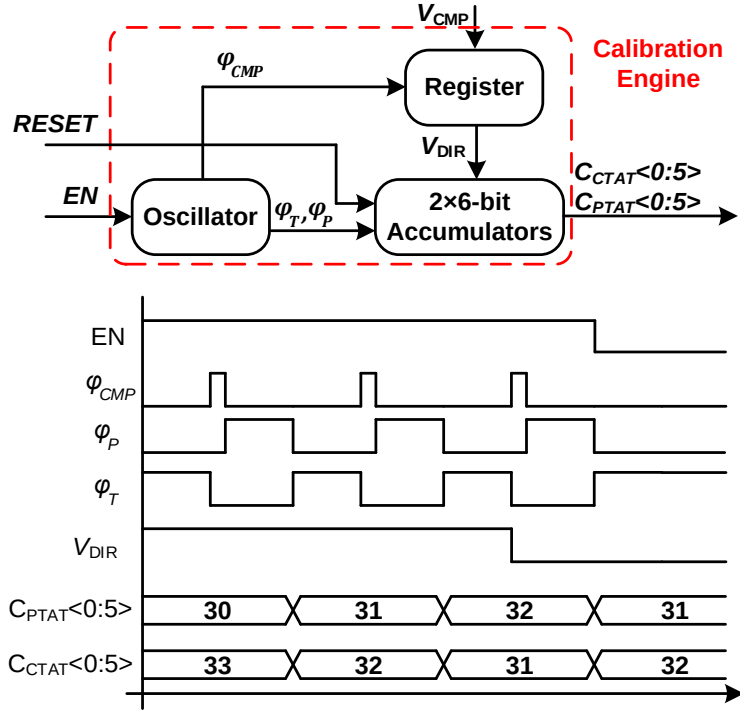


Figure 4-12 Block diagram and key operational waveforms of the calibration engine.

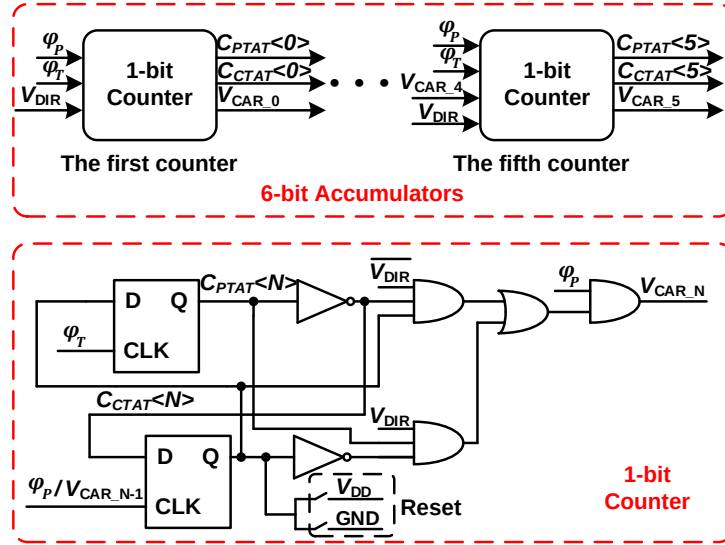


Figure 4-13 Circuit configuration of the 6-bit accumulators.

If the accumulators are in plus mode, $C_{PTAT}<0:5>$ increases by one and $C_{CTAT}<0:5>$ reduces by one, and vice versa for minus mode. Moreover, an external signal (RESET) can initialize $C_{PTAT}<0:5>$ to 32 and $C_{CTAT}<0:5>$ to 31 such that half of the trimming

bits are active. Figure 4-13 shows the circuit configuration of the customized 6-bit digital accumulator to reduce both the power and area consumption.

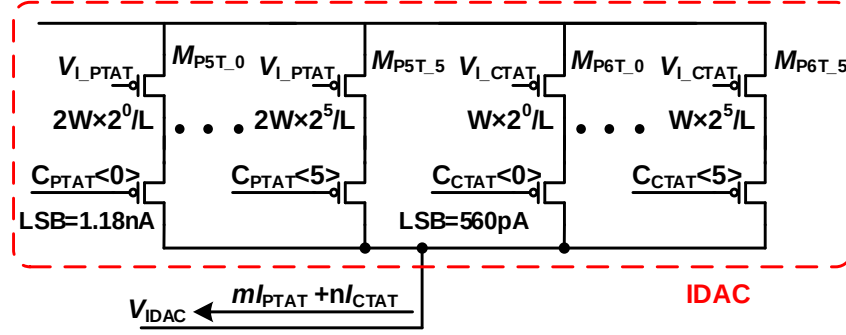


Figure 4-14 Circuit configuration of the current digital-to-analog converter (IDAC).

Figure 4-14 shows the circuit configuration of the proposed IDAC. To achieve a 0.25% inaccuracy from -40°C to 120°C , the LSB of the trimming voltage should be $\sim 0.5\text{mV}$ [4.4]. Using the 65nm standard CMOS process as introduced in Section 4.2.1, we should employ a trimming ratio $\Delta m/\Delta n = -2$. In TT corner at 27°C , we set the unit current for I_{PTAT} as 590pA and the unit current for I_{CTAT} as 560pA. With reference to the $2\times$ relationship between Δm and Δn , we proposed an IDAC where the transistor widths of $M_{\text{P5T}_0} - M_{\text{P5T}_5}$ (current mirror for copying I_{PTAT}) are two times larger than that of $M_{\text{P6T}_0} - M_{\text{P6T}_5}$ (current mirror for copying I_{CTAT}) as shown in Figure 4-14. Due to mismatch between PMOS devices, the actual trimming ratio $\Delta m/\Delta n$ typically varies from -2.015 to -1.975 according to simulation results. This introduces an additional TC of $\sim 0.3\text{ppm}/^{\circ}\text{C}$ in V_{REF} . By using the proposed IDAC, when $V_{\text{REF}} < V_{\text{CAL}}$ (i.e., in FF corner), $C_{\text{PTAT}}\langle 0:5 \rangle$ increases by one and $C_{\text{PTAT}}\langle 0:5 \rangle$ decreases by one. This activates two units of I_{PTAT} while disable a unit of I_{CTAT} . Similarly, when $V_{\text{REF}} >$

V_{CAL} (i.e., in SS corner), $C_{\text{PTAT}}\langle 0:5 \rangle$ decreases by one and $C_{\text{PTAT}}\langle 0:5 \rangle$ increases by one. This disables two units of I_{PTAT} while activating a unit of I_{CTAT} . In this case, we can adjust both the drift in V_{REF} and its TC.

4.4 MEASUREMENT RESULTS

Figure 4-15 presents the proposed design as fabricated in 65nm standard CMOS, occupying a chip area of 0.04 mm^2 .

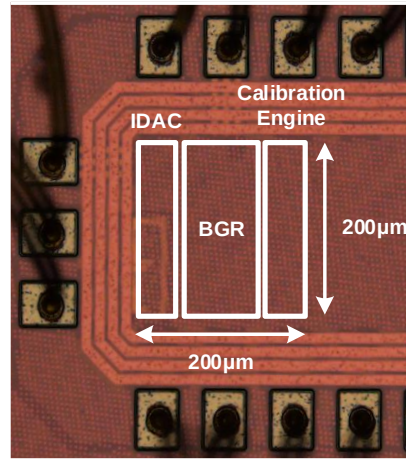
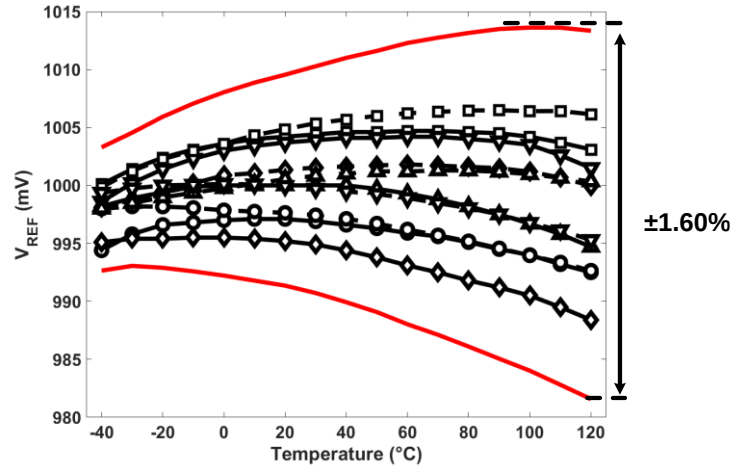


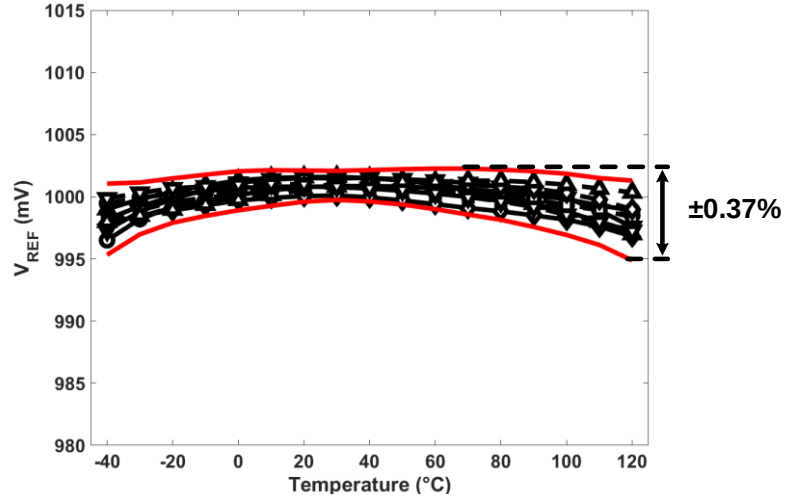
Figure 4-15 The die photo of the proposed BGR.

The BGR contributes to half of the area, while the calibration loop and IDAC occupied the remaining half. Figure 4-16 plots the measured V_{REF} of 10 chips at a supply voltage $V_{\text{DD}} = 1.2\text{V}$ from -40°C to 120°C . From Figure 4-16(a), the measured V_{REF} for 10 chips varies from 988.4mV to 1006.5mV before trimming, with a 1σ inaccuracy of $\pm 0.53\%$. Figure 4-16(b) shows the measured V_{REF} for 10 chips after applying the proposed auto-calibration technique. It varies from 996.5mV to 1002.2mV, with the 1σ inaccuracy improved to $\pm 0.12\%$. Considering both the voltage drift of V_{CAL} and the quantization error of the trimming circuit, $V_{\text{REF}|27^\circ\text{C}}$ varies at most

from 999.8mV to 1003.0mV ($\pm 1.6\text{mV}$).



(a)



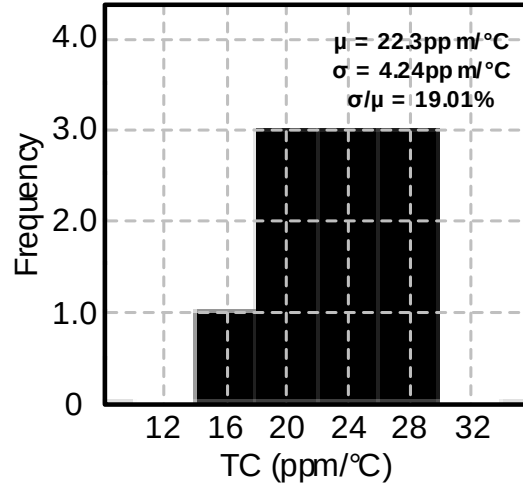
(b)

Figure 4-16 Measured (a) untrimmed BGR V_{REF} , and (b) trimmed BGR V_{REF}

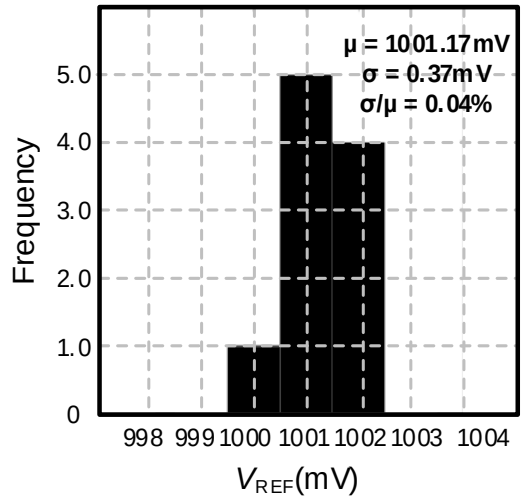
from -40 to 120°C at $V_{DD}=1.2\text{V}$.

In this case, instead of an intersection point, $V_{REF|27^{\circ}\text{C}}$ falls within a range (from 1000.5mV to 1001.7mV or $\pm 0.6\text{mV}$) as shown in Figure 4-16(b) after applying the proposed calibration method. Compare with the simulation result as shown in Fig. Figure 4-6, the obtained performance is worse mainly due to the nonideal effect such as mismatch. Figure 4-17(a) shows the measured TC variation for 10 chips, ranging from 18.6ppm/ $^{\circ}\text{C}$ to 28.8ppm/ $^{\circ}\text{C}$. This corresponds to an average TC of 22.3ppm/ $^{\circ}\text{C}$

with $\sigma/\mu = 19.01\%$. If a much smaller TC is desired, the curvature correction technique in [4.23] can be applied to the proposed auto-calibration BGR. Besides, two-point or even three-point auto-calibration should be developed and applied as β_F can lead to residual TC after applying the proposed one-point trimming.



(a)



(b)

Figure 4-17 Distribution of (a) the trimmed BGR TC, and (b) the trimmed BGR

V_{REF} across 10 chips at 27°C and $V_{DD}=0.5V$.

Figure 4-17(b) shows the variation in V_{REF} for 10 chips at 27°C, demonstrating a standard deviation of 0.37mV and a σ/μ variation of 0.04%. Figure 4-18 shows the

LNR of the proposed BGR at 27°C. The reference voltage increases by 1.64mV as the supply voltage increases from 1.2V to 2.5V, corresponding a LNR of 0.126%/V.

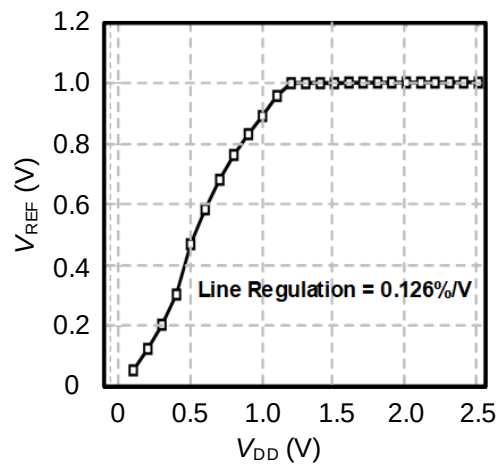


Figure 4-18 Measured LNR of a typical trimmed chip at 27°C.

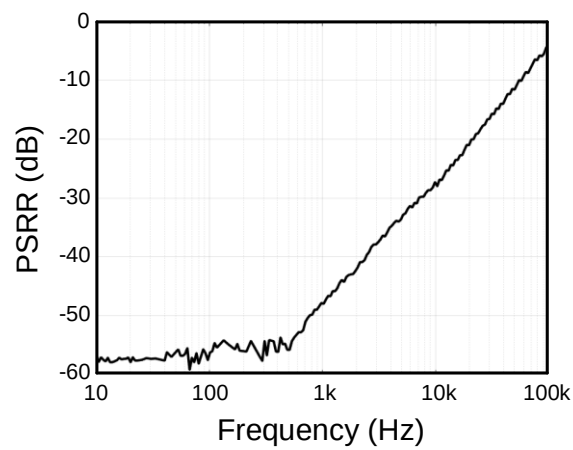


Figure 4-19 PSRR of the proposed BGR from 10Hz to 100kHz.

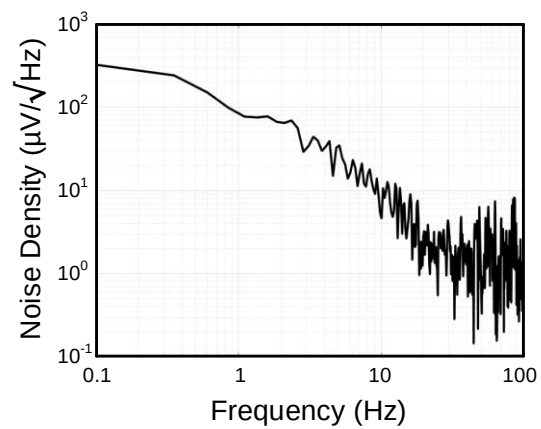


Figure 4-20 Noise spectrum of V_{REF} from 0.1Hz to 100Hz.

Table 4-2 Comparison of BGR Metrics with Other Low Power and Low Supply Voltage designs

Specification	This work	[4.7]	[4.14]	[4.15]	[4.16]	[4.17]	[4.3]	[4.4]	[4.5]	[4.6]	[4.12]
Year	2022	JSSC 2021	TCAS-I 2017	JSSC 2021	TCAS-I 2015	JSSC 2013	JSSC 2021	JSSC 2011	TCAS-I 2019	TCAS-I 2018	JSSC 2017
Technology	65nm	0.13 μ m	0.18 μ m	0.18 μ m	90nm	0.18 μ m	0.18 μ m	0.16 μ m	0.13 μ m	0.5 μ m	65nm
Supply Voltage (V)	1.2 – 2.5	3.3	1.3 – 1.8	1.3 – 1.8	1.15 – 1.3	1.2 – 1.8	1.8 $\pm$ 10%	1.8 $\pm$ 10%	2.6 – 5	2.1 – 5	0.8 – 1.4
Current Consumption (μ A)	0.81	120	28	0.15	0.5	0.08	17	116	94	38	16.25
Average V_{REF} (mV)	1001.1	1160	547	1170	723	1090	1141.9	1087.5	2471	1196	428
Auto Calibration	Yes	No	No	No	No	No	No	No	No	No	No
Trimming Method	Auto 1-point	Untrim	Untrim	Untrim	Untrim	Untrim	Manual 1-point	Manual 1-point	Manual 1-point	Manual 1-point	Manual 2-point
Estimated 1σ inaccuracy @Entire Temp. Range (%)	$\pm$ 0.12	$\pm$ 0.54	N.A.	N.A.	N.A.	N.A.	(+0.01, -0.4)	$\pm$ 0.05	N.A.	N.A.	N.A.
Estimated 1σ inaccuracy @Room Temp. (%)	0.04	N.A.	0.16*	0.33	1.3	0.73	N.A.	N.A.	0.2	0.65	0.39
Trimmed TC (ppm/ $^{\circ}$ C)	18.6-28.8 (μ :22.3)	5.8 – 13.5 (μ :8.75)	1.7 – 10.6 N.A.	12.2 – 41 (μ :26.3)	43.5 – 60.7 (μ :53.1)	N.A. (μ :147)	3.2 – 5.5 (μ :4.3)	5 – 12 N.A.	0.9 – 7.52 (μ :3.0)	3.98 – 8.29 (μ :5.87)	3.2 – 9.8 (μ :5.6)
Curvature Correction	No	Yes	Yes	No	No	No	Yes	Yes	Yes	Yes	Yes
Temp. Range ($^{\circ}$ C)	-40 – 120	-40 – 150	-40 – 140	-40 – 140	0 – 100	-40 – 120	-40 – 125	-40 – 125	-45 – 125	-5 – 125	-40 – 125
PSRR	58 dB @10Hz	82 dB @ 10Hz	62 dB @DC	52 dB @100Hz	50 dB @DC	62 dB @100Hz	76 dB @DC	74 dB @DC	83 dB @DC	84 dB @100Hz	87 dB @10Hz
Active Area (mm ²)	0.04	0.08	0.0094	0.0082	0.028	0.0294	0.38	0.12	0.0616	0.053	0.0104
# of sample	10	7	8	15	5	9	18	61	10	3	50

* Estimated from figure

Figure 4-19 shows the PSRR of the proposed BGR from 10Hz to 100kHz. The value at 10 Hz is around -58 dB, and at 1 kHz, the PSRR is around -48 dB. Figure 4-20 shows the noise spectrum of V_{REF} from 0.1Hz to 100Hz, with a measured noise integrated up to 10Hz of 457 μ V. The current consumption of the BGR with the calibration engine activated is 2.18 μ A. After the calibration engine is shut down, the current consumption reduces to 810nA. Table 4-2 compares the proposed BGR with the state of the arts. For the BGRs without trimming in [4.7], [4.14]–[4.17], this work can achieve a much smaller 1σ inaccuracy. Compared with those works without curvature correction [4.15]–[4.17], this work achieves the best TC, the smallest σ/μ , and the widest supply voltage range. Compared with the design in [4.15], this work can achieve 89% smaller σ/μ despite the low power and area consumption. Moreover, this work can achieve $1.6\times$ wider temperature range than [4.16]. Even though the works with curvature correction [4.7] and [4.14] can achieve a smaller TC, they consume about $150\times$ and $35\times$ more current, respectively. For the BGRs with manual trimming in [4.3]–[4.6], and [4.12], this work achieves a comparable or even better σ/μ after applying our proposed auto-calibration technique. In addition, the proposed auto-calibration technique can relax the costly labor calibration process. The measurement setup is shown in Figure 4-21.

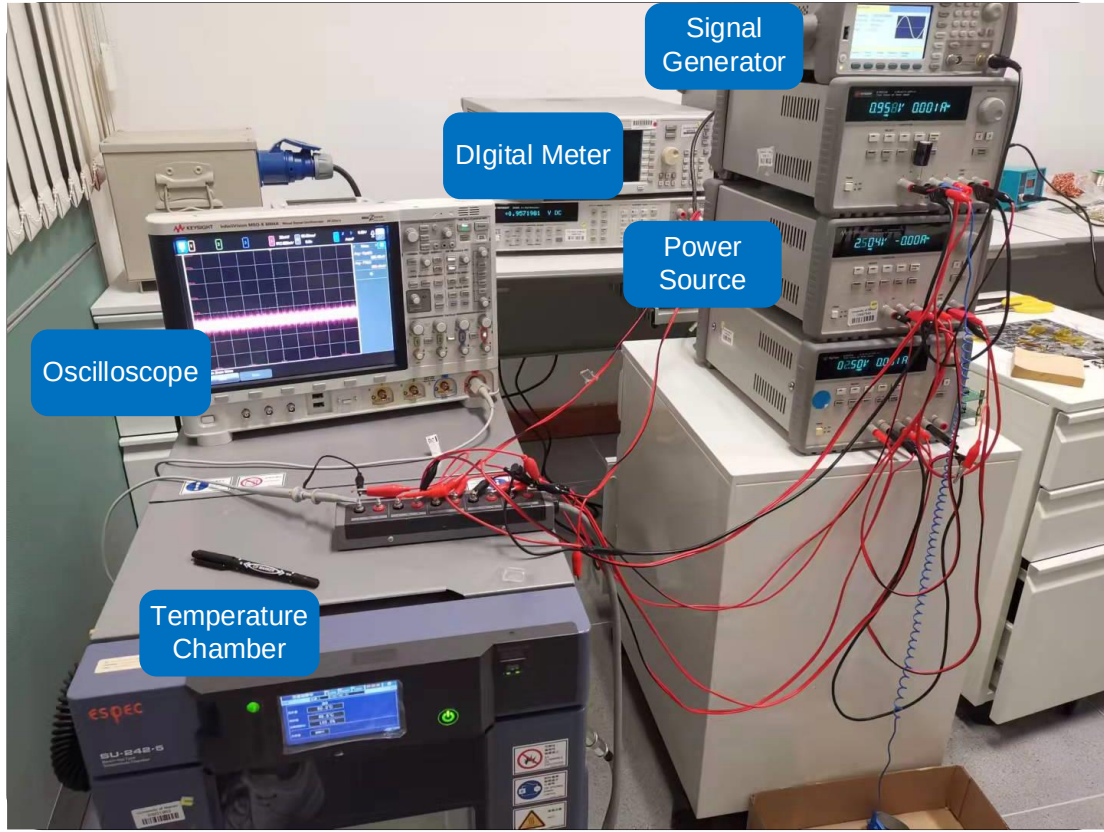


Figure 4-21 Testing environment.

4.5 SUMMARY

This work presented an auto-calibration technique for current-based BGR, which can reduce the expensive labor calibration cost usually required after fabrication to ensure V_{REF} accuracy in industrial applications. The proposed digital-assisted auto-calibration loop can relax the costly calibration process. First, we explored the relationship between the process corner variation on I_S , η_F , and β_F of the BJT and the V_{EB} variation. Utilizing this relationship, we theoretically developed a trimming ratio $\Delta m/\Delta n$ of the PTAT and CTAT coefficients, which is utilized in the auto calibration circuit to compensate both the drift in V_{REF} and its TC simultaneously. We also

proposed an IDAC for auto trimming. Finally, measurement results confirmed that the proposed auto-calibrated BGR obtains a measured average TC of 22.3 ppm/°C at 1.2V supply from −40°C to 120°C. The LNR is 1.26mV/V or 0.13%/V from 1.2V to 2.5V. Based on 10-chip measurement results, this work can improve the σ/μ variation in V_{REF} from $\pm 0.53\%$ before trimming to $\pm 0.12\%$ after applying the proposed auto-calibration technique at 27°C.

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Chapter 5 CONCLUSIONS AND PERSPECTIVE FOR FUTURE WORKS

5.1 CONCLUSIONS

This thesis proposes different BGR circuits for IoT application to achieve low power consumption and high accuracy. The key contributions and results of this thesis are summarized in below:

- 1) In Chapter 2, we present a SCN BGR that can achieve low power consumption without using a large amount of resistance or large chip area consumption issue. We proposed a second-order effect analysis for relaxing SCN non-ideal effect. A PTAT current source is employed to reduce V_{EB} nonlinearity, and a low-leakage design is implemented in the voltage divider SCN, thus improving both TC and operating temperature range.
- 2) In Chapter 3, we present a SCN BGR using a dual clock path topology to reduce the capacitor size and area. A fast PTAT clock assists in reducing the output voltage ripple of the dual-branch interleaved $2\times CP$. A multi-branch V_{EB} generation is employed to relax the settling error of series-parallel SCN in order to reduce the TC. A slow PTAT clock is employed to relax the settling error of the voltage divider SCN to reduce power consumption and improve the TC.
- 3) In Chapter 4, we present an auto-calibration technique for BGR that can adjust both V_{REF} and its TC simultaneously by using only one-point trimming. A

systemically analysis of the process variation induced BJT voltage drift is also presented for designing the trimming circuit.

5.2 FUTURE WORKS

The voltage reference is a critical building block of the ICs for IoT application, as it may limit the system's efficiency and accuracy. To achieve low power consumption and high accuracy properties, the SCN BGR circuits and the trimming technique are presented in this thesis. At the time of completing the thesis, the minimum supply voltage for the BGR remained limited to 0.5V within an operating temperature range of -40 to 120°C. Besides, Designing an auto-trimmed bandgap reference with curvature compensation remains a challenging problem. Based on the presented research works, the suggested future works are listed as follows:

- 1) Due to the charge redistribution process in the current SCN BGR topology, it requires a large charging current to relax the settling error. The SCN using flying capacitor may be a promising solution to solve this problem, potentially reducing power consumption and minimum supply voltage ($<0.35\text{V}$).
- 2) A leakage-based compensation method that requires almost no power can be applied to the SCN BGR for TC suppression ($<10\text{ ppm}/^\circ\text{C}$ and $<50\text{nW}$). The nonlinear gate-induced leakage current or the nonlinear drain-source leakage current can be a potential solution.
- 3) An automatic multi-point trimming (two-point or even three-point) can be developed by using the proposed analysis of process variation induced BJT voltage drift in this thesis, potentially reducing the TC and σ inaccuracy ($<5\text{ ppm}/^\circ\text{C}$ and $<\pm 0.05\%$).

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